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The self-consistent field energy landscape and its application to describing electron correlation.

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THE SELF-CONSISTENT FIELD ENERGY LANDSCAPE AND ITS APPLICATION TO DESCRIBING ELECTRON CORRELATION

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B.S. in ACS Certificate, Western Kentucky University, 2018
M.S. in Chemistry, University of Louisville, 2021

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DEDICATION

This work is dedicated for my curiosity for quantum mechanics and coding. I received different trainings and experience in both fields and this curiosity has converted into the passion during this PhD. I am looking forward to my growth in the next journey.

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ABSTRACT

THE SELF-CONSISTENT FIELD ENERGY LANDSCAPE AND ITS APPLICATION TO DESCRIBING ELECTRON CORRELATION

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Chemical reaction mechanisms are determined by the change in electron-electron interactions at different nuclear geometries. Owing to the electron mass, electron-electron interactions must be described using quantum mechanics. To further pursue the accurate description of electron interactions, it is necessary to determine the electronic wavefunction through development of computationally efficient techniques. In this thesis, we have investigated the construction of wavefunctions composed of separately optimized self-consistent field solutions. In order to enable such computational approaches, developing efficient searching algorithm among high-dimensional wavefunction parameter space is necessary to locate unique self-consistent field (SCF) solutions, which serve as basis for the reference wavefunction. When using the self-consistent field solutions as a basis to construct a wavefunction, it is necessary to understand how each self-consistent field solution captures the different electron interactions present in a system, as well as how specific linear combinations represent different forms of electron correlation. Thus this work consists of one part in which techniques for the identification of self-consistent field solutions is discussed, and another part in which the connections of self-consistent field solutions to electron correlation is discussed. Using the developments in these first two projects, in the

third project we seek to determine the extent to which wave functions based on linear-combinations of self-consistent field solutions improve performance over other well-used wave function approximations, by studying the field response in real-time time-dependent simulations.

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INTRODUCTION

Method development has been extremely crucial part in the quantum mechanics field. Many algorithms have been proposed to solve the Schrödinger equation of studied systems. Hartree Fock (HF) is the fundamental wavefunction algorithm which serves as the starting point for the advanced computational methods. Since the accurate understanding of chemical reactions is based on the precise description of electronic states, capturing correct amount of correlations for each electronic state matters. Post-HF methods such as perturbation theory, coupled cluster, complete-active-space SCF (CASSCF) and configuration interactions (CI) are beyond HF method by capturing the missed correlations. According to the requirements of correlations from different molecular systems, these post-HF methods can be combined, dissected or adding the new flavors to the wavefunction to achieve the better accuracy with reasonable computational cost. An alternative post-Hartree Fock approach that we investigate in this work, is the construction of wavefunctions from linear combinations of independently optimized self-consistent field solutions. Such an approach requires development of techiques for locating solutions, and development of understanding how different self-consistent field solutions capture electron-electron interactions.

A preliminary for constructing wavefunctions formed from linear-combinations of self-consistent field solutions is identification of the set of solutions to the Hartree-Fock equations (global elucidation). Global elucidation of SCF solution among high-dimensional wavefunction parameter space is a NP-hard (non-polynomial) problem, and few methodologies are proposed so far to address this issue, such as grid search (1) using the Lie algebraic description of wavefunction parameters, homotopic method(2) and metadynamics(3). Basin hopping method is originated from the biology field(4), and we take advantage of this global optimization method to do the global elucidation of SCF solutions in this case which allows the Lie-algebraic description of wavefunction applying to the larger molecular systems compared with grid search method. Since Lie-algebraic description can determine the wavefunction type, the properties of different wavefunction classes are explored for the single Slater determinant, and these identified SCF solutions can be used as the basis for the post-HF calculations. Besides, the recent proposed active space decomposition methodology(5) reveals that the SCF solutions serve as the indicator for the correlation mechanism. As demonstrated in the subsequent chapters, there are several applications for the identified SCF solutions.

In the second chapter, basic quantum mechanics concepts, HF wavefunction method, post-HF wavefunction methods and orbital entropy in quantum information are discussed. The orbital entropy and orthogonal/nonorthogonal configuration interactions are detailed in methodology part in chapter four and five, respectively. Chapter three shows the work regarding to the global elucidation of basin hopping method, H_4, benzene and NO_2 are used to demonstrate the novelty and efficiency of our proposed methodology. Chapter four applies the disconnectivity graph to describe the correlation mechanism and orbital entropy is introduced to further differentiate the multiple correlation mechanism of SCF solutions from orbital levels. H_2 in STO-3G basis set is used to set the rules for the disconnectivity graph, and ozone is selected to explore the multiple correlation mechanism using the quantum information concepts. Chapter five explores the nonorthogonality within real-time time-dependent configuration interactions (CI) expansions in polarizability, second-order hyperpolarizability, and HHG spectrum in three molecules, and the comparison between non-orthogonal CI (NOCI) and orthogonal CI (OCI) are discussed in different basis sets. Since finite-field method (6) is introduced to extract the electronic responses without performing the Fourier transfrom, the balance of convergent field and fitting percentage is the major technical issue to address.

QUANTUM CHEMICAL AND COMPUTATIONAL THEORY

Hartree Fock (HF) is the fundamental wavefunction method proposed to solve the time-independent Schrödinger equation. This method introduces the spin orbital picture to describe each electron within the molecular system and best set of spin orbitals can be obtained via diagonalization to approximate desired electronic state. Besides, HF method is also the starting point for many advanced approximations which can capture the electron correlations, (7) since the accurate description of all chemical reactions is highly dependenpent on the quality of the electron correlations. In addition, the quantum information is illustrated since the orbital entropy is used to differentiate the multiple correlation mechanisms between different SCF solutions in chapter 4. In this chapter, the basic concepts involved in the quantum chemistry, the mechanism of HF, symmetry of HF wavefunctions, HF optimizations, post-HF and quantum information theory are described below.

2.1 Basic Quantum Mechanics

Before diving into the quantum chemistry theory, it is necessary to introduce the basic concepts involved in any theoretical methods. In this case, operator, wavefunction, time-dependent and time-independent Schrödinger equation, antisymmetry, Slater determinants, variational principle and Born-Oppenheimer approximation are chosen to discuss in this section.

Operator and Wavefunction

An operator is a mapping rule that maps given values to the new values.(8) If changing the order of multiple operators do not affect the final mapping result, one can say these operators commute. Besides operator, wavefunction is another basic concept in quantum chemistry. The wavefunction is the solution of the Schrödinger equation, and its square represents the density probability of the electron. If one applies operator $\hat{A}$ to the function f(x), the result is simply a constant k multiplying by the same function f(x), and f(x) is the eigenfunction of operator $\hat{A}$ with eigenvalue k. For example, the exact wavefunction is the eigenfunction for Hamiltonian operator $\hat{H}$, and the corresponding eigenvalue is the energy E.

Time-Dependent and Time-Independent Schrödinger Equation

For the sake of simplicity, the time-dependent Schrödinger equation (TDSE) for a one-particle in one-dimensional system is chosen to discussed here, which can be written as:

$$-\frac{\hbar}{i}\frac{\partial\Psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2} + V(x,t)\Psi(x,t) \tag{2.1.1}$$

where $\hbar$ is defined as $\hbar \equiv \frac{h}{2\pi}$, h is Planck constant, m is the mass of the particle, and V(x,t) is the potential function based on the position and time. $\Psi(x,t)$ is time-dependent wavefunction, which depends on position and time. Since time-dependent (orthogonal and nonorthogonal) configuration interactions are used to solve (2.1.1) in chapter 5, the details of the relevant proposed algorithms are not illustrated here.

Time-independent Schrödinger equation (TISE) can be drived from TDSE if allows the wavefunction only depends of the position x, and the time-dependent wavefunction can be decompose based on position x and time t:

$$\Psi(x,t) = f(t)\psi(x) \tag{2.1.2}$$

and use (2.1.2) to replace the wavefunction in TDSE, one obtains:

$$-\frac{\hbar}{i}\frac{df(t)}{dt}\psi(x) = -\frac{\hbar^2}{2m}f(t)\frac{d^2\psi(x)}{dx^2} + V(x)f(t)\psi(x) \tag{2.1.3}$$

If dividing right hand side of (2.1.2) from both sides:

$$-\frac{\hbar}{i}\frac{1}{f(t)}\frac{df(t)}{dt} = -\frac{\hbar^2}{2m}\frac{1}{\psi(x)}\frac{d^2\psi(x)}{dx^2} + V(x) \tag{2.1.4}$$

Since left side of this equation is depends on the time and right side is depends on the position, the only way to satisfy the equation is to let both sides equal to a constant. As a result, one has time-independent Schrödinger equation below:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x) \tag{2.1.5}$$

where E is the constant, which represents energy. Since the probability density given by wavefunction $\psi(x)$ from TISE does not change along time, it is defined as the stationary state.

Antisymmetry and Slater determinants

The antisymmetry principle requires all many-electron wavefunction must be antisymmetric if the interchange of the coordinate (both spatial and spin) of any two electrons due to the Pauli Exclusion principle in general chemistry.(7) Since the Slater determinants can satisfy the antisymmetry requirement easily, many-electron wavefunction uses it as the common representation. For example, if one considers a two-electron system:

$$\Phi_{12}^{HP}(x_1, x_2) = \psi_i(x_1)\psi_j(x_2) \tag{2.1.6}$$

where x_1 is the coordinates of electron 1 and x_2 is the coordinates of electron 2. If one switches two electrons, the wavefunction can be written as:

$$\Phi_{21}^{HP}(x_1, x_2) = \psi_i(x_2)\psi_j(x_1) \tag{2.1.7}$$

Neither wavefunction satisfies the antisymmetry condition, but the linear combination of the two can reach the antisymmetry requirement:

$$\Phi(x_1, x_2) = 2^{-1/2}(\psi_i(x_1)\psi_j(x_2) - \psi_i(x_2)\psi_j(x_1)) \tag{2.1.8}$$

in which $2^{-1/2}$ is the normalization factor and minus sign between two wavefunction is to ensure the antisymmetry. This antisymmetric function can also be rewritten as a determinant:

$$\Phi(x_1, x_2) = 2^{-1/2} \begin{vmatrix} \psi_i(x_1) & \psi_j(x_1) \\ \psi_i(x_2) & \psi_j(x_2) \end{vmatrix} \tag{2.1.9}$$

The generalization of N-electron system can be rationalized as:

$$\Phi(x_1, x_2, \ldots, x_N) = (N!)^{-1/2} \begin{vmatrix} \psi_i(x_1) & \psi_j(x_1) & \ldots & \psi_k(x_1) \\ \psi_i(x_2) & \psi_j(x_2) & \ldots & \psi_k(x_2) \\ \vdots & \vdots & & \vdots \\ \psi_i(x_N) & \psi_j(x_N) & \ldots & \psi_k(x_N) \end{vmatrix} \tag{2.1.10}$$

The constant $(N!)^{-1/2}$ is a normalization factor. In this N electron in N spatial orbital system, each row represents the electron and each column symbolizes spatial orbital. If interchanging any two rows, the sign of the determinant will change which follows the Pauli Exclusion principle. This generalized Slater determinant can also be written as:

$$\Phi(x_1, x_2, \ldots, x_N) = |\psi_i(x_1)\psi_j(x_2) \ldots \psi_k(x_N)\rangle \tag{2.1.11}$$

in which only the diagonal terms are shown. The spatial orbitals within the Slater determinant are orthogonal to each other, but spatial orbitals between different Slater determinants are not necessary satisfy this condition. Besides, the antisymmetry also leads to the formation of the exchange integral, and it is interpreted that the motion of two electrons with parallel spins is correlated. The specific form of exchange integral is shown in Hartree Fock section below.

Variational Principle

The variational principle allows us to obtain the upper boundary for the system's ground state energy(8):

$$\int \Phi^* \hat{H} \Phi d\tau \geq E_0 \tag{2.1.12}$$

where E_0 is the exact ground state energy we try to approach, $\hat{H}$ is the Hamiltonian operator and Φ is the normalized approximate wavefunction. Φ can be written as the linear combination of exact set of stationary wavefunctions which forms a complete set:

$$|\Phi\rangle = \sum_i c_i |\Psi_i\rangle \tag{2.1.13}$$

If plugging this expansion into Rayleigh ratio:

$$\begin{aligned}
E =& \frac{\langle|\Phi|\hat{H}|\Phi\rangle}{\langle\Phi|\Phi\rangle} = \frac{\langle(\sum_i c_i\Psi_i)|\hat{H}|(\sum_j c_j\Psi_j)\rangle}{\langle(\sum_i c_i\Psi_i)|(\sum_j c_j\Psi_j)\rangle} \\
=& \frac{\sum_{ij} c_i^* c_j \langle\Psi_i|\hat{H}|\Psi_j\rangle}{\sum_{ij} c_i^* c_j \langle\Psi_i|\Psi_j\rangle} = \frac{\sum_{ij} c_i^* c_j E_j \langle\Psi_i|\Psi_j\rangle}{\sum_{ij} c_i^* c_j \langle\Psi_i|\Psi_j\rangle} \\
=& \frac{\sum_{ij} c_i^* c_j E_j \delta_{ij}}{\sum_{ij} c_i^* c_j \delta_{ij}} = \frac{\sum_i |c_i|^2 E_i}{\sum_j |c_j|^2} \geq E_0
\end{aligned} \tag{2.1.14}$$

in which δ_{ij} is Kronecker delta that satisfies the orthonormal requirement for each molecular orbital pair i&j, E_i is the obtained approximated ground state energy, and E_0 is the exact energy for the ground state. The last step of this equation holds true because E_0 is the lowest energy and the square of the coefficient of (2.1.14) is a positive number.

Born-Oppenheimer Approximation

In Born-Oppenheimer approximation, electronic and nuclei motions are separated.(8) Since the mass of nuclei is much larger than the electron which causes its slow moving speed, one ignores the motion of nuclei but still counting their repulsion effects for the total energy. The molecular energies are using the electronic Hamiltonian $\hat{H}_{el}$ below:

$$\hat{H}_{el} = -\frac{\hbar^2}{2m_e}\sum_i \nabla_i^2 - \sum_\alpha\sum_i \frac{Z_\alpha e^{'2}}{r_{i\alpha}} + \sum_j\sum_{i>j}\frac{e^{'2}}{r_{ij}} \tag{2.1.15}$$

where $\hbar$ is the reduced Planck constant, m_e is the mass of the electron, Z_α is the charge of the nuclei, $e^{'}$ is the charge of the electron, $r_{i\alpha}$ is the distance between nuclei and electron, and r_{ij} is the distance between two electrons. And corresponding electronic energy can be obtained through:

$$\hat{H}_{el}\Psi_{el} = E_{el}\Psi_{el} \tag{2.1.16}$$

in which $\hat{H}_{el}$ is electronic Hamtonian, Ψ_{el} is the electronic wavefunction, and E_{el} is the electronic energy in theory. The last step is to add nuclei repulsion energy back:

$$E_{total} = E_{el} + \sum_{\alpha}^{N_{atoms}} \sum_{\beta>\alpha}^{N_{atoms}} \frac{Z_\alpha Z_\beta}{R_{\alpha\beta}} \tag{2.1.17}$$

where Z_α and Z_β are two nuclei charges, and R_{AB} is the distance between two nuclei. Since this section is the simple explanation, the details of how to compute total energy is illustrated in Hartree Fock Theory chapter.

2.2 Hartree Fock Theory

Hartree Fock (HF) is a simple approximated wavefunction method, which serves as the starting point for many advanced computational methods. Each HF Slater determinant represents an electronic state which contains a corresponding set of spin orbitals. The single molecular orbital energy can be determined by diagonalizing the Fock matrix, which are discussed below. The more specific form of spin orbitals can be described through the basis set idea, thanks to the Roothaan. The spatial part of spin orbitals (molecular orbital)will be written as the linear combination of atomic orbitals, and diagonalization of Fock matrix updates the coefficient in front of each molecular orbital until it converges. The details of this self-consistent-field (SCF) procedure is discussed below. After the SCF procedure, each molecular orbital energy is obtained but the summation of them are not the total electronic energy of the molecule. In order to get total molecular energy, one-electron and two-electron integrals are computed, as well as the nuclear repulsion energy.

Fock Operator

Fock operator is a one-electron operator, which contains core-Hamiltonian operator $\hat{h}$, coulomb potential $\hat{J}$ and exchange potential $\hat{K}$:

$$\hat{f}(1) = \hat{h}(1) + \sum_{j}^{N_{elec}} [\hat{J}_j(1) - \hat{K}_j(1)] \tag{2.2.1}$$

in which the core-Hamiltonian includes the kinetic energy and potential energy respect to the nuclei for each single electron,

$$\hat{h}(i) = -\frac{1}{2}\nabla_i^2 + \sum \frac{Z_A}{r_{iA}} \tag{2.2.2}$$

where Z_A is nuclei A's charge and r_{iA} is the distance between electron i to the nuclei A. Coulomb potential shows the coulomb repulsion between spin orbitals and exchange potential exists due to the antisymmetry properties required in Slater determinant. N number of one-electron HF equations can be written as:

$$[\hat{h}(1) + \sum_{i \neq j}^{N_{elec}} \hat{J}_j(1) - \sum_{i \neq j}^{N_{elec}} \hat{K}_j(1)]\psi_i(1) = \varepsilon_i \psi_i(1) \tag{2.2.3}$$

where this whole equation can be viewed as the eigenequation, ε_i is the eigenvalue that is orbital energy for spatial orbital ψ_i (eigenfunction) obtained after diagonalizing the fock matrix. If we consider Roothaan's equation, the spatial orbital can be written as the linear combination of atomic orbitals:

$$\psi_i(1) = \sum_{\mu=1}^{N_{AO}} C_{\mu i}\phi_\mu(1) \tag{2.2.4}$$

where μ is atomic orbital symbol and $C_{\mu i}$ is coefficient in front of each atomic orbitals, which is called molecular orbitals' (MOs) coefficients. If inserting (2.2.4) into the fock operator and multiply by ϕ_ν^*,

$$\sum_{\nu}^{N_{AO}} C_{\nu i} \int d_1 \phi_\nu^*(1)\hat{f}(1)\phi_\mu(1) = \varepsilon_i \sum_{\nu}^{N_{AO}} C_{\nu i} \int d_1 \phi_\nu^*(1)\phi_\mu(1) \tag{2.2.5}$$

where $\int d_1\phi_\nu^*(1)\hat{f}(1)\phi_\mu(1)$ can be written as $F_{\nu\mu}$ which is fock matrix and $\int d_1\phi_\nu^*(1)\phi_\mu(1)$ is $S_{\nu\mu}$ which is overlap matrix. The Roothaan equation above is also the eigenfunction, and its eigenvalue is orbital energy relative to the selected basis set. If we consider the wavefunction type, for example, the real restricted (RR) HF method, the fock matrix element is shown as:

$$\hat{f}_{\mu\nu} = h_{\mu\nu} + \sum_{i}^{N_{elec}/2} [2(\mu\nu|ii) - (\mu i|i\nu)] \tag{2.2.6}$$

The specific forms of one-electron and two-electrons integrals involved in the formulas above are shown below, respectively:

$$(\mu|\hat{h}|\nu) = \int dx_1 \psi_\mu^*(x_1)\hat{h}(r_1)\psi_\nu(x_1) \tag{2.2.7}$$

$$(\mu\nu|ii) = \int dx_1 dx_2 \psi_\mu^*(x_1)\psi_\nu(x_1)\frac{1}{r_{12}}\psi_i^*(x_2)\psi_i(x_2) \tag{2.2.8}$$

$$(\mu i|i\nu) = \int dx_1 dx_2 \psi_\mu^*(x_1)\psi_i(x_1)\frac{1}{r_{12}}\psi_i^*(x_2)\psi_\nu(x_2) \tag{2.2.9}$$

And if inserting eq.(2.2.4) into the eq.(2.2.6), one obtains:

$$f_{\mu\nu} = h_{\mu\nu} + \sum_{i}^{N_{elec}/2} \sum_{\lambda\sigma}^{N_{AO}} C_{\lambda i}^* C_{\sigma i}[2(\mu\nu|\lambda\sigma) - (\mu\sigma|\lambda\nu)] \tag{2.2.10}$$

where $C^*_{\lambda i}C_{\sigma i}$ is density matrix, and this is the fock matrix being diagonalized in the SCF procedure. If one considers the unrestricted HF wavefunction, the fock matrix will be constructed for alpha and beta spin:

$$f^{\alpha}(1) = h(1) + \sum_{a}^{N^{\alpha}}[J^{\alpha}_{a}(1) - K^{\alpha}_{a}(1)] + \sum_{a}^{N^{\beta}} J^{\beta}_{a}(1) \tag{2.2.11}$$

where the last coulomb interaction occurs between alpha and beta spin, and same term also exist in fock matrix for beta spin:

$$f^{\beta}(1) = h(1) + \sum_{a}^{N^{\beta}}[J^{\alpha}_{a}(1) - K^{\beta}_{a}(1)] + \sum_{a}^{N^{\alpha}} J^{\alpha}_{a}(1) \tag{2.2.12}$$

in which two fock matrices need to be solve simultaneously for each SCF iteration. In order to simplify the interpretation, real-restricted (RR) wavefunction will be used as the example to explain the HF energy and SCF procedure in the following section.

Hartree Fock Energy

The total electronic energy is not the summation of all orbital energies after diagonalizing the fock matrix. The energy formula requires to compute one-electron, two-electron energies and nuclei repulsion energy. One-electron energy can be derived through:

$$\begin{aligned}
E_0 &= \langle\Psi_{SD}|\sum_i \hat{h}_i|\Psi_{SD}\rangle = \sum_i(\langle\prod_k \psi_k|\hat{A}^{\dagger}\sqrt{^{N}elec!})\hat{h}_i(\sqrt{^{N}elec!}|\hat{A}\prod_j \psi_j\rangle)\\
E_0 &= \sum_i N_{elec!}\langle\prod_k \psi_k|\hat{h}_i|\hat{A}\prod_j \psi_j\rangle = \sum_i\langle\prod_k \psi_k|\hat{h}_i|\sum_{\pi\in S_N}\epsilon_\pi\hat{\pi}\prod_j \psi_j\rangle\\
E_0 &= \sum_i\sum_{\pi\in S_N}\epsilon_\pi\langle\prod_k \psi_k|\hat{h}_i|\hat{\pi}\prod_j \psi_j\rangle = \langle\psi_1|\hat{h}_1|\psi_1\rangle + \langle\psi_2|\hat{h}_2|\psi_2\rangle + \cdots = \sum_i\langle\psi_i|\hat{h}_i|\psi_i\rangle
\end{aligned} \tag{2.2.13}$$

in which Ψ_{SD} is the exact wavefunction, ψ is the spatial orbitals (i, j, k,1,2 are subscripts), $\hat{A}^{\dagger}(\hat{A}^{\dagger}{=}\hat{A})$ is antisymmetry operator, $\hat{h}_i$ is one-electron operator, $\sqrt{^{N}elec!}$ is normalization constant, ϵ_π is the parity of permutation (±1) and $\hat{\pi}$ is the permutation operator. The two-electron energy can be obtained in a similar way:

$$\begin{aligned}
E_1 &= \langle\Psi_{SD}|\sum_{i<j} r_{ij}^{-1}|\Psi_{SD}\rangle = \sum_{i<j}(\langle\prod_k \psi_k|\hat{A}^{\dagger}\sqrt{^{N}elec!})r_{ij}^{-1}(\sqrt{^{N}elec!}|\hat{A}\prod_l \psi_l\rangle)\\
E_1 &= \sum_{i<j} N_{elec!}\langle\prod_k \psi_k|r_{ij}^{-1}|\hat{A}\prod_l \psi_l\rangle = \sum_{i<j}\langle\prod_k \psi_k|r_{ij}^{-1}|\sum_{\pi\in S_N}\epsilon_\pi\hat{\pi}\prod_l \psi_l\rangle\\
E_1 &= \sum_{i<j}\sum_{\pi\in S_N}\epsilon_\pi\langle\prod_k \psi_k|r_{ij}^{-1}|\hat{\pi}\prod_l \psi_l\rangle = \sum_{i<j}\langle\psi_i\psi_j|r_{ij}^{-1}|\psi_i\psi_j\rangle - \langle\psi_i\psi_j|r_{ij}^{-1}|\psi_j\psi_i\rangle
\end{aligned} \tag{2.2.14}$$

where r_{ij}^{-1} is two-electron operator, $\langle\psi_i\psi_j|r_{ij}^{-1}|\psi_i\psi_j\rangle$ is Coulomb interactions and $\langle\psi_i\psi_j|r_{ij}^{-1}|\psi_j\psi_i\rangle$ is exchange interactions. The last term is the nuclei repulsion energy that can be computed as:

$$V_{NN} = \sum_{A}^{N_{atom}} \sum_{B>A}^{N_{atom}} \frac{Z_A Z_B}{R_{AB}} \tag{2.2.15}$$

in which Z_A Z_B are nuclei charge, and R_{AB} are the distance between nuclei A and nuclei B. Since all the terms are derived so far, the total electronic energy can be generated now:

$$E_{HF} = E_0 + E_1 = \sum_{i}^{Nelec} (i|\hat{h}|i) + \sum_{i>j}^{Nelec} [(ii|jj) - (ij|ji)] + \sum_{A}^{N_{atom}} \sum_{B>A}^{N_{atom}} \frac{Z_A Z_B}{R_{AB}} \tag{2.2.16}$$

in which $(i|\hat{h}|i)$ is one-electron molecular orbital integral, that symbolizes the average properties for one electron. (ii|jj) and (ij|ji) are coulomb integral and exchange integral for the molecular orbitals, which are the most computational expensive step in fock matrix. The last term is the coulomb repulsion between different nuclei. Since we apply the Born-Oppenheimer approximation, this term is treated as the constant.

Self-Consistent-Field Procedure

The diagram below introduces the basic idea of the self-consistent-field (SCF) procedure for RR (real-restricted) wavefunction in Figure 2.1. There are several details need to mention here: 1).This is a nonlinear process, since fock matrix is built upon molecular orbital coefficients (MOs) and diagonalize this fock matrix provides new set of MOs and the new set of molecular orbital energies. 2).Diagonalizing fock matrix should be in orthogonal basis and other steps should be done in non-orthogonal basis. 3).Density is constructed from MOs for each iteration and it also determines the end of this process if the convergence threshold is reached. The transformation matrix resulting from diagonalizing the overlap matrix can convert between non-orthogonal and orthogonal basis. This transformation is either obtained from symmetry orthogonalization or canonical orthogonalization. The convergence threshold is normally around 10^{-9} for density matrix. Finally, SCF method facilitates the computing of total molecular energy by using matrix multiplications for the electron integrals, which further accelerates the method developments in computational chemistry field.

2.3 Symmetries of Hartree-Fock wavefunctions

Lie algebraic approach is the fundamental algorithm to generate initial guess of SCF solutions in chapter 3, the details of how to characterize the SCF solution space using the Lie algebra and how to map back to the Lie group to obtain the initial guess are discussed in this section. The simplified version of this method is restate in the methodology part in chapter 3.

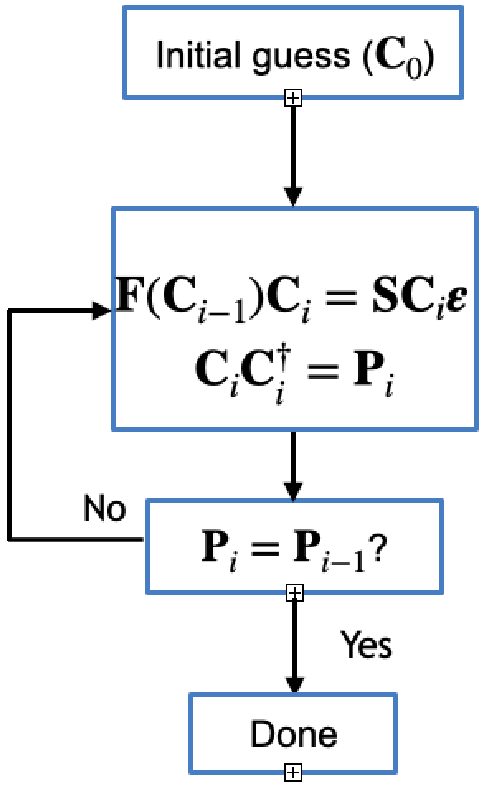


Figure 2.1. Self-Consistent Field Method.

Lie Algebraic Description of SCF Solution Space

Based on the Thouless' Theory(9):

$$|\Psi\rangle = e^{\hat{K}}|\Phi\rangle \tag{2.3.1}$$

that any Slater determinant $|\Psi\rangle$ can be transformed from another Slater determinant $|\Phi\rangle$ through the proper transformation matrix $e^{\hat{K}}$. However, the search of different SCF solution is nonlinear process. We can apply Lie algebra idea to provide a linear vector space (red surface in 2.3) in which the perturbation of molecular orbital coefficients (MOs) can be obtained easily. If one describes two SCF solutions by a similarity transformation of the creation operators based on the Baker-Campbell-Hausdorff expansion(1), MOs from one SCF solution can be related to another MOs of second SCF solution:

$$\tilde{C} = Ce^{\hat{K}} \tag{2.3.2}$$

where C is unperturbed MOs, $\tilde{C}$ is new MOs, $e^{\hat{K}}$ is the transformation matrix between two MOs and K is the Lie algebra. Since $\tilde{C}$ and C belong to the unitary group, which enforce $e^{\hat{K}}$ also belong to the unitary group. Thus, K is the anti-Hermitian matrix, and can be decomposed into:

$$K = iD + iS + A \tag{2.3.3}$$

in which D is diagonal matrix, S is the symmetric matrix, and A is the anti-symmstric matrix. Since one wants the wavefunction to be positive and real, the diagonal matrix D is projected out, and the remaining two terms form the special unitary groups ($\mathfrak{su}(2)$). In Figure 2.2, generalized Pauli matrices form basis vectors for each generator on the $\mathfrak{su}(2)$ linear vector space, eight generators transform the unique occupied-virtual orbital pairs through the rotation matrices such that they preserve different symmetries ($\hat{S}^2$,$\hat{S}_z$,$\hat{\theta}$,$\hat{C}$,$\hat{1}$) whose relations are further demonstrated in Table 2.1.

$$\mathbf{k}^{A,00} = ik^{A,00}\begin{bmatrix} \boldsymbol{\sigma}_y^{pq} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\sigma}_y^{pq} \end{bmatrix} \quad \mathbf{k}^{A,z} = ik^{A,z}\begin{bmatrix} \boldsymbol{\sigma}_y^{pq} & \mathbf{0} \\ \mathbf{0} & -\boldsymbol{\sigma}_y^{pq} \end{bmatrix} \quad \mathbf{k}^{A,x} = ik^{A,x}\begin{bmatrix} \mathbf{0} & \boldsymbol{\sigma}_y^{pq} \\ \boldsymbol{\sigma}_y^{pq} & \mathbf{0} \end{bmatrix} \quad \mathbf{k}^{A,y} = k^{A,y}\begin{bmatrix} \mathbf{0} & \boldsymbol{\sigma}_y^{pq} \\ -\boldsymbol{\sigma}_y^{pq} & \mathbf{0} \end{bmatrix}$$

$$\mathbf{k}^{S,00} = ik^{S,00}\begin{bmatrix} \boldsymbol{\sigma}_x^{pq} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\sigma}_x^{pq} \end{bmatrix} \quad \mathbf{k}^{S,z} = ik^{S,z}\begin{bmatrix} \boldsymbol{\sigma}_x^{pq} & \mathbf{0} \\ \mathbf{0} & -\boldsymbol{\sigma}_x^{pq} \end{bmatrix} \quad \mathbf{k}^{S,x} = ik^{S,x}\begin{bmatrix} \mathbf{0} & \boldsymbol{\sigma}_x^{pq} \\ \boldsymbol{\sigma}_x^{pq} & \mathbf{0} \end{bmatrix} \quad \mathbf{k}^{S,y} = k^{S,y}\begin{bmatrix} \mathbf{0} & \boldsymbol{\sigma}_x^{pq} \\ -\boldsymbol{\sigma}_x^{pq} & \mathbf{0} \end{bmatrix}$$

Figure 2.2. Eight generators on the linear vector space.

Figure 2.3 from grid search method (1) is used to better interpret the Lie algebraic approach for searching SCF solutions. In (a), it demonstrates the searching on the linear Lie algebra (red) and mapping back to the Lie group manifold (blue) to obtain the initial guess. The Lie algebra forms a tangent space form the original solution which lies on the z axis, and grid points are mapped back using truncated at second-order Zassenhaus' formula, which is discussed below. The large deviation of this mapping only observed when the rotation angle is larger than $\frac{\pi}{2}$ which are not considered in this case. However, the grid points increase exponentially as the size of the molecule, which lead to the original idea regarding using the basin hopping to do the global elucidation for SCF solutions in chapter 3.

After perturbing the MOs, we truncate Zaussenhaus' formula at the second order when mapping from linear vector space to Lie group:

$$e^{\sigma_1+\sigma_2} = e^{\sigma_1}e^{\sigma_2}e^{-\frac{1}{2}[\sigma_1,\sigma_2]} \tag{2.3.4}$$

where σ_1 and σ_2 represent first order perturbation matrix, and second order term can be constructed from first order terms. since it takes infinite terms to get the exact answer and higher-order terms do not contribute too much compared with the first two terms. If the first two order terms cause a non-negligible error during mapping, higher-order terms can be constructed by recursively applying the truncated second-order Zassenhaus' equation.

The next focus is to discuss how different generators preserve the certain symmetry of each wavefunction. In Table 2.1, the combination of generators that give rise to one symmetry-adapted method and seven broken spin-symmetry are shown, along

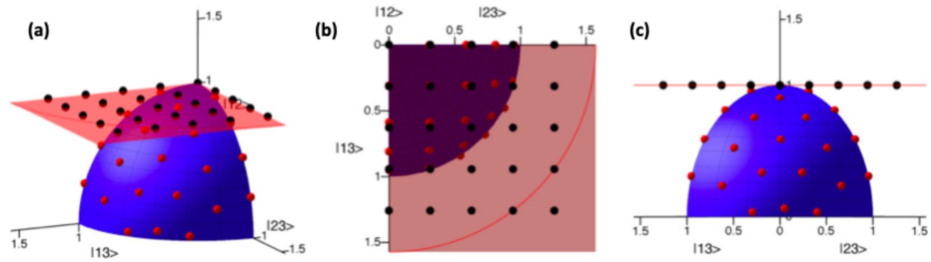


Figure 2.3. Lie algebraic description of SCF solution space. In (a), red surface is Lie algebra vector space, and blue sphere is corresponding Lie group. In (b), points near origin on Lie algebra linear vector space map closely for the points in Lie group, and points beyond red line will be discarded whose vector norms exceed $\frac{\pi}{2}$. In (c), through linearize SCF solution space on the Lie algebra linear vector space, different SCF guess can be generated easily and mapped back to Lie group.(1)

with the preserved symmetries of the exact wavefunction that are preserved.(10) The generators $\hat{k}$ are invariant upon action of a symmetry operator $\hat{g}$ which is an element of the symmetry group G ($\hat{g}\hat{k}\hat{g}^{-1} = \hat{k}$). To construct a wavefunction that belongs to a particular subgroup, $\hat{K}$ ($\hat{K} = \sum \hat{k}$) is built only from that are unchanged the relevant symmetry operations.(1) Through breaking different symmetries of the wavefunction, different types and amount of correlations will be captured for the same molecular system. $\hat{S}_z$ is the operator that measures spin angular momentum projected along the principle spin axis z. $\hat{S}^2$ is the total spin squared operator, $\hat{C}$ is the complex conjugation operator, $\hat{\theta}$ is the time-reversal operator and $\hat{1}$ is the identity operator. The first wavefunction type is the well-known closed-shell real restricted wavefunction. If one lifts the constraint towards the coefficient, the MOs can be complex which leads to the complex restricted (CR) wavefunction. CR can describe the electronic degeneracy without requiring the multiple Slater determinants.(11) When one use the open-shell wavefunction, it further broaden the ability to describe bond dissociations since alpha and beta have separate spatial orbitals. The general HF (GHF) method can describe the spin frustration since the spin is no longer constrained along with z axis. Based on the selected generators, one can obtain the corresponding wavefunction solution after the local optimization.

2.4 Hartree-Fock Wavefunction Optimization Algorithms

This section is included since the stability code in chapter 4 uses steepest descent (SD) and Newton-Raphson (NR) methods when searching nearby SCF solutions on the wavefunction parameter space (WPS). Before the discussion of these two methods, the concepts of gradient and Hessian are introduced since the SCF optimizations here are directly using the gradient and Hessian to locate the local minimum on the WPS. In addition, direct inversion in the iterative subspace (DIIS) is also discussed

Table 2.1. Wavefunction symmetries constructed from generators $\hat{K}$

Wavefunction class	Preserved symmetries	Generators
real restricted (RR)	$\hat{S}^2, \hat{S}_z, \hat{\Theta}, \hat{C}, \hat{1}$	$\hat{k}^{A,00}$
complex restricted (CR)	$\hat{S}^2, \hat{S}_z, \hat{1}$	$\hat{k}^{A,00} + \hat{k}^{S,00}$
real unrestricted (RU)	$\hat{S}_z, \hat{C}, \hat{1}$	$\hat{k}^{A,00} + \hat{k}^{A,z}$
paired unrestricted (PU)	$\hat{S}_z, \hat{\Theta}, \hat{1}$	$\hat{k}^{A,00} + \hat{k}^{S,z}$
complex unrestricted (CU)	$\hat{S}_z, \hat{1}$	$\hat{k}^{A,00} + \hat{k}^{S,00} + \hat{k}^{A,z} + \hat{k}^{S,z}$
real general (RG)	$\hat{C}, \hat{1}$	$\hat{k}^{A,00} + \hat{k}^{A,z} + \hat{k}^{A,x} + \hat{k}^{S,y}$
complex-paired general (CPG)	$\hat{\Theta}, \hat{1}$	$\hat{k}^{A,00} + \hat{k}^{S,z} + \hat{k}^{S,x} + \hat{k}^{S,y}$
complex general (CG)	$\hat{1}$	$\hat{k}^{A,00} + \hat{k}^{S,00} + \hat{k}^{A,z} + \hat{k}^{S,z} + \hat{k}^{A,x} + \hat{k}^{S,x} + \hat{k}^{A,y} + \hat{k}^{S,y}$
real-paired general (RPG)	$\hat{\Theta}, \hat{C}, \hat{1}$	$\hat{k}^{A,00} + \hat{k}^{S,y}$

This table shows one symmetry-adapted method and seven broken-spin symmetry methods. Each method has their generators to preserve the certain symmetry of wavefunction. The real-paired general (RPG) is collinearly aligned with y axis and can be rotate to paired unrestricted (PU), so itself is not a new broken-spin symmetry method. (1)

since it is the default in Gaussian 16 to accelerate convergence during SCF optimizations.

Gradient and Hessian

Points on the WPS that define electronic structures of chemical interest are stationary points:

$$\frac{\partial E_{WPS}}{\partial P} = 0 \tag{2.4.1}$$

where E_{WPS} is the electronic energy and P is the arbitrary parameter inside wavefunction. The stationary points are used to rationalize reactants, products and intermediates which are defined by local minimum on the WPS. However, the WPS is a high-dimensional space, which indicates the efficient algorithms are required to locate those stationary points. Before approaching the searching algorithm, one needs to describe WPS first. WPS can be sketched as a multivariate smooth function using Taylor series:

$$f(P) = f(p_0) + \frac{1}{1!}\frac{d}{dp}f(p_0)(p-p_0) + \frac{1}{2!}\frac{d^2}{dp^2}f(p_0)(p-p_0)^2 + \frac{1}{3!}\frac{d^3}{dp^3}f(p_0)(p-p_0)^3 + \dots \tag{2.4.2}$$

Since the approximated wavefunction parameter space is usually truncated to the second oder, we can define the first order derivative as gradient and derivative for second order as Hessian, which can be expressed in a matrix form below:

$$g = \begin{bmatrix} \frac{\partial f}{\partial p_1} \\ \frac{\partial f}{\partial p_2} \\ \vdots \\ \frac{\partial f}{\partial p_n} \end{bmatrix} \qquad H = \begin{bmatrix} \frac{\partial^2 f}{\partial p_1 p_1} & \frac{\partial^2 f}{\partial p_1 p_2} & \cdots & \frac{\partial^2 f}{\partial p_1 p_n} \\ \frac{\partial^2 f}{\partial p_2 p_1} & \frac{\partial^2 f}{\partial p_2 p_2} & \cdots & \frac{\partial^2 f}{\partial p_2 p_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial p_n p_1} & \frac{\partial^2 f}{\partial p_n p_2} & \cdots & \frac{\partial^2 f}{\partial p_n p_n} \end{bmatrix} \tag{2.4.3}$$

And f(P) can be rewritten as:

$$f(P) = f(p_0) + g^T(p - p_0) + \frac{1}{2}(p - p_0)^T H(p - p_0) + \dots \tag{2.4.4}$$

where g is gradient and H is Hessian represented in the matrix form.

Steepest Descent (SD)

The idea of steepest descent (SD) can be understood by its equation below:

$$d = -g \tag{2.4.5}$$

where d is step size and g is gradient direction. The variation of SD is scaled steepest descent (SSD) that scales gradient by the step size: $d = -sg$, where s is a step size. Minimization can be achieved by stepping along the opposite direction of the gradient in both methods. Once the energy increases, an approximated minimum can be computed by interpolation between two calculated points. This interpolated point is used to calculate the gradient and used for the next line search. Normally. the line search can guarantee to approach the minimum if SD works accurately. SD is very simple algorithm and requires only to store the gradient vector. However, if the explored surface has long narrow valleys, the SD path oscillates around the minimum path and the convergence rate becomes slow as approaching the minimum. Besides, the step size varies dynamically based on the change function values in the line search. Due to its simplicity, SD is mainly applied to relax the poor starting point before more advanced methods used and complex minimization where high-level algorithm is not working to achieve the minimization goal.

Newton-Raphson Methods (NR)

Newton-Raphson (NR) method improves the convergence through the curvature of the WPS. If we set initial energy and the end gradient to zero in (2.4.4), the formula can be obtained for NR:

$$\Delta x = -gH^{-1} \tag{2.4.6}$$

where Δx is the change in WPS, g is direction of step, and H is the step size. Since NR is exact to second order, so a simple harmonic oscillation can be optimized in one step. Since computing Hessian at every step is really expensive, quasi-Newton-Rhapson method such as Broyden-Fletcher-Goldfarb-Shanno (BFGS) is proposed to approximate Hessian from previous point, so that it decrease the computational cost and improve the convergence at the same time.

Direct Inversion in the Iterative Subspace (DIIS)

During SCF cycles, before reaching the convergence, an error vector $\mathbf{e}$ is defined in direct inversion in the iterative subspace (DIIS) method(12):

$$\mathbf{e} = \sum_{i=1}^{k-1} \mathbf{e}_i = \sum_{i=1}^{k-1} (\mathbf{F}_i\mathbf{P}_i\mathbf{S} - \mathbf{S}\mathbf{P}_i\mathbf{F}_i) \tag{2.4.7}$$

where i is step index during the optimization, $\mathbf{e}_i$ is the error for each step,$\mathbf{F}$ is fock matrix, $\mathbf{S}$ is atomic orbital overlap matrix, and $\mathbf{P}$ is density matrix. Once the error vector is constructed for each iteration, the scalar product $B_{ij} = \mathrm{Tr}(\mathbf{e}_i \cdot \mathbf{e}_j^\dagger)$ is evaluated for building the linear equation to minimize the error vector with least-squares criteria:

$$\begin{pmatrix} 0 & -1 & -1 & \cdots & -1 \\ -1 & B_{11} & B_{12} & \cdots & \\ \vdots & \vdots & \ddots & \vdots & \\ -1 & \cdots & \cdots & B_{ij} & \\ \vdots & & & & \end{pmatrix} \begin{pmatrix} -\lambda \\ c_1 \\ \vdots \\ c_i \\ \vdots \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ \vdots \\ 0 \\ \vdots \end{pmatrix} \tag{2.4.8}$$

where λ is Lagrangian multiplier. Coefficient c_i constrained to $\sum_k c_k = 1$ while optimizing the error vector. If this equation is ill-conditioned, omitting the first, second ,etc equations until the condition is acceptable. The next step is to create fock matrix using the optimal coefficients in error vectors:

$$\mathbf{F}^{'} = \sum_{i=1}^{k} c_i\mathbf{F}_i \tag{2.4.9}$$

and then diagonalize $\mathbf{F}^{'}$ to determine the new density matrix $\mathbf{P}$ and fock matrix $\mathbf{F}$ to be added inside error vector list. This process continues until the convergence criteria meets.

Avoiding Variational Collapse in Hartree-Fock Wavefunction Optimization

Since HF is chosen to obtain the excited states, the variational collapse occurs which indicates the necessarily to implement the efficient local optimization algorithms. Maximum overlap method (MOM), initial maximum overlap method (IMOM), and stage-targeted energy projection method (STEP) are introduced here to locate interested low-lying excited states for chapter 3 and 4. In addition, σ-SCF constructs the energy target function and minimize the variance to obtain the excited states which is not used in the later chapters.

Maximum Overlap Method (MOM)

The maximum overlap method (MOM) states that the new occupied orbitals are selected based on the overlap with the span of the old occupied orbitals. The overlap orbital can be defined as(13):

$$O = (C^{\text{old}})^{\dagger} S C^{\text{new}} \tag{2.4.10}$$

If O_{ij} represents the overlap between the ith old orbital and jth new orbital, and the projection of the jth new orbital onto the old occupied space is:

$$p_j = \sum_{i}^{n} O_{ij} = \sum_{\nu}^{N} [\sum_{\mu}^{N} (\sum_{i}^{n} C_{i\mu}^{\text{old}}) S_{\mu\nu}] C_{\nu j}^{\text{new}} \tag{2.4.11}$$

p_j becomes the selection criteria for the occupied orbitals for the diagonalization to obtain the new MO coefficients, and this procedure continues until reaching the convergence threshold. The advantage for this method is that it can easily converged to the solutions closest to the initial guess without knowing the detailed knowledge of targeted electronic structure and its computational cost is negligible.

When cases arise the SCF converges into an undesired solution with MOM, especially common in systems with near-degeneracies, the modified MOM is proposed (initial maximum overlap method, IMOM) and overlap matrix can be revised as(14):

$$O = (C^{\text{initial}})^{\dagger} S C^{\text{current}} \tag{2.4.12}$$

and projection p_{ij} can be computed with:

$$p_j = (\sum_{i} O_{ij}^2)^{1/2} \tag{2.4.13}$$

which serves as the selection criteria for the occupied orbitals to be diagonalized. The iterative procedure is exactly same as MOM's discussed above. IMOM prevents the MOs to drift away from initial guess and enhances the usefulness over the MOM in certain cases.

Stage-Targeted Energy Projection Method (STEP)

Stage-targeted energy projection (STEP) puts the constraints through the level shifting the energy of all virtual orbitals. (15)The level shifting is widely applied to the degenerate or near-degenerate frontier orbitals to improve the convergence, since it limits the allowed occupied-virtual rotations. The projector in atomic orbital basis can be constructed as:

$$\hat{Q} = \sum_{\mu\nu} \sum_{a}^{virt} |\mu_{\mu a} C_{\nu a}^{*} \langle \nu| = \sum_{\mu\nu} |\mu\rangle Q_{\mu\nu} \langle \nu| \tag{2.4.14}$$

where the matrix Q is the virtual MOs block of $CC^{\dagger}$. The modified Fock matrix F can be obtained:

$$F^{'} = F + \eta SQS \tag{2.4.15}$$

where η is a parameter, and the level shift ηSQS is to elevate the energy for all virtual orbitals by η Hartree. η can be computed through:

$$\eta = |\varepsilon_{\mathrm{HOMO}} - \varepsilon_{\mathrm{LUMO}}| + \epsilon^{'} \tag{2.4.16}$$

where $\epsilon^{'}$ is a small empirical parameter. In the paper we cite, they found $\epsilon^{'} = 0.1$ Har is the optimal to prevent the variational collapse with same SCF cycles.

The modified STEP is proposed in chapter 4 to ensure the local optimization only occurs inside the initially defined active space. Two energy gaps are introduced: inactive occupied orbitals & active space, and active space & inactive virtual orbitals. The distance metric is computed for each SCF cycle and function based on the distance metric is used to control two gaps to prevent any inactive occupied or virtual orbitals being swapped inside active space. The relevant details can be found in the methodology part in chapter 4.

σ-SCF Method

σ-SCF method contains two steps: constructing the energy target function and minimizing the variance(16). The functional for the first step can be built through:

$$W[\Psi](\omega) \equiv \langle\Psi|(\omega - \hat{H})^2|\Psi\rangle \tag{2.4.17}$$

in which Ψ is a fermionic and normalized wavefunction. If expanding $|\Psi\rangle$ in the eigenbasis, $|\Psi\rangle = \sum_n c_n|n\rangle$, ($|n\rangle$ is a orthonormal molecular orbital basis) one can obtain a lemma:

$$W[\Psi](\omega) = \sum_n |c_n|^2(E_n - \omega)^2 \geq (E_m - \omega)^2 \sum_n |c_n|^2 = (E_m - \omega)^2 \tag{2.4.18}$$

where E_m is the energy closest to ω in the ordered energy spectrum for all n. One can define explicitly an energy-targeting function:

$$\Omega(\omega) = \min_{\Psi} W[\Psi](\omega), \quad \text{subject} \quad \text{to} \quad \langle\Psi|\Psi\rangle = 1 \tag{2.4.19}$$

After applying the Hellmann-Feynman theorem to $\Omega(\omega)$:

$$\frac{\partial\Omega}{\partial\omega} = 2(\omega - \langle\hat{H}\rangle) \tag{2.4.20}$$

and stationary point is given by $\omega = \langle\hat{H}\rangle$, and the energy is the function of ω:

$$E(\omega) \equiv \langle\Psi(\omega)|\hat{H}|\Psi(\omega)\rangle) \tag{2.4.21}$$

whose plateaus form a one-to-one mapping with the energy levels. Since the Hellmann-Feynman theorem still holds true inside mean-field picture, the variance minimization step can be proceeded through:

$$\sigma_H^2 = \min_{\Phi} W[\Psi](\omega), \quad \text{subject} \quad \text{to} \quad \langle \Phi | \Phi \rangle = 1 \tag{2.4.22}$$

and the energy variance functional is defined by:

$$S[\Phi] \equiv W[\Phi](\omega)|_{\omega = \langle \hat{H} \rangle} = \langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2 \tag{2.4.23}$$

Since the variance minimization does not depend on the ω, the energy-targeting information is maintained if using the same $\Phi(\omega)$ as the initial guess. To generalize these two steps, the fock matrix can be defined as:

$$F_{\mu\nu}^s[\{P^s\}] = \frac{\partial E_{el}[\{P^s\}]}{\partial P_{\mu\nu}^s} \tag{2.4.24}$$

Due to the purpose of minimizing W(ω) and S, thus two generalized fock matrices are defined as:

$$f_{\mu\nu}^s[\{P^s\}](\omega) \equiv \frac{\partial W[\{P^s\}](\omega)}{\partial P_{\mu\nu}^s} \tag{2.4.25}$$

$$F_{\mu\nu}^s[\{P^s\}] \equiv \frac{\partial S[\{P^s\}]}{\partial P_{\mu\nu}^s} \tag{2.4.26}$$

for energy targeting and variance minimization, respectively. Therefore, σ-SCF method can be concluded in three steps: 1). choose an appropriate value for ω. 2).self-consistently perform the energy-targeting calculation with f; obtain the solution $\Phi(\omega)$ in terms of P(ω). 3). Using P(ω) generated above as initial guess, self-consistently perform the variance minimization with F.

2.5 Post Hartree Fock Methods

In this section, post-HF methods are proposed to improve the accuracy of results by capturing the electron correlations. The mechanism of configuration interaction (CI), perturbation theory and multiconfigurational SCF (MCSCF) are discussed below.

Configurational Interactions

Configuration interaction (CI) can capture the correlation energy through diagonalizing the N-electron functions constructed by the linear combination of trial wavefunctions. One can use the many-electron wavefunction as a basis to expand the exact many-electron wavefunction $|\Psi_0\rangle$. If we assume $|\Phi_0\rangle$ as the reasonable approximation, we can write exact wavefunction $|\Psi_0\rangle$ as:

$$|\Psi_0\rangle = c_0|\Phi_0\rangle + \sum_{ra} c_a^r|\Phi_a^r\rangle + \sum_{a<b,r<s} c_{ab}^{rs}|\Phi_{ab}^{rs}\rangle + \sum_{a<b<c,r<s<t} c_{abc}^{rst}|\Phi_{abc}^{rst}\rangle + ... \tag{2.5.1}$$

in which a,b,c,... are the occupied orbitals, and r,s,t,... represent the virtual orbitals. This equation is full CI wavefunction. If we consider to remove the restriction from (2.5.1) to obtain the formal form:

$$|\Psi_0\rangle = c_0|\Phi_0\rangle + (\frac{1}{1!})^2 \sum_{ra} c_a^r|\Phi_a^r\rangle + (\frac{1}{2!})^2 \sum_{a<b,r<s} c_{ab}^{rs}|\Phi_{ab}^{rs}\rangle + (\frac{1}{3!})^2 \sum_{a<b<c,r<s<t} c_{abc}^{rst}|\Phi_{abc}^{rst}\rangle + ... \tag{2.5.2}$$

where the factor of $(1/n)^2$ is included. If we assume 2K spin orbitals (N is occupied, 2K-N is virtual) and n electrons inside the molecule, the total number of excited state determinants are $\binom{N}{n}\binom{2K-N}{n}$. Even for a small molecule, the FCl matrix is large. The lowest energy serves as the upper bound for the exact ground-state energy based on the variational principle. The energy difference between lowest CI energy and HF energy is called basis set correlation energy. For a given basis set, the FCI gives the best result but really expensive to compute. In order to develop a better understanding of the FCI matrix, we can write the linear expansion above as:

$$|\Psi_0\rangle = c_0|\Phi_0\rangle + \sum_S c_S|\Phi_S\rangle + \sum_D c_D|\Phi_D\rangle + \sum_T c_T|\Phi_T\rangle ... \tag{2.5.3}$$

where $|\Phi_S\rangle$ represents all the singly-excited determinants, $|\Phi_D\rangle$ includes all the doubly-excited determinants, and so on. There are few properties related to the FCI matrix: 1). There is no coupling between HF ground state and single excitations based on the Brillouin's Theorem. 2).There is no coupling between $|\Phi_0\rangle$ and triples and quadruples based on the Slater-Condon rules. 3). Single excitations contribute to double excitations directly and contribute ground state indirectly. 4). Double excitations play an important role in contributing the $|\Psi_0\rangle$.

The correlation energy can be described when the intermediate normalization is enforced to (2.5.1):

$$\langle\Psi_0|\Psi_0\rangle = 1 + \sum_{ra}(c_a^r)^2 + \sum_{a<b,r<s}(c_{ab}^{rs})^2|\Phi_{ab}^{rs} + ... \tag{2.5.4}$$

while (2.5.4) is not normalized, but has property: $\langle\Phi_0|\Psi_0\rangle$=1. We can rewrite linear variational method by multiply $\langle\Psi_0|$:

$$\langle\Phi_0|H - E|\Psi_0\rangle = E_{corr}\langle\Phi_0|\Psi_0\rangle = E_{corr} \tag{2.5.5}$$

If we replace Ψ_0 with (2.5.1):

$$\begin{aligned}\langle\Phi_0|H - E_0|\Psi_0\rangle =& \langle\Phi_0|H - E_0(|\Phi_0\rangle + \sum_{ct} c_c^t|\Phi_c^t\rangle + \sum_{c<d,d<u} c_{cd}^{tu}|\Phi_{cd}^{tu}\rangle + ...)\\ =& \sum_{c<d,t<u} c_{cd}^{tu}\langle\Phi_0|H|\Phi_{cd}^{tu}\rangle\end{aligned} \tag{2.5.6}$$

Thus the correlation energy is determined mainly on the double excitations, and other excitations such as singles and triples contribute double excitations directly as well. Since the FCI is computational expensive, the truncated CI methods are widely used, such as doubly excited CI (CID) and singly-doubly excited state CI (CISD).

Perturbation Theory

Rayleigh-Schrödinger perturbation theory provides another perspective to obtain the correlation energy for one-particle or N-particle systems. This method desires to systematically improve the eigenfunctions and eigenvalues of H_0 so they become closer to the eigenfunctions and eigenvalues of total H. We can introduce λ to construct the total H:

$$H = H_0 + \lambda V \tag{2.5.7}$$

where H_0 is zero-order hamiltonian, and V is the perturbation. The exact eigenfunctions and eigenvalues in a Taylor series in λ can be written as:

$$|\Psi_i\rangle = |i\rangle + \lambda|\Phi_i^{(1)}\rangle + \lambda^2|\Phi_i^{(2)}\rangle + ... \tag{2.5.8}$$

$$E_i = E_i^{(0)} + \lambda E_i^{(1)} + \lambda^2 E_i^{(2)} + ... \tag{2.5.9}$$

where $E_i^{(n)}$ is the nth-order energy. If we apply the intermediate normalization by multiplying $\langle i|$ to (2.5.8), we have:

$$\langle i|\Psi_i\rangle = \langle i|i\rangle + \lambda\langle i|\Phi_i^{(1)}\rangle + \lambda^2\langle i|\Phi_i^{(2)}\rangle + ... = 1 \tag{2.5.10}$$

If we plug (2.5.7), (2.5.8) and (2.5.9) into Schrödinger equation:

$$\begin{aligned}&(H_0 + \lambda V)(|i\rangle + \lambda|\Phi_i^{(1)}\rangle + \lambda^2|\Phi_i^{(2)}\rangle + ...)\\ &= (E_i^{(0)} + \lambda E_i^{(1)} + \lambda^2 E_i^{(2)} + ...)(|i\rangle + \lambda|\Phi_i^{(1)}\rangle + ..)\end{aligned} \tag{2.5.11}$$

and sorting terms of eigenvalues and eigenfunctions based on their λ^n order:

$$H_0|i\rangle = E_i^{(0)}|i\rangle \tag{2.5.12}$$

$$H_0|\Phi_i^{(1)}\rangle + V|i\rangle = E_i^{(0)}|\Phi_i^{(1)}\rangle + E_i^{(1)}|i\rangle \tag{2.5.13}$$

$$H_0|\Phi_i^{(2)}\rangle + V|\Phi_i^{(1)}\rangle = E_i^{(0)}|\Phi_i^{2}\rangle + E_i^{(1)}|\Phi_i^{(1)}\rangle + E_i^{(2)}|i\rangle \tag{2.5.14}$$

and so on. The next step will be solving the wavefunction and its related energy. We can take first order correction as the example here. It is easy to write (2.5.13):

$$(E_i^{(0)} - H_0)|\Phi_i^{(1)}\rangle = (V - E_i^{(1)})|i\rangle \tag{2.5.15}$$

If both side multiply by $|i\rangle$ which is same as $|\Phi_i^{(0)}\rangle$ and simplify both sides, the first-order energy can be obtained:

$$E_i^{(1)} = \langle i|\hat{H}^{(1)}|i\rangle \tag{2.5.16}$$

The first-order wavefunction can be expanded in terms of the eigenfunction of H_0:

$$|\Phi_i^{(1)}\rangle = \sum_n c_n^{(1)}|n\rangle \tag{2.5.17}$$

Since the eigenfunctions of H_0 are orthonormal, multiplying this equation by $\langle n|$ to get:

$$\langle n|\Phi_i^{(1)}\rangle = c_n^{(1)} \tag{2.5.18}$$

so the first-order wavefunction is:

$$|\Phi_i^{(1)}\rangle = \sum_n |n\rangle\langle n|\Phi_i^{(1)}\rangle \tag{2.5.19}$$

and the second-order energy can be easily obtained if the first-order wavefunction is known with same strategy applied for computing the first-order energy above:

$$E_i^{(2)} = \sum_n \frac{|\langle i|V|n\rangle|^2}{E_i^{(0)} - E_n^{(0)}} \tag{2.5.20}$$

The same principle is applied to the higher-order energy and wavefunctions. We can also treat total Hamiltonian as the one-particle Hamiltonian:

$$H = H_0 + V = \sum_i h_0(i) + \sum_i v(i) \tag{2.5.21}$$

If we only consider the zeroth and first-order terms, and it becomes the fock operator. If one wants to capture the correlation energy, it needs to start at second-order at least. The second-order energy can be computed either from the many-electron perspective or the one-electron perturbation theory, in which all offer the same result as (2.5.20). The perturbation method is size-consistent but not variational and it scales roughly N^5.

Besides, there are many applications using the perturbation theory, for example, the resonance energy of cyclic polyene can be studied with one-particle perturbation theory. As for the many-electron perturbation theory, it is widely applied to describe molecular features in electric field, magnetic field, anharmonic oscillator, spin-orbital coupling and relativistic correction for the kinetic energy, etc. Finally, there are some computational methods built based on the perturbation theory, Moller-Plesset method (MPn), doubly-hybrid method that using perturbation theory to capture the correlation inside functionals, and CASSCF method to including the dynamic correlation with perturbation theory.

Multiconfigurational Self-Consistent-Field (MCSCF)

If one optimizes both orbitals and CI coefficients, the wavefunction is described by the multiconfigurational self-consistent-field (MCSCF):

$$|\Psi_{MCSCF}\rangle = \sum_I c_I|\Phi_I\rangle \tag{2.5.22}$$

If the "multi configuration" is only one, it is simply HF. If we take H_2 at its ground state, the closed-shell configurations can be written as:

$$|\Psi_{MCSCF}\rangle = c_\sigma|\psi_\sigma\bar{\psi}_\sigma\rangle + c_{\sigma^*}|\psi_{\sigma^*}\bar{\psi}_{\sigma^*}\rangle \tag{2.5.23}$$

in which the orthonormal orbitals ψ_σ and ψ_{σ^*} can be expanded in atomic basis:

$$|\psi_i\rangle = \sum_\mu C_{\mu i}\phi_\mu \qquad i = \sigma, \sigma^* \tag{2.5.24}$$

The MCSCF energy minimization is subject to $\delta_{\sigma\sigma^*}$ and $c_\sigma^2 + c_{\sigma^*}^2 = 1$ to determine the orbitals and CI coefficients. In this case, the Ψ_{MCSCF} is FCI. If the larger basis is used, the MCSCF energy will be higher than FCI but lower than two configurations using CI methods. Besides, MCSCF is much harder to converge, and tends to converge to the non-minima solution. This issue normally solved by expanding the energy to the second order for both orbitals and configurational coefficients and further applying the Newton-Raphson methods to enforce the convergence to a minimum. MCSCF method is also applied to the large molecular system by using the active space: complete active space self-consistent-field (CASSCF) or restricted active space self-consistent-field method (RASSCF). Even though the selection of the correct active space is not trivial question, CASSCF/RASSCF methods have their own rules to addressing this issue when dealing with the complex and large molecular systems. In CASSCF, the MOs are partitioning into active and inactive spaces. The inactive MOs have either 2 or 0 electrons, i,e. either doubly occupied or empty. Full CI is performed in the active space and all proper symmetry-adapted configurations are contained in MCSCF optimizations. While in RASSCF, active space orbitals are further divided into three sections: RAS1, RAS2, and RAS3, in which each section has allowed occupation numbers. Specifically, RAS1 contains doubly occupied electrons, RAS3 contains empty orbitals, and full CI or certain level excitations is generated using the orbitals in RAS2. The highest excitation from RAS1 to RAS3 is only limited to the double excitations. The occupation numbers in natural orbitals can also help to determine the active space used in unrestricted natural orbital complete active space method(UNOCAS), and orbitals with fractional numbers (between 0.02 and 1.98) are normally selected as active space orbitals. In later section, quantum orbital entropy and mutual information for paired orbitals are used to discern the different correlation mechanism from the orbital levels for given SCF solutions. Since the intention of chapter 4 is to improve the nonorthogonal MCSCF method,(5) it will be useful to understand the multiple correlation mechanisms for each SCF solution and between SCF solutions via the orbital entropy and mutual information discussed in the following section.

2.6 Orbital Correlations in Quantum Information

Quantum information theory has been applied in many body quantum physics, solid-state physics and quantum chemistry recently, such as quantum phase transition(17) and site reordering of lattice(18), etc. With the entanglement information regarding the locality of lattice model, density-matrix renormalization group (DMRG) achieves high efficiency for the convergence rate in the solid-state physics,(19; 20) which also benefits the studies who uses DMRG regarding the magnetization process of spin chains,(21) polymer phonons in nonadiabatic case,(22) and one-particle quantum mechanics.(23) Besides, the mutual information can also be used as a tool for understanding the bonding structure of molecular systems.(24; 25; 26)

In chapter 4, we adopt generalized correlation functions (27; 25) to express one and two-orbital reduced density matrix by simply introducing a nonvanishing matrix element of a single-site states within a many-body system. The single-site and two-site density matrix for computing the mutual information can be described by the eigenvalues of corresponding reduced density matrix of the transition operator. The detail of the transition operator and reduced-density matrix form are illustrated in the methodology part in chapter 4.

The one and two orbital density matrix can be obtained from the many-electron states and its general wavefunction can be divided into "system" and "environment"(28):

$$|\Psi\rangle = \sum_{i,j} C_{i,j}|i\rangle|j\rangle \tag{2.6.1}$$

where $|i\rangle$ denotes the 'system' and $|j\rangle$ represents the 'environment'. The reduced density matrix can be express with the density operator:

$$\rho^{sys}_{i,i'} = \sum_{j} \rho_{i,j,i',j} = \sum_{j} \langle j|\langle i|\Psi\rangle\langle\Psi|i^{'}\rangle|j\rangle \tag{2.6.2}$$

where the density operator $\hat{\rho} = |\Psi\rangle\langle\Psi|$. The system i can be a single orbital or the orbital pair which provides the corresponding one-orbital and two orbital reduced density matrix.

Orbital Entropy and Mutual Information

A novel way to describe the electron correlation effects is to explore the entanglement among molecular orbitals. (25)The study of iron nitrosyl complexes demonstrates that the orbital entropy can classify the static, nondynamic and dynamic correlations of the complex electronic structures.(26)The evaluation of one orbital with all other orbitals is called single orbital entropy, which can be computed with the equation below:

$$s(1)_i = -\sum_{\alpha} w_{\alpha,i} ln w_{\alpha,i} \tag{2.6.3}$$

where α denotes four possible occupation of single spatial orbital i, and $w_{\alpha,i}$ is the eigenvalue of the one-orbital reduced density matrix. Another orbital evaluation applied here is mutual information, which can reflect the entanglement of chosen paired orbitals (i,j) embedded in the remaining orbital environment. The mutual information $I_{i,j}$ can be computed with:

$$I_{i,j} = \frac{1}{2}(s(1)_i + s(1)_j - s(2)_{i,j})(1 - \delta_{i,j}) \tag{2.6.4}$$

in which s(1) is single orbital entropy for orbital i and orbital j, $s(2)_{i,j}$ is the two-orbital entropy calculated from the two-orbital reduced density matrix, which contains sixteen possible states for any selected orbital pair. The details of two-orbital entropy are shown in methodology part in chapter 4. $\delta_{i,j}$ is Kronecker delta. The magnitude of one orbital entropy and mutual information corresponds to different correlation effect qualitatively.(29) In chapter 4, we apply this orbital entropy idea to differentiate the SCF solutions located in the nearby regions on the WPS.

GLOBAL ELUCIDATION OF SCF SOLUTIONS USING BASIN HOPPING METHOD

3.1 Introduction

Through solving the Schrödinger equation, it enable us to obtain the electron's wavefunction using the Hartree Fock (HF) method. Wavefunction preserves the certain symmetries based on the different categories it belongs to. It is common to break the spin symmetry to describe bond dissociation condition, in which the energy is correctly described but leads to the spin contamination after the Coulson-Fisher point. (30) The studies of further breaking S_z and complex conjugate symmetries are also essential to identify and investigate certain properties by taking advantage of flexibility of those wavefunctions. (31; 11) Symmetry-breaking wavecfunction does not have good quantum number, and it is necessary to propose the efficient methods to characterize the solutions.(32) Besides breaking the symmetry, there are interests arising in understanding the instability of wavefunction and the related symmetry breaking behaviors.(33; 34; 35; 11) The goal of these studies is meant to interpret why the wavefunction has to break certain symmetries to locate the solutions in which either disappearing or discontinuous solutions occurs using the symmetry preserved wavefunction's description.After developing a good understanding of symmetry-breaking wavefunction, research in restoring the symmetry of the wavefunction turns out be inevitable due to the benefits already brought by the symmetry-broken wavefunction. Various symmetry-projected schemes are proposed to restore the spin symmetry in most cases. (36; 37; 38) One also sees the potential of symmetry-projected wavefunction, and use it as the basis to do the multireference calculations,(39; 40; 41) and modifications of symmetry-projected multireference methods are also explored to capture the dynamic correlations as well.(42; 43; 44) Last but not the least, general Hartree Fock (GHF) is a crucial broken symmetry method being recognized gradually due to its flexible description of the spin. This advantage is used to describe the spin frustration(45) and magnetic properties of studied molecule sucessfully.(46)

HF wavefunction has many attractive properties and revised potentials illustrated above. Revised single reference wavefunction methods can also be applied to describe the complicated correlation within the systems. (47; 48; 49) Single reference wavefunction in HF can break different symmetries to capture various correlation, but certain conditions require the multireference methods, such as bond breaking(50),avoid crossing(51)and conical intersection.(52) Scaling problem is another issue to be addressed associated with method development, since one always wants to extend methods to the larger molecular systems. Either decreasing the scaling(53) when constructing the mechanism or wisely using the active space(54) can result in the reduction of the computational cost. In addition, dynamic correlation can be included in the multireference methods to better describe the systems.(55) The utilization of the density matrix within wavefunction method (56; 57) allows one to simplify the complexity of the molecular integral transformation and fractional occupation inside active space, respectively.

In order to demonstrate the efficiency of the multireference wavefunction methods, it is essential to identify unique SCF (self consistent field) solutions and use them to provide the suitable basis sets for post-HF calculations, since the accurate descriptions of the complex chemical reactions are beyond Hartree-Fock(HF) single determinant's ability. However, the nonlinear problem we need to consider here, such as global optimization, is a NP-hard problem, which have non-deterministic polynomial time complexity. (58) All global optimization strategies proposed so far are efficient towards some specific situations, and they require an iterative process, which cannot guarantee to find the optimal answer all the time.(59) In our case, global elucidation of SCF solutions is an extension of the global optimization problem in which all solutions are sought in the high-dimensional potential energy surface(PES), and thus also a NP-hard problem, and only a few global elucidation algorithms were proposed so far due to its complexity.(2; 1; 3) Besides, the original HF method cannot describe certain features of electronic states, such as strong correlations and near-degeneracy cases. It also suffers the variational collapse when describing excited states. There are some optimization methods proposed to avoid variational collapse. The maximum overlap method (MOM) by Gill and co-workers determines the occupied orbitals through computing the greatest overlap with previously occupied orbitals or initial MOM.(13; 14) This method is already implemented inside our code to be able to optimize to the higher SCF solution on the PES. The σ-SCF method of Voorhis and co-workers use energy-target and variance minimization to prevent variational collapse.(16) Additionally, Neuscamman and co-workers have utilized Lagrangian minimization to construct excited state variational formula to target the desired excited states.(60) In this chapter, the Lie algebraic method is used to linearize SCF solution space and can simplify searching algorithms. Lie algebra forms a linear vector space tangent to the origin of MOs space, which is described by the Lie group and can be mapped back to the group space through Zaussenhaus' formula. Different approximated wavefunctions (perturbed MOs) can be generated via applying infinitesimal change on the linear vector space followed by exponentiation to obtain a transformation matrix.(1) At the same time, the black-box method, basin hopping algorithm(4) searching algorithm, is proposed to locate the unique SCF solutions on the PES and the stochastic Metropolis-Hasting criteria determines the starting point within each iteration.

3.2 Computational Methodology

Normally, excited states are described via a linear combination of substituted determinants achieved by swapping occupied-virtual orbitals pairs of the reference state. Multireference methods are the natural choice of method because of the ability to capture strong correlations. However, it is also possible for a single HF determinant to capture some correlation energy through broken symmetry methods. These single Slater determinants can be identified later as a suitable basis for the post-HF calculation to compactly describe the wavefunction. To simply the description of SCF solution space, Lie algebra can be borrowed here. This idea already received a great success in global elucidation of H_4 and made the connection between Lie group and various wavefunction classes through the generator on the Lie algebra surface in Figure 2.3.(1) The grid searching algorithm on the Lie algebra surface is illustrated in Figure 2.3, but the scaling of generated guesses increases exponentially for large molecular system. In this case, basin hopping algorithm(4) is proposed to improve the scaling issue by the stochastic selecting the starting SCF solutions, so that different guesses are generated on the Lie algebra surface (red) and mapped back to the Lie group (blue sphere) for the local optimizations. The sections below cover the mechanisms of Lie algebra and basin hopping algorithm in detail.

Lie Algebraic Approach

The Lie algebra helps to linearize the SCF solution space, so that it facilitate the generation of rotation matrices for each pair of occupied and virtual orbitals. The truncated at second order Zassenhaus' formula is used to exponentially map the perturbed MOs to the Lie group to obtain the initial guess, and executing the local optimization to obtain different SCF solutions. The details of Lie algebra mechanism are shown in chapter 2.

Basin Hopping Algorithm

Using the Lie algebraic approach, stochastic basin hopping was implemented to perform global elucidation of SCF space. Basin hopping method was first proposed by Wales to locate the global minimum structures. (4) In this case, we apply this searching algorithm to do the globally locating the SCF solutions without caring they are global minimum electronic state. In the basin hopping approach, the function surface is transformed to the value obtained by local optimization in Figure 3.1. Therefore, local optimization of energy functional E(K) is transformed as:

$$\tilde{E}(K) = opt(E(K)) \tag{3.2.1}$$

The transformed surface is then explored by stochastically sampling the transformed surface around an initial point (denoted previous) and determining if the initial point should be updated to a new point (denoted current) using the Metropolis-Hastings (MH) criteria:

$$e^{E_{previous} - E_{current}/k_B T} > R \quad R \in (0, 1) \tag{3.2.2}$$

in which k_B is Boltzmann constant, T is simulation temperature, $E_{previous}$ is previous energy, and $E_{current}$ is current energy. If optimized energy is lower than the previous energy, it will be directly selected as a new starting point for the next iteration.

The random number R between zero and one is generated for each iteration. The corresponding procedures are posted on the right-hand side of Figure 3.1 and each color indicates each steps on the left-hand side action.

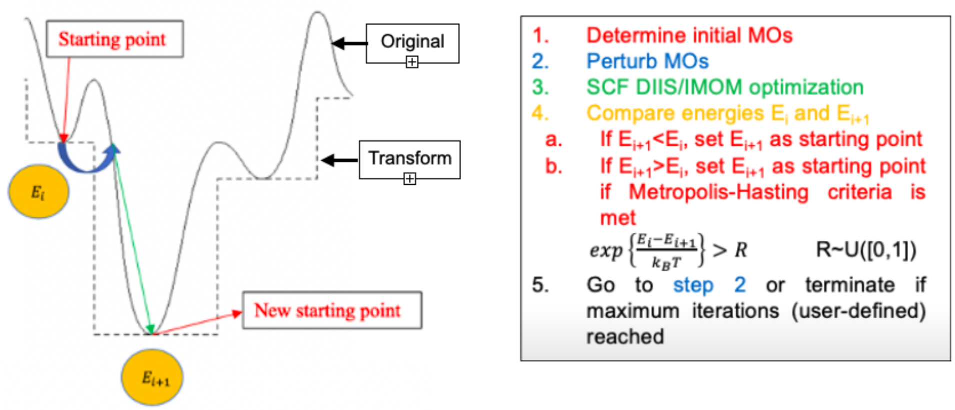


Figure 3.1. Basin hopping algorithm is shown on the left-hand side graph and the rules to decide the starting point for each iteration is listed on the right-hand side. Each step inside graph is corresponding to the rules with same color.

In Figure 3.2, the probability of accepting the solution which has higher energy than the previous starting solution is sketched. A, B, C points represent the all conditions occur in this case. The further away from the y axis, means the current energy is lower than the previous solution. However, the random number R is generated between 0 and 1 for each iteration, so that even though the solution like C with lower probability can also have chance to be selected as the new starting point for the next iteration. The intention of R is taking the advantage of the randomness in order to access all possible energy states.

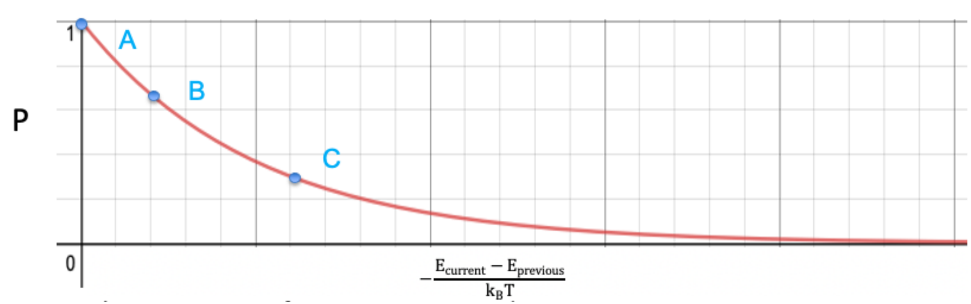


Figure 3.2. Probability of accepting different solutions are illustrated in A,B,C points if energy is higher than the previous starting solution.

3.3 Results and Discussion

To demonstrate the utility of basin hopping in exploration of the SCF solution space, in this section, we present the applications of the proposed algorithm to three systems. First, we study H4 to examine how the choice of parameters (magnitude of perturbation vector, maximum number of iterations, and temperature) affects the types of solutions identified, the number of solutions identified, and the efficiency with which those solutions are identified. Subsequently, we examine the low-lying SCF solutions of benzene and NO_2 with double and triple zeta basis sets including polarization functions to demonstrate that the code can be applied to chemically relevant problems. The algorithm was implemented in a stand-alone in-house code that utilizes the MQCPack library(61) as well as a modified version of Gaussian 16.(62) Comparative post-HF calculations were performed using implementations within the Molpro package.(63)

Parameter test in H4

H_4 in C_{2v} geometry with the minimal basis of Huzinaga (64) is chosen to be the molecule to evaluate the parameters inside basin hopping algorithm, since this molecule's wavefunction is well studied from the previous research.(3; 2; 1) The parameters we want to investigate are number of iterations, temperature, spin-symmetry wavefunction class, perturbation vector magnitude and initial starting MO coefficients. We first investigate how the number of solutions in different wavefunction classes change in various iteration number and temperature, and then analyze the properties of the identified SCF solutions. In Figure 3.3 (a) shows the averaged result over 5 runs with 100, 225, 500 and 1000 iterations under the temperature 1×10^4,1×10^5, and 1×10^6 K. The temperature is directly related the Metropolis-Hastings criteria which controls the acceptance of located solutions. The number of SCF solutions identified does not follow a linear trend respect to the iterations but reaches a constant between 100 and 225 iterations. Such results indicate the basin hopping algorithm has sampled all dimensions in the SCF solution space and further improvements is depends on the particular stochastic trajectory in the simulation. The temperature effect is not that obvious maybe due to the small solution space we pick. Figure 3.3 (b) shows the mean energy of all identify solutions in different wavefunction classes. For all the symmetry wavefunctions considered here, the mean energy increases as iteration number increases, and mean energy of solutions in first 100 iterations are lower than higher iterations' energy. The reasons can be the energy of starting point is lowest energy, and higher energies are bias against by the Metropolis-Hasting criteria.

Having examined the effect of different parameters on the basin hopping simulation, we now look at how the simulation progresses depending on the wavefunction symmetry requirements imposed on the search. Due to the hierarchy of symmetry-broken method(65), searches in the RG generator subspace will also find RU and

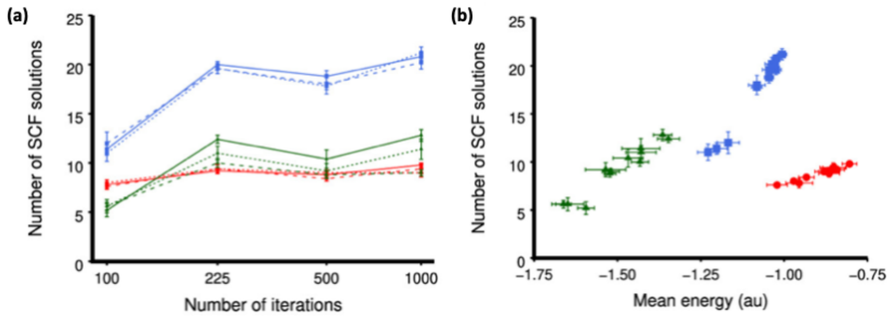


Figure 3.3. Effect of temperature parameter and total iterations on real restricted (red circles), real unrestricted (blue squares), and real general (green triangles) HF solutions obtained from minimal basis H_4 (not counting for degeneracy). Each point shown is an average of five runs where the error bars indicate the standard error. Panel (a) shows the mean number of solution identified with 100,225,500 and 1000 total iterations for temperatures 1×10^4 K (solid line),1×10^5 K (dashed line), and 1×10^6 K (dotted line). Panel (b) shows the dependence of the mean of mean energies on the mean number of solutions identified.

(RR) solutions. In Figure 3.4, the trajectory of the basin hopping algorithm is shown for six different symmetry classes–RR, CR, RU, CU, RG, and CG. The lower symmetry method like RG tends to locate lower energy solutions due to its flexibility of the wavefunction. Besides, the real wavefunctions find more solutions compared with complex wavefunction in H_4 model.

Molecular Tests

In this section, we examine the SCF solutions of benzene obtained from basin hopping simulations performed using the 6-31G(d) basis set with 1000 iterations. First, we examine two simulation input parameters that were not possible to study with minimal basis H4 owing to the small number of basis functions. In Figure 3.5, we explore how the inclusion or exclusion of core orbitals to affect the number of SCF solutions, as well as the perturbation vector L. In (a), we found exclusion of core orbitals tend to locate more low-lying solutions compared with including all the orbitals. The mean energy with and without core orbitals has huge energy difference since the full orbital space involves the core excitations. In (b), the different magnitude of perturbation vectors are tested, and it turns out this factor does not change the solutions a lot. In the later contents, we use $1.0\times$L as the perturbation vector and excluding the core orbitals for all simulations.

Figure 3.6 shows how the low-lying singlet and quasi-singlet HF solutions of benzene in the 6-31G(d) basis determined using the basin hopping algorithm relate to the excited states obtained by post-HF methods and experiment. Owing to the absence of diffuse functions in the basis set, only valence excitations are obtained in the calculated electronic states, as low-lying Rydberg states are missing. Within a 10.0 eV energy window, four different classes of SCF solution symmetry were located. The

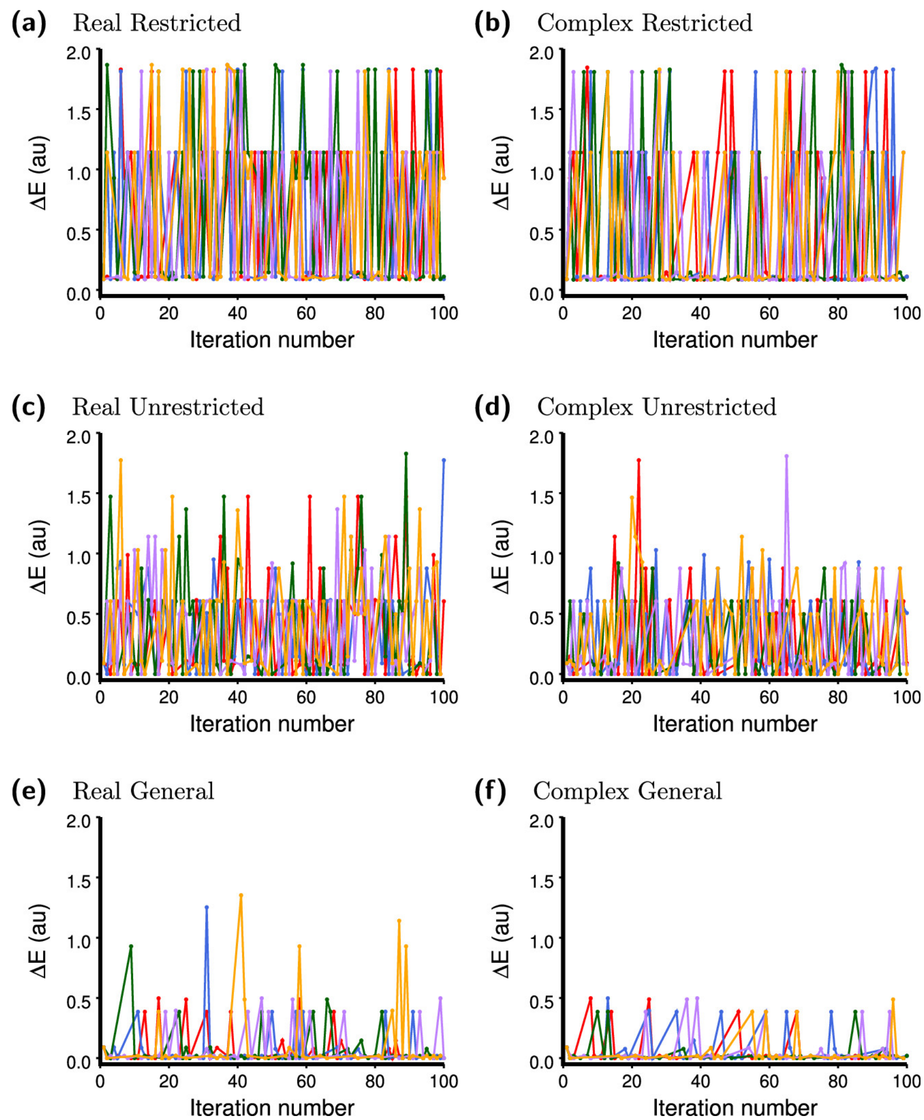


Figure 3.4. Energy of solution located after local optimization from perturbed electronic structure at each iteration of basin hopping simulation over (a) real restricted, (b) complex restricted, (c) real unrestricted, (d) complex unrestricted, (e) real general, and (f) complex general wavefunction symmetry classes in minimal basis H4 for the first 100 iterations using a simulation temperature of 10^7 K. Five different simulation runs are shown with different color lines and only those points where locally optimized solutoon changes from the previous identity solution are shown. Energy is relative to lowest identified solution.

only (RR) solution identified corresponds to the unstable ${}^1A_{1g}$ ground state, which connects degenerate RU ${}^1A_{1g}$ ground-state solutions via $\pi \rightarrow \pi^*$ orbital rotations. All other RU solutions found could be identified with a particular electronic state, although all doubly degenerate wavefunctions were found to be nondegenerate when orbital optimization at the HF level of theory was permitted, as indicated by the prime and double prime labels. In addition to RU and (RR) solutions, a number

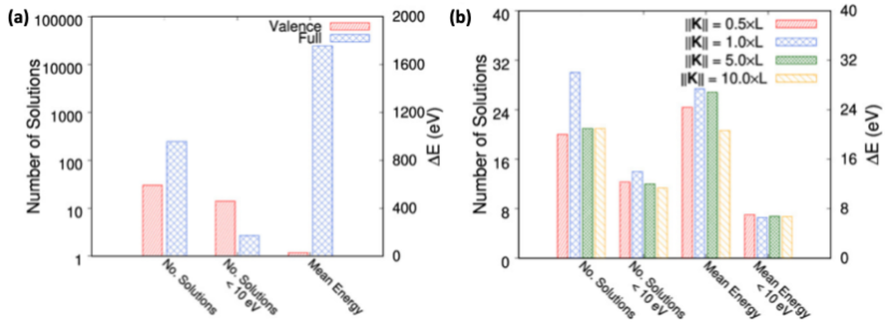


Figure 3.5. Effect of (a) core orbitals in perturbation vector and (b) magnitude of perturbation vector in determining number and type of real unrestricted HF solutions in benzene using the 6-31G(d) basis set averaged over three runs of 1000 iterations. Energy is relative to lowest identified solution.

of lower symmetry HF solutions were identified (4 RG, 1 CPG, and 4 CU). The atom-localized magnetic moments of the lowest RG and CG solutions are shown as insets in Figure 3.6.

The lowest RG solution (Figure 4†) describes the $^1B_{2u}$ wavefunction and is obtained from the unstable RU $^1B_{2u}$ solution by allowing the β spins (shown pointing down at right upcorner in Figure 3.6, although in nonrelativistic studies, the absolute spin orientation is arbitrary) to move off the z axis into the xz plane. The same solution was also identified enforcing CPG symmetry, where the β spins break into the yz plane rather than the xz plane. Three other RG solutions at 5.70, 5.88, and 5.89 eV have qualitatively the same spin density but with noncollinear sites having reduced symmetry across the σ_v plane. Although interpretation of RG solutions in terms of (RR) orbitals is complicated due to mixing of spin components,(66) the RG solutions appear to be approximations of the other $\pi \rightarrow \pi^*$ electronic states ($^1B_{1u}$and $^1E_{1u}$). Similar behavior is observed in the set of CU solutions where the real and imaginary components contain different $\pi \rightarrow \pi^*$ transitions. The lowest identified CG solution that showed noncoplanar electronic structure (right corner in Figure 3.6) displayed a Möbius-like periodicity around the ring. Such electronic structure is typically observed in the ground state of odd- numbered rings (cyclopropene, cyclopentane, etc.), as it allows the magnetic coupling to be maximized.(35) This behavior is reminiscent of (although different from) the inversion of the Hückel aromaticity rule in the lowest excited singlet and triplet states(67), and to our knowledge has not previously been observed. Like the lower energy RG solutions, the CG solution is also described using CPG symmetry, but in contrast displays the torsional spin density resulting from magnetization vectors that are no longer fixed to the yz plane.

The next studied system is NO_2. NO_2 plays an important role in atmospheric and combustion chemistry and so understanding its photophysics is important to correctly model ozone depletion, hydroxyl radical formation, air pollution, and climate change.(68; 69) The photophysics of NO2 is famously complicated, with an odd num-

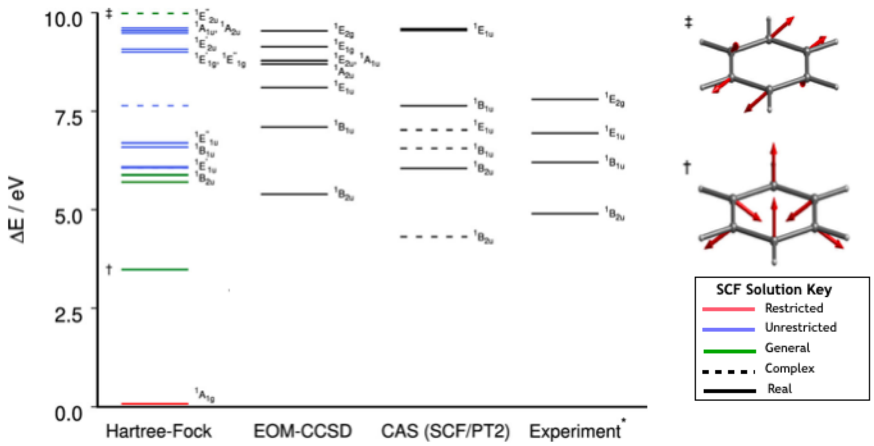


Figure 3.6. Low-energy (<10.0 eV) singlet (or quasi-singlet) HF solutions located via basin hopping simulation using the 6-31G(d) basis set, relative to the global minimum $^1A_{1g}$ real unrestricted solution (red, restricted; blue, unrestricted; green, general; solid, real; dashed, complex). HF ΔE values are compared to EOM-CCSD, 4 state-averaged CASSCF(6,6) and XMS-CASPT2(6,6), and experimental values. Irreducible representation labels indicate electronic states which corresponds to real restricted and real unrestricted HF wavefunctions, in addition to experimental and post-HF methods. † Atomic magnetic moments of lowest energy real general HF solution. ‡ Atomic magnetic moments of lowest energy complex (paired) general HF solution.

ber of electrons and doublet and quartet states among its low-lying solutions (70) and numerous state crossings in the excited-state manifold. The large number of state crossings implies the presence of low symmetry SCF solutions.(32) Figure 3.7 (a) shows the low-lying HF/cc-pVTZ solutions identified by a basin hopping simulation at the B3LYP/cc-pVTZ $^1A_1$ minimum energy geometry. While a large number of the solutions found were real unrestricted wave functions, complex unrestricted and real general solutions were also identified. The PES changes along O-N-O bending angle using the wavefunction computed at 134.4° as initial guess. Among all the states crossing, we focus on understanding the nature of the RG solution as it changes from quartet to doublet, where the relevant PESs are shown in Figure 3.7 (b). Although quartet states arising from spin-forbidden transitions are proposed to contribute to the visible absorption spectrum, the absence of phosphorescence in NO2 implies intersystem crossing facilitated by unusually large spinorbit coupling.(71; 72) Indeed, the absence of long-lived quartet states has been invoked in rationalization of the observed kinetics of hydroxyl radical formation from electronically excited NO_2 and H_2O.(18) The nature of the intersystem crossing can be examined adiabatically using RG solutions to provide a description of spin-crossing processes. The RG solution identified at 134.4° coalesces with the RU approximation of the 2B1/2u state at large angles, while at small angle, it coalesces with the RU $^4B_2$ state, and so models the $^4B_2 \rightarrow {}^2B_1$ transition.

The nature of the change in electronic structure in the $^4B_2 \rightarrow {}^2B_1$ transition can be ascertained by examining Ŝz of the atomic and molecular magnetic moments at different points across the PES in Figure 3.7 (c). At small ONO angles, the solution is an eigenfunction of Ŝz and has an unpaired electron located on each atomic center. A CoulsonFischer point where RU and RG solutions diverge is found at 104°; although interestingly, the RG solution is higher in energy than the 4B2 RU solution, implying a contribution from the spatial- symmetry-broken $^2B_1$ RU solution (labeled as $^2B_1'$) rather than the spatial-symmetry-preserving $^2B_1$ RU solution to which the RG solution eventually coalesces. As the RG solution moves further in energy from the $^4B_2$ state and closer to the spatial-symmetry-preserving $^2B_1$ solution, the molecular Ŝz decreases until spin polarization localizes to one side of the molecule, giving an unpaired electron of the nitrogen and one of the oxygen atoms. As the angle increases past 125°, the $^4B_2$ state reaches a minimum before increasing in energy toward the RG solution, increasing the quartet character of the wave function and causing spin polarization to become symmetric over the molecule. At 135°, the $^4B_2$ and $^2B_1$ states cross and the RG solution is the lowest in energy of the three SCF solutions. The RG solution then rapidly approaches the spatial- symmetry-preserving $^2B_1$ RU solution before coalescing to give a single unpaired electron that is almost completely localized to the nitrogen center. Therefore, the ability to identify SCF solutions of different symmetries that represent low-lying excited states offers a description of how electronic structure evolves to permit the $^4B_2 \rightarrow ^2B_1$ radiationless transition. The findings warrant further investigation into how SCF solutions identified can be used to provide a quantitative description of interaction energies.

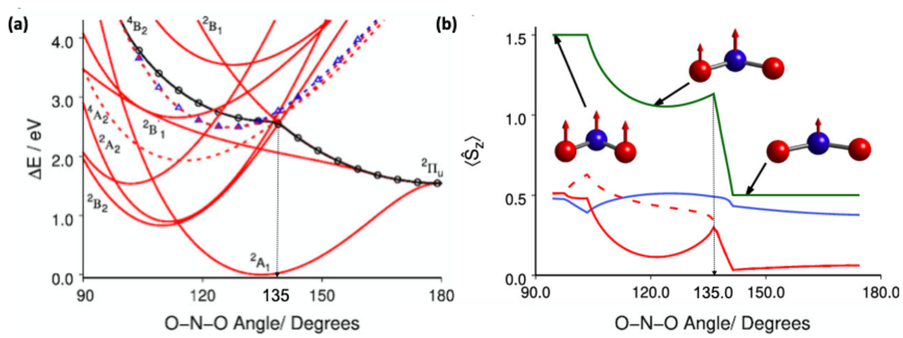


Figure 3.7. Energies, spin expectation values, and magnetic moments of NO_2. Panel (a) shoes low-lying potential energy surfaces of NO_2 of B3LYP/cc-pVTZ solutions identified at a bond angle of 134.4° ($^2A_1$ minimum geometry) obtained by scanning along the O-N-O bond angle (red line, real unrestricted; blue triangles, complex unrestricted; black circles, real general; solid, doublet; dashed, quartet.) Irreducible representation labels indicate electronic states which corresponds to real unrestricted HF wavefunction. Panel (b) shows extracted potential energy surfaces from panel (a) discusses in the next. Panel (c) shows the real general solution atomic and molecular $\hat{S}_z$ expectation values (blue,$\langle \hat{S}_N \rangle$;solid red,$\langle \hat{S}_{O_{right}} \rangle$;dashes red, $\langle \hat{S}_{O_{left}} \rangle$; green, $\langle \hat{S}_{\mathrm{NO}_2} \rangle$) with inset showing atomic magnetic moments along the NO_2 bending coordinate.

3.4 Conclusions

The problem of exploring SCF solution space through the application of basin hopping algorithms has been introduced. Linearization of the SU($2N_{basis}$) group through a Lie algebraic approach allows straightforward implementation of global optimization methods to the many-electron problem. The method presented represents the first application of a stochastic approach to elucidating SCF solution space and was demonstrated to efficiently locate low-lying SCF solutions by biasing toward low-energy regions of wavefunction parameter space.

The parameter tests using H_4 and benzene demonstrate the most important parameter is the number of iterations. We also found lower symmetry wavefunction or lower iterations lead to the lower energy solutions in the early stage of the simulation. Such findings imply that the basin hopping algorithm can be used to partition SCF solution space and target low-energy SCF solutions within a finite number of iterations, although we expect continued application of the methodology to larger and more disparate systems will yield improved understanding as to the role of temperature, initial starting point, and structure of perturbation vector. Future development directions include testing modifications to the acceptance criteria to determine whether basin hopping can be used more generally to divide SCF solution space into energy regions and run multiple simulations to identify a broader set of solutions.

The basin hopping algorithm was applied to identify and examine the low-lying HF solutions of benzene using the 6- 31G(d) (96 basis functions) and NO_2 with cc-pVTZ basis sets (42 basis functions). Although, because of the nature of the global elucidation problem, it is not possible to definitively claim that all HF solutions were identified; comparison with (EOM-CCSD) results did not reveal any electronic states for which the simulation did not identify a solution. Therefore, the approach presented is demonstrated to be reliable and readily applicable to medium-sized systems of up to 100 basis functions, although there is no reason to suspect that larger systems are not also amenable to study by the proposed method. The study of HF solutions in benzene showed that there are three sets of HF descriptions of the $E_{1g} \rightarrow E_{2u}$ transition (RU, CU, and RG). Additionally, analysis of the magnetic structures indicated an excited state displaying Möbius periodicity typically observed in spin-frustrated systems. In NO_2, an RG solution describing the ${}^4B_2 \rightarrow {}^2B_1$ transition was identified and used to compute the interaction energy between the two states. While the local magnetic moments gave a clear and lucid description of the qualitative change in the electronic structure, the interaction energy was computed as being an order of magnitude larger than that computed by other methods.

Finally, we discuss how the approach can be further developed and applied to large systems. Recalling that global elucidation of stationary points in a function belongs to the set of NP-hard problems, extending the algorithm to better obtain solutions of interest in larger systems requires the development of heuristics to reduce the size of the space that must be explored. Ultimately, defining subspaces of the N-electron Hilbert space over which to search depends on the particular solutions which are of interest. However, well-known features from electronic structure theory that could be exploited include orbital active spaces, for when the desired solutions are known to be contained in a subset of (ov) rotations, as well as limits on the number of dimensions along which the vector K can perturb, as symmetry breaking in SCF solutions is connected to electron correlation for which low-order substitutions are most important (cf. truncated configuration interaction). Additionally, as identified solutions appear to be clustered around a particular reference energy, it may be possible to run simulations in parallel which are biased toward different reference energies and thus divide the problem space into energy windows. More generally, the most expensive part of the algorithm is the local SCF optimization and adaptations of the algorithm that reduce the number of minimizations may prove effective.

CONNECTIONS BETWEEN THE SELF-CONSISTENT FIELD ENERGY LANDSCAPE AND ELECTRON CORRELATION MECHANISMS USING A SYMMETRY-BASED GRAPHICAL APPROACH

4.1 Introduction

Peter Pulay proposes the unrestricted natural orbital complete-active-space (UNOCAS) to insist selecting the active space according to the occupation number (73), and later applying UNOCAS to interpret the multiple correlation mechanism using the averaged density.(74) In order to develop better idea of multiple correlation mechanism to further improving our active space decomposition methodology(5), the disconnectivity graph is introduced to connect the nearby solutions located by the stability test.(11) Since we have developed a good understanding of the wavefunction's characters from the previous chapter, this chapter will focus on the interpretation of located SCF solutions from the PES. The first thing we need to consider here is whether the located solution is stable or not, since the wavefunction's symmetry can always be broken to obtain the lower energy. The mechanism of the stability test was proposed long ago (75; 76) which specify the internal and external instability under the matrix context. This idea is also the core of our stability test which will be discussed in the methodology section below. The next thing we need to think about is how each SCF solution connected with the nearby solutions on the PES. One uses H_2 as the example to analyze its solutions' topology with hypersphere model and demonstrate the saddle point in hessian within different wavefunction methods.(77) The disconnectivity graph is widely used in modeling protein folding behavior.(78; 79; 80) Thermodynamics and kinetics properties through canonical partition functions and rate constants are used to gain further insight to sketch the disconnectivity graph. Except for describing biomolecules' behaviors, PES arising from wavefunction parameters' changes can also be illustrated through the disconnectivity graph. One also uses this idea to sketch the electronic structure of H_4 and analyzed them using the disconnectivity graph.(81) Through plotting the disconnectivity graph, we are able to observe the connectivity and hierarchy of different basins (SCF solutions) generated by different broken symmetry methods and symmetry-adapted method. We can interpret how the symmetry-broken energy surfaces are relate and how SCF solutions are related to different correlation structures and correlation mechanisms.

The treatment of electron correlation has been the essential topic for decades to infinitely approach the exact results. (82) Many methods designed to capture the strong correlations within the active space,(83; 84), adding the dynamic corrections,(85; 86), and study the nonadiabatic coupling between states,(87) etc. Besides various computational methods to address the correlations, modifications can also be done using different orbitals.(88; 89) Correlations for larger molecular systems do not limit by the scaling of the general active space methods,(90; 91) and the selection of active space can be guided by density matrix renormalization group (DMRG) method with polynomial scaling(92) or the first-order perturbation method.(93) Some orbitals are modified to preserve the symmetry and degeneracy, which improves the quality of the correlations to some extent. (94) In addition, the combinatorial method under the Monte Carlo context is applied to the electron correlation problem,(95) and variational quantum eigensolver (VQE) is used to located the optimal geometries by incorporating the correlations outside the active space.(96) One also explore the electron-proton correlations using the nuclear-electronic orbitals,(97) but this chapter is mainly focuses on the electronic correlations can be obtained.

Orbital entropy and mutual information are basic concepts from the quantum information. (98) These concepts can be used to evaluate the entanglement patterns in many-body systems, (27) specify the active space orbitals in DMRG methods,(28), accelerate the DMRG convergence by optimize the orbitals' order,(99) and analyze the quantum chemistry problems.(29) Bond formation processes (25) and orbital entanglement within the active space (26) are two typical examples to analyze the correlation using the orbital entropy and mutual information. In this case, we adopt this orbital entropy idea to interpret the correlation mechanisms among the located SCF solutions within the active space, which is demonstrated using the disconnectivity graph based on the energy. Three small molecules within the active space are evaluated in this chapter.

4.2 Computational Methods

In this section we describe the collection of theories and algorithms that are required to identify and determine connectivity between SCF solutions. First we describe the methods for local optimization and global elucidation of SCF solutions, including the new developments for performing global elucidation within an active orbital subspace. Second, we describe the tools used for characterizing and describing individual SCF solutions that are required for determining connectivity and between SCF solutions and establishing the extent to which SCF solutions represent different correlation mechanisms. Finally, we describe the approach for drawing and analyzing connections between SCF solutions, in which the theories and algorithms described in the preceding section are used.

Local optimization and global elucidation of self-consistent field solutions

Local self-consistent field optimization algorithms

A plethora of local optimization routines have been developed in recent years, designed to determine the stationary point in wavefunction parameter space closest to the initial guess determinant. In this work we utilize two different schemes – the state-targeted energy projection (STEP) procedure,(15) and the maximum overlap method (MOM) and initial maximum overlap method (IMOM) schemes.(13; 14)

In MOM/IMOM, the occupied set of orbitals in each iteration is chosen according to the overlap with either the occupied orbitals of the previous iteration (MOM) or of the initial guess (IMOM). In our global elucidation algorithm we use a slightly modified of the equation for computing the overlap:

$$P_p = \max_{i'}(abs(\sum_{\mu\nu} C_{\mu i'} S_{\mu\nu} C_{\nu p})) \tag{4.2.1}$$

where p is an orbital in the new iteration and i' is an occupied orbital in the previous (or initial) iteration, μ, ν . . . are atomic orbitals, $S_{\mu\nu}$ is the atomic orbital overlap matrix and Cp is an molecular orbital (MO) coefficient matrix element. The modified version of the overlap calculation is designed to avoid incorrect placement of core orbitals in the virtual space, due to the fact that the cumulative small overlap of diffuse virtual orbitals can be larger than the single relatively large overlap of core orbitals. Experience with MOM and IMOM has demonstrated that the methods perform well at avoiding variational collapse, especially when a non-minima stationary point is desired, but that the converge of the modified SCF optimization can be problematic.

Rather than use overlap as the metric to determine which orbitals are occupied in each iteration, the STEP procedure performs level shifting to ensure the virtual orbitals remain energetically above the occupied orbitals. In the STEP approach, the virtual orbitals are shifted in energy by an amount η:

$$\eta = |\varepsilon_{HOMO} - \varepsilon_{LUMO}| + \epsilon \tag{4.2.2}$$

where ε_{HOMO} and ε_{LOMO} are is the energy gap between the highest occupied and lowest unoccupied virtual orbitals, and ε is an empirical parameter that controls the local convergence performance of the STEP procedure. The level-shift ensures that the lowest energy virtual orbital in each iteration has an energy of at least above that of the highest energy occupied orbital. Larger values of are better at preventing variational collapse, while smaller values improve the convergence behavior. A problem with STEP is that very large level shifts are required in order to ensure convergence to non-minima SCF solutions.

Active-orbital subspace constrained global elucidation of self-consistent field solutions

In order to enforce local optimization within an orbital active space, a constraint is applied to the SCF energy (fig. 1). The notation for density matrices is as follows: P is the occupied orbital density matrix, Q is the virtual orbital density matrix, and R is the active orbital density matrix. The superscripts A and I indicate the active or inactive orbital subset respectively. The SCF energy is modified by defining the Lagrangian:

$$\tilde{E}_i = E_i + \lambda\alpha d_{0i}^2 ln(\alpha d_{0i}^2 + 1) \tag{4.2.3}$$

where E_i is the unconstrained SCF energy in the current iteration i, λ is an undetermined multiplier, α is an empirical parameter that controls the restoring force of the constraint, and d_{0i}^2 is the square distance between the guess active space density R_0 and the current iteration active space density R_i. The distance is obtained from:(3)

$$d_{0i}^2 = N_{orbs} - \langle R_i S R_0 S \rangle \tag{4.2.4}$$

where S is the atomic orbital overlap matrix, N_{orbs} is the number of occupied and virtual active space orbitals, and $\langle ... \rangle$ is the trace. The value of d_{0i}^2 corresponds to the number of orbitals that are different between the two densities.

The constraint Fock matrix can be obtained by taking the energy derivative with respect to R_i. Additionally, a level shift κ is applied to the initial guess inactive virtual orbital density Q_0^I to improve convergence to a solution within the active space and avoid convergence to spurious minima in the constraint space, where the d_{i0} is large enough to push active orbitals higher in energy than the inactive virtual orbitals. To optimize to different SCF solutions within the constrained space, a local convergence method must be employed, as discussed in section 2.1. Use of IMOM or MOM requires minor modifications to ensure the active and inactive occupied orbital subsets are treated separately. Where STEP is used the virtual orbital density matrix (including both active and inactive virtual orbitals) in the current iteration Q_i is also level shifted, in which case the resulting effective Fock matrix is formed as:

$$\tilde{F}_i = F_i + \gamma S R_0 S + (\gamma + \eta) S Q_i S + (\gamma + \eta + \kappa) S Q_0^I S \tag{4.2.5}$$

where F_i is the unconstrained Fock matrix, and γ is

$$\gamma = \lambda \{ \frac{\alpha d_{i0}^2}{\alpha d_{i0}^2 + 1} + \alpha In(\alpha d_{i0}^2 + 1) \} \tag{4.2.6}$$

which comes from taking the derivative of (4.2.3). Additionally, constraints can be applied to prevent convergence to previously identified solutions, based on the SCF metadynamics approach, where

$$\tilde{F}_i' = \tilde{F}_i + \sum_x P_x \lambda_x \alpha_x e^{-\alpha_x d_{ix}^2} \tag{4.2.7}$$

These additional constraints form the basis of a taboo algorithm, in which reconvergence to solutions already in the database can be inhibited for a period of time in order to drive the search to new regions of SCF space within the active space.

Theories for characterization of self-consistent field solutions

In this section, we introduce several concepts that are used in order to determine connectivity between SCF solutions and to analyze the extent to which the SCF solutions represent correlation mechanisms. We first briefly derive and explain the

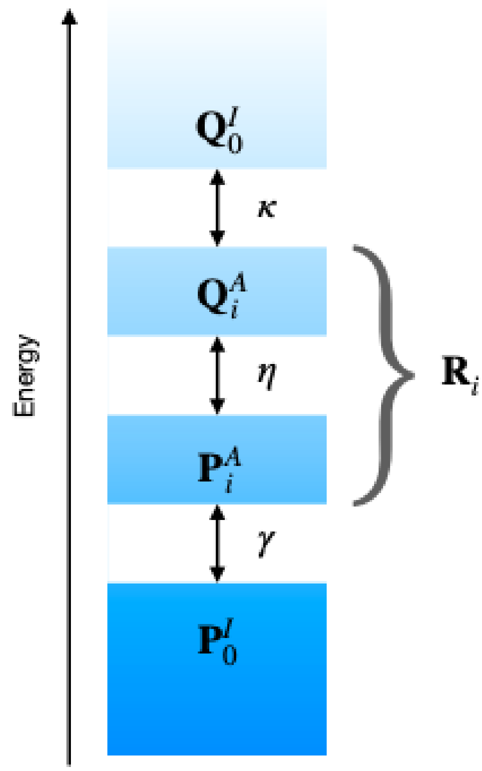


Figure 4.1. Scheme depicting the level-shifts applied in the active space global self-consistent field elucidation algorithm. The figure illustrates the optimization is constrained to the active space by level-shifting all virtual and active occupied orbitals by an amount dependent on the difference between the active space density of the guess orbitals and those of the current iteration $f(d^2)$. To prevent variational collapse within the active space, all virtual orbitals are shifted by $\eta = |E_{HOMO} - E_{LUMO}| + \varepsilon$ as defined by Carter-Frenk and Herbert's STEP procedure. The guess inactive virtual orbitals are additionally shifted by a constant C to prevent mixing into the active space.

different spin symmetries in SCF solution space, which is required as spin-symmetry forms the basis of the approach for determining connectivity between solutions. Second, we describe the HF orbital rotation Hessian, which is required for determining the directions to follow to identify a connected solution. Third, we then describe an approach that allows any SCF solution to be written as an orthogonal determinant expansion obtained from occupied-virtual (ov) substitutions from a reference determinant, which is useful for understanding how different SCF solutions encode correlation with respect to a given reference determinant. Finally, we de- scribe how to compute the orbital entanglement entropy and the mutual pair information in order to establish the extent to which SCF solutions partition the wavefunction into correlation mechanisms, and how well the proposed connectivity diagrams describe the partitioning of the wavefunction into correlation mechanisms.

Hartree-Fock orbital rotation Hessian and eigenvalues

The solutions identified in the SCF global elucidation algorithm are not necessarily min- ima in SCF parameter space, but may be higher order saddle points. To establish the form of the surface around the SCF stationary point, analysis of the occupied-virtual (ov) orbital rotation Hessian matrix indicates the changes in electronic structure that reduce the energy. Although, the calculation and analysis of the orbital rotation Hessian was reported over ffity years ago,(75) the extension and continued investigation of the stability of the SCF wave- function has continued to provide new insights.(34; 35; 81) The ov orbital rotation Hessian has the form:

$$H = \begin{bmatrix} A & B \\ B^* & A^* \end{bmatrix} \tag{4.2.8}$$

where the matrices A and B correspond to the energy derivative with respect to the action of any two ov orbital rotation generators:

$$A_{iajb} = F_{ab}\delta_{ij} - F_{ij}\delta_{ab} + \langle ai||jb\rangle \tag{4.2.9}$$

$$B_{iajb} = \langle ai||bj\rangle \tag{4.2.10}$$

where F_{ab} is a virtual-virtual (vv) element of the MO Fock matrix, F_{ij} is a occupied-occupied (oo) element of the MO Fock matrix. In general, A is Hermitian while B is symmetric. Diagonalization of H provides eigenvalues indicating the curvature of the energy in wavefunction parameter space along the associated orbital-rotation eigenvectors. Where A and B are real, it is possible to separately obtain the coordinates that preserve and break complex conjugation symmetry by diagonalizing $(A + B)$ and $(A - B)$ respectively. To correspond to orbital rotations, the eigenvectors of H must be put in the form:

$$H = \begin{bmatrix} D \\ D^* \end{bmatrix} \tag{4.2.11}$$

which satisfies the vanishing η-norm condition $V^\dagger \eta V = 0(34)$, where

$$\eta = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \tag{4.2.12}$$

The eigenvalues of H are real and, as well a positive and negative values indicating uphill and downhill curvature respectively, include zero eigenvalues. These zero-modes can be characterized as proper and improper, depending respectively, on whether they arise from degeneracy in the wavefunction or whether they arise from symmetry breaking and indicate the Goldstone manifold.(34)

Evaluation of Thouless parameterization of self-consistent field solutions

Single-reference wavefunctions, in which the N-electron wavefunction is described by a set of orthonormal molecular orbitals, can be represented using a unitary matrix. As a result, the set of all possible single-reference wavefunctions can be described by the $SU(2N)$ Lie group. Transformation between members of the $SU(2N)$ group can be described by exponential mapping from the corresponding $\mathfrak{su}(2N)$ Lie algebra, where the transformation is performed within a linear vector space. As a result, Thouless' theorem can be used to express any determinant $|\Psi'\rangle$ starting from a specific reference determinant $|\Psi\rangle$

$$|\Psi'\rangle = \exp\left\{\sum_{ia} k_{ia} a_a^\dagger a_i\right\}|\Psi\rangle \tag{4.2.13}$$

where $a_a^\dagger$ and a_i are creation and annihilation operators respectively, and k_{ia} is a (possibly imaginary) coefficient. Often, linear combinations of terms in eq.4.2.13 are taken in order to ensure that the transformation preserves spin or spatial symmetry. Expanding out the exponential as an infinite series leads to an expression that is equivalent to the configuration interaction expansion from the reference determinant $|\Psi\rangle$

$$|\Psi'\rangle = |\Psi\rangle + \sum_{ia} k_{ia} a_a^\dagger a_i |\Psi\rangle + \frac{1}{2}\sum_{ijab} k_{ia}k_{jb} a_b^\dagger a_a^\dagger a_j a_i |\Psi\rangle + \mathcal{O}(N^3) \tag{4.2.14}$$

Left-multiplying by a test determinant $\langle\Phi|$ provides the coefficients $\{k\}$ multiplied by the overlap of the test determinant with each term in the Thouless expansion of $|\Psi\rangle$

$$\langle\Phi|\Psi'\rangle = \langle\Phi|\Psi\rangle + \sum_{ia} k_{ia}\langle\Phi| a_a^\dagger a_i |\Psi\rangle + \frac{1}{2}\sum_{ijab} k_{ia}k_{jb}\langle\Phi| a_b^\dagger a_a^\dagger a_j a_i |\Psi\rangle + \mathcal{O}(N^3) \tag{4.2.15}$$

If the test determinant $\langle\Phi|$ is chosen to be equal to the reference determinant $\langle\Psi|$ or some orbital substitution from the reference determinant, e.g. $\langle\Psi|a_i^\dagger a_a$, then due to orthonormality of Slater determinants, all terms are zero in the expansion except the term that contains $\langle\Phi|$. For example, taking $\langle\Phi| = \langle\Psi|a_i^\dagger a_a$ gives

$$\langle\Psi|a_i^\dagger a_a|\Psi'\rangle = \cancelto{0}{\langle\Psi|a_i^\dagger a_a|\Psi\rangle} + k_{ia}\cancelto{1}{\langle\Psi|a_i^\dagger a_a a_a^\dagger a_i|\Psi\rangle} + \sum_{j\neq i, b\neq a} k_{jb}\cancelto{0}{\langle\Psi|a_i^\dagger a_a a_b^\dagger a_j|\Psi\rangle} + \ldots = k_{ia} \tag{4.2.16}$$

so that the coefficient k_{ia} can be determined as the overlap $\langle\Phi|\Psi'\rangle$, which is obtained from the determinant of the occupied-occupied overlap matrix.

Evaluation of determinant description of correlation mechanisms

Since the quantum information is illustrated in chapter 2 already, the methodology discussed here is simply showing the basic formulas of orbital entropy and the matrix forms for transition operators for single and orbital pairs. The single orbital entropy can be computed with the equation below:

$$s(1)_i = -\sum_{\alpha} w_{\alpha,i} ln w_{\alpha,i} \tag{4.2.17}$$

where α denotes four possible occupation of single spatial orbital i, and $w_{\alpha,i}$ is the eigenvalue of the one-orbital reduced density matrix. The mutual information $I_{i,j}$ can be computed with:

$$I_{i,j} = \frac{1}{2}(s(1)_i + s(1)_j - s(2)_{i,j})(1 - \delta_{i,j}) \tag{4.2.18}$$

in which s(1) is single orbital entropy for orbital i and orbital j, $s(2)_{i,j}$ is the two-orbital entropy calculated from the two-orbital reduced density matrix, which contains sixteen possible states for any selected orbital pair. $\delta_{i,j}$ is Kronecker delta. The magnitude of one orbital entropy and mutual information corresponds to different correlation effect qualitatively. (29)

Measurement of Orbital Entropy

The reduced density matrix of one and two orbitals can be constructed based on the generalized correlation function.(27; 25) The single electron basis can be empty, occupied with alpha spin or beta spin, or the doubly occupied condition. The one orbital reduced density matrix ρ_i is shown below:

$$\rho_i = \begin{pmatrix} \langle O_i^{(1)} \rangle & 0 & 0 & 0 \\ 0 & \langle O_i^{(6)} \rangle & 0 & 0 \\ 0 & 0 & \langle O_i^{(11)} \rangle & 0 \\ 0 & 0 & 0 & \langle O_i^{(16)} \rangle \end{pmatrix} \tag{4.2.19}$$

where $\langle O_i^{(n)} \rangle$ is single orbital operator that is the simple creator operator acts on the specific site and annihilate the rest sites:

$$\langle O_i^{(1)} \rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \langle O_i^{(6)} \rangle = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \langle O_i^{(11)} \rangle = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, etc \tag{4.2.20}$$

The structure of one-orbital operator $O^{(m)}$ contains a single element that has same occupation at position i, j, where i is the initial state $|i\rangle$ and j is the final state $|j\rangle$. It can be interpreted as the transition from state $|i\rangle$ to $|j\rangle$. In the case of two-orbital reduced density matrix, ρ_{ij} can be computed with the product of two single operators,$O_i^{(m)} O_i^{(n)}$. This can be interpreted as the transition from state $|i, j\rangle$

to state $|k,l\rangle$, in which $|i\rangle$ and $|k\rangle$ are initial states for two electrons, and $|j\rangle$ and $|l\rangle$ are the final states. The non-zero matrix elements in two-orbital reduced density matrix have same quantum number n and s_z, so that only 36 elements need to be calculated. (See Table 4.1 below) The final two orbital reduced density matrix is renormalized before the evaluation of two orbital entropy.

Table 4.1. All possible situation when computing two-orbital entropy. Each number represents the location index which corresponds to the occupation within one orbital entropy matrix.

ρ_{ij}	n=0,$s_z = 0$	n=1,$s_z = -\frac{1}{2}$	n=1,$s_z = 1_{\overline{2}}$	n=2, $s_z = -1$	n=2,$s_z = 0$	n=2, $s_z = 1$	n=3,$s_z = -1_{\overline{2}}$	n=3,$s_z = 1_{\overline{2}}$	n=4, $s_z = 0$
	--	- ↓ ↓ -	-↑ ↑ -	↓ ↓	- ↑↓ ↓↑ ↑↓ ↑↓ -	↑ ↑	↓ ↓↑	↓↑ ↓ ↑ ↓↑	↓↑ ↑	↓↑ ↓↑
--	1/1								
- ↓		1/6 2/5							
↓ -		5/2 6/1							
- ↑			1/11 3/9						
↑ -			9/3 11/1						
↓ ↓				6/6					
- ↓↑					1/16 2/15 3/14 4/13				
↓↑					5/12 6/11 7/10 8/9				
↑↓					9/8 10/7 11/6 12/5				
↓↑ -					13/4 14/3 15/2 16/1				
↑ ↑						11/11			
↓ ↓↑							6/16 8/14		
↓↑ ↓							14/8 16/6		
↑ ↓↑								11/16. 12/15	
↓↑ ↑								15/12. 16/11	
↓↑ ↓↑									16/16

4.3 Results and Discussion

Graphical depiction of the topology of self-consistent field solution space

In order to provide insight as to how mean field solutions reflect correlation mechanisms, and how correlation mechanisms relate to symmetry breaking, it is necessary to investigate the topology of the mean-field surface. In particular, it is necessary to know the position of stationary points on the mean-field surface, as well as the connectivity between these stationary points. The high-dimensional topology of the mean-field surface can be reduced to a graph in which all stationary points are written as vertices and the stationary points that are connected are drawn together using edges. It is also possible to go further and draw directed edges that indicate the direction in which the electronic energy decreases. In one dimension, the connectivity can be defined by moving in each of the two possible directions until another stationary point is found. Only stationary points that are connected to each other without passing through a third stationary point are shown as connected by edges. Figure 4.2 illustrates how such an approach works in one dimension.

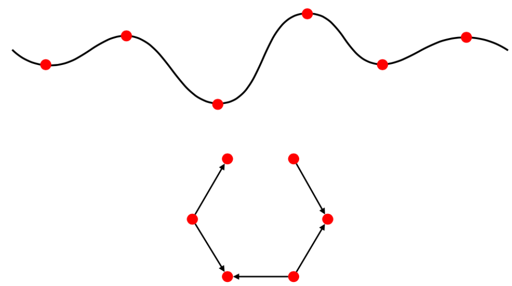
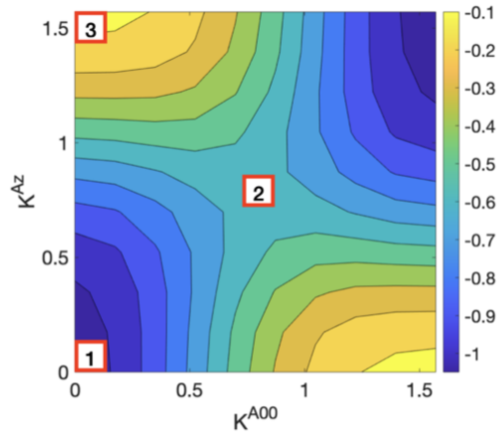


Figure 4.2. Graph construction scheme from one-dimensional surface. At top, the one-dimensional surface is shown with red points indicating stationary points. At bottom, directed graph representing one-dimensional surface is shown, where shape of pictorial representation is unimportant.

While in one-dimension, it is straightforward to determine the connectivity between vertices as there are a finite number of directions (two) in which it is possible to transverse the surface, the addition of a second degree of freedom leads to an infinite possible number of directions in the plane which makes defining connectivity more difficult, as without symmetry it is likely that all stationary points are connected to all others, with the problem increasing with the number of dimensions. Therefore, it is necessary to define connectivity in a meaningful way for the electronic structure problem. In the similarly related chemical reaction network edges are drawn between vertices representing chemical species that are interconverted by a first-order saddle point. However, in the many-electron problem, higher-order saddle points are important as they describe symmetry adaption and excited-state approximations and so must be included in the graph.Additionally, saddle-point order can change unexpectedly so it is not possible to describe Nth order saddle points as being connected by N+1th order saddle points, and symmetry breaking leads to modes that corresponds to zero eigenvalues. To illustrate how saddle point order can change in unintuitive ways between stationary points in SCF solution space, figure 4.3 shows the energy surface of H_2 computed in a minimal basis set (STO-3G) using two generators (k_{12}^{A00} and k_{12}^{Az}) that enable all collinear solutions to be determined. Searches outside of the two generator space lead to different but equivalent solutions corresponding to arbitrary rotation of spin axis. The table to the right details the number of negative and zero modes of the orbital rotation Hessian. As can be seen, while using two real unrestricted (RU) generators gives stationary points with all saddle point orders between zero and the number of dimensions, the four real general (RG) generators or the eight complex general (CG) generators introduce zero modes, coplanarity and noncoplanarity, such that some saddle point orders are absent, and nearby stationary points have very different saddle point orders. For example, in CG with 8 generators, only zeroth, fourth and eighth order saddle points are present. Therefore, it is not possible to use eigenvector following, in which a mode is followed uphill from the minimum while all all other modes are minimized, as this approach assumes that nearby stationary points are connected by a ± 1 change in the saddle-point order.

In order to determine the connection between stationary point, we propose to use an approach in which each eigenvector is followed until either 1) the energy change when stepping along the eigenvector changes sign, 2) the eigenvector is close to perpendicular to the gradient difference (we use within 5 degrees). Once either of these conditions is reached, damped Newton steps are taken to follow the eigenvector end point to the nearest stationary point. We first illustrate the method on minimal basis H_2 at 2 a_0 bond distance following the two eigenvectors that remain in the plane of the k_{12}^{A00} and k_{12}^{Az} generators from the lowest energy real restricted (RR) and the RU stationary points (red line in Figure 4.4). The eigenvectors from the RR minimum connect with RR maximum along either k_{12}^{A00} or the k_{12}^{Az} paths, where k_{12}^{A00} path gas higher curvature around the RR minimum. The same analysis with four generators that give all real wavefunctions result in two additional pathways, one of which is equivalent to the k_{12}^{Az} path but proceeds via wavefunctions collinear with y spin axis instead of the z spin axis along k_{12}^{Sy}, and the second which also leads to the RR maximum but along a triplet pathway defined by k_{12}^{Ax}. Using eight generators reveals a third route equivalent to the k_{12}^{Az} path that leads to the RR maximum but rotating around y axis into the complex plane, an additional route to the RR maximum along a triplet pathway but through the complex plane, and two equivalent paths that lead to the RU solution. However, the two paths in the complex plane that lead to the RU solution are interesting because they involve degenerate eigenvectors, so any linear combination of these pathways is also a valid connection, including the path that leads to the RR maximum. Therefore, unlike in the real wavefunction parameter space where the eigenvectors are not degenerate, the connection between solutions in the colinear complex space is not well defined. The paths from the RU solutions in the plane of k_{12}^{A00} and k_{12}^{Az} lead to the RR minimum and maximum. Analysis of figure 4.4 reveals why it is not possible to connect to the RU solution by following eigenvectors of either RR solution in the real plane, as the eigenvector towards RU requires displacement along both eigenvectors. Of the other two generators in the four-generator real wavefunction space, one eigenvector connects the RU and triplet solutions, while the other corresponds to an improper zero-mode indicating the collective rotation of the total spin axis. In the full space, the remaining four eigenvectors in the complex plane mirror those of the real space. In terms of connectivity, the above analysis suggests that RR minimum and maximum are connected, while the RU solution does not connect in any simple way, and so corresponds to a distinct state(the singly excited state), although following the degenerate eigenvectors in the complex plan can reveal the two determinants required to obtain a spin-adapted wavefunction.

While the previous example is useful for examine how stationary points in mean-field wavefunction parameter space can be connected by following eigenvectors, the RR solutions are good approximations to the exact wavefunctions, with only dynamic correlation absent, such that there are no strong correlation mechanism present. To illustrate how the connectivity based on following orbital rotation Hessian eigenvec-



Solution #	Symmetry	# Generators	# Negative modes	# Zero modes
		8	0	0
1	RR	4	0	0
		2	0	0
		8	4	2
2	RU	4	2	1
		2	1	0
		8	8	0
3	RR	4	4	0
		2	2	0

Figure 4.3. Energy surface of H_2 computed using Hartree-Fock with STO-3G basis set along generators k_{12}^{A00} and k_{12}^{Az}, corresponding to possible solutions with RU or higher symmetry. Three labelled points correspond to different stationary points on surface (stationary points on right axis equivalent with stationary points diagonally opposite). Table to right shows the number of negative modes and number of zero modes of the orbital rotation Hessian for 2 generator space (k_{12}^{A00} and k_{12}^{Az}), four generator space (k_{12}^{A00}, k_{12}^{Az},k_{12}^{Ax} and k_{12}^{Sy}) corresponding to all real wavefunctions, and eight generator space (k_{12}^{A00}, k_{12}^{Az},k_{12}^{Ax},k_{12}^{Sy},k_{12}^{S00}, k_{12}^{Sz},k_{12}^{Ay} and k_{12}^{Sx}) corresponding to all real and complex wavefunctions. While additional solutions can be found in 4 and 8 generator spaces, they are all degenerate with solutions found in 2 generator space.

tors reveals correlation mechanisms in a simple system, we now analyze the same minimal basis H_2 molecule, but at a slightly stretched geometry of 1.7 Å(4.5). For the sake of brevity, in this section we will only discuss the solutions following paths in the plane of the k_{12}^{A00} and k_{12}^{Az} generators. At 1.7 Å symmetry broken solutions arise as both a new minimum along k_{12}^{Az} (RU spin symmetry broken), and a new maximum along k_{12}^{A00} (RR spatial symmetry broken). As a result, the RR are first-order saddle points instead pf minima/maxima like at the equilibrium geometry. As a result, following the eigenvectors from the RR solutions leads to the RU minimum and the spatial broken RR maximum, such that the symmetry adapted RR solutions are no longer directly connected. Following the eigenvectors from the symmetry-broken solutions reveals that the presence of a connecting pathway depends on which stationary point is used as the starting point. For example, from the RU minimum at $k_{12}^{A00} = 0$ and $k_{12}^{Az} > 0$, following the eigenvector that initially proceeds along the k_{12}^{A00} generator leads to the spatial symmetry broken RR maximum. However, starting from the spatial symmetry broken RR maximum, it is not possible to retrace the same pathway, although it is possible to connect to a minimum corresponding to the same solution at $k_{12}^{A00} = \frac{\pi}{2}$ and $k_{12}^{Az} < 0$. As a result, regardless of the starting point, the same connectivity is obtained, even though the specific connecting pathways are different. Finally, as at the equilibrium geometry, the RU first-order saddle point cannot be found by searching from any other solution, even though it connects to the two symmetry broken solutions.

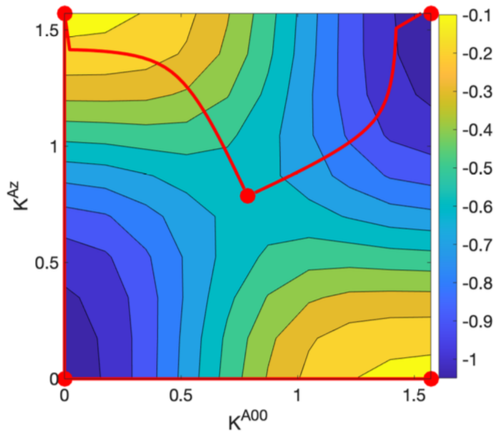


Figure 4.4. Connecting paths in the k_{12}^{A00} and k_{12}^{Az} plane from the lowest energy RR solution and the RU solution computed through following eigenvectors as described in the text. Red points indicate the location of stationary points. Only the forward pathways are shown.

Having computed which stationary points are connected, the next challenge is to translate this information into a graph that express the topology of self-consistent field solution space in a useful way. Recent work by Burton and Wales (81) used disconnectivity graphs to indicate where minima in self-consistent field solution space are connected by first-order saddle points. As discussed above, even though disconnectivity graphs are a useful approach for simplifying the description of the SCF topology, the limitation of only describing minima limits the utility of the approach. However, we propose that the disconnectivity graph is still a useful representation if the criteria for drawing it are modified, where the process is outlined in Figure 4.6. First (Figure.4.6b), stationary points are represented as points and connections are only drawn between solutions that are connected by following an eigenvector downhill in energy. Second (Figure.4.6c), all but one equivalent solutions (zero distance between solutions) which are shown as separate vertices, due to periodicity (compactness) of the $\mathfrak{su}(2\mathrm{N})$ Lie algebra that defines the topology, are removed and the connections redrawn to the remaining vertex. Third (Figure.4.6d), the vertices are arranged in order of energy of the associate stationary point. Fourth (Figure.4.6e), the spin and spatially adapted restricted solutions are identified, and all edges that do not originate of end at the restricted vertices are removed. Fifth (Figure.4.6f), if multiple edges are drawn from a symmetry broken solution, all edges are removed except for those that connect to the closest restricted solution, preferably downhill in energy. In this step there is some arbitrariness, and the particular choice as to which solutions are retained or deleted depends on the specific concept the disconnectivity graph is trying to convey. The choice put forward here is designed to highlight the symmetry breaking processes that lead to a reduction in the energy, as these are associated with the correlations that stabilize the system. Finally. if restricted solutions are connected if they are not already. Interestingly, it can be seen that, at least in the example shown in Figure.4.6, the symmetry-broken solutions are shown as being connected to the closest spin and spatially adapted restricted solutions in both energy and orbital rotation space.

In practice, finding all solutions of all symmetries in order to establish connections can be a challenge and complicate visualization. Instead, a more tractable scheme can be found using the hierarchy espoused in the subgroup inclusion diagram (fig.4.7). In this approach, a particular spin symmetry is chosen such that solutions of that spin symmetry and any symmetries included in that group (higher in the subgroup inclusion diagram) are represented by nodes in the graph, while the end points represent the spin symmetries of lower symmetry that are directly connected to the symmetry used to specify the nodes. As an example, if the RU spin symmetry is of interest, the nodes will be solutions of RU and RR spin symmetry, while the end points will be solutions of RG and complex unrestricted (CU) spin symmetry. Spatial symmetry can be treated in a similar approach if desired by using the subgroup inclusion diagram formed from the direct product group of spin and spatial symmetries. Additionally, assuming only low energy states are of interest, the search can be performed starting from solutions identified from searching within a subset of orbitals using the active space approach outlined above and/or using an energy threshold for determining which solutions to search from, and only searching in downhill directions.

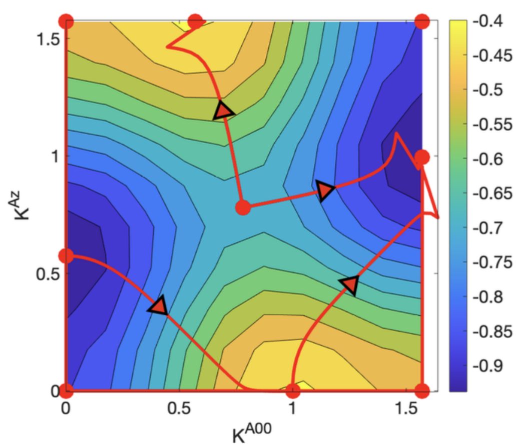


Figure 4.5. Connecting paths in the k_{12}^{A00} and k_{12}^{Az} plane computed through following eigenvectors as described in the text. Red points indicate the location of stationary points and arrowheads indicate pathways that can only be followed in one direction. Only pathways from the five unique solutions in the bottom right if the figure are drawn.

Assessing SCF solutions as descriptors of correlation mechanisms in ozone

Ozone is an example of a molecule that is known to have multiple correlation mechanisms acting. The role of the fractionally occupied natural orbitals in determin-

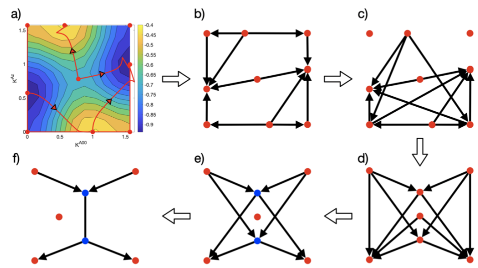


Figure 4.6. Process for building disconnectivity graph from SCF topology and connectivity between solutions.

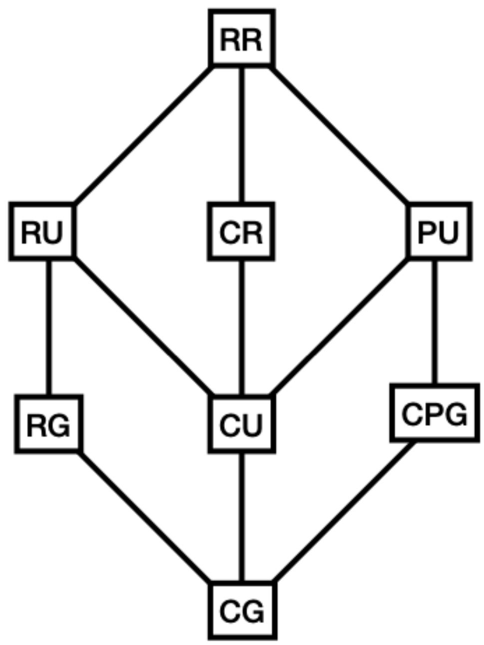


Figure 4.7. Process for building disconnectivity graph from SCF topology and connectivity between solutions.

ing the relevant correlated orbital space was first highlighted by Pulay and coworkers in the original UNOCAS investigation,(54) although the presence of multiple correlation mechanisms was not understood until much later, where the averaged density UNOCAS was found to give results in agreement with very large active space density matrix renormalization group calculations.(74) Subsequently is was found that the correlated space could be partitioned according to the fractionally occupied natural orbitals belonging to each of the two SCF solutions using the nonorthgonal active space decomposition approach.(5) In this section, we reexamine the SCF solutions using one-orbital entropy and pair mutual information metrics to establish whether the SCF solutions really describe different correlation mechanisms. The one orbital entropy measures entanglement of an orbital with all other orbitals, and so should be high for all the fractionally occupied natural orbitals identified in each SCF solution. The pair mutual information describes the entanglement of two orbitals, and so if the SCF solutions describe a partitioning of the correlated space, this measure should be high for orbitals in the same correlation mechanism (fractionally occupied natural orbitals belonging to the same SCF solution) and low for orbitals belonging to different correlation mechanisms (fractionally occupied natural orbitals belonging to different SCF solutions).

Figure 4.8 shows the fractionally occupied natural orbitals from the two RU SCF solutions, along with their corresponding occupation number. Solution A has a σ type orbital constructed from p_x orbitals of each oxygen in the plane of the molecule and the π antibonding orbital formed from p_z orbitals of each oxygen perpendicular to the plane. Solution 2 contains the same π antibonding orbital, but also contains the π nonbonding orbital. Thus one SCF solution describes correlation between σ and π^* orbitals, and the other describes correlation between nonbonding and π^* orbitals. The fact that π^* is correlated with two different partner orbitals means there are two correlation mechanisms and so, as a SCF solution can only describe a single correlation mechanism, there are two SCF solutions. Table 4.2 shows the one-electron orbital entropies on the diagonal and pair mutual information metric on the off-diagonal. All orbitals show relatively high one-orbital entropies that are substantially higher than the other natural orbitals, although the nonbonding π orbital (B1) has a much lower one-orbital entropy than the other two orbitals. Examining the pair mutual information, the correlation between σ and π^* orbitals and between nonbonding π and π^* is shown to be larger than the correlation between σ and nonbonding π. This finding is in agreement with the hypothesis that SCF solutions partition the correlation mechanism and explains the good performance of the nonorthogonal active space decomposition method compared to the averaged density UNOCAS approach.

Table 4.2. One orbital entropies (diagonal) and pair mutual information (off-diagonal) of orbitals with labels given in Figure 4.8.

	A1	A2	B1	B2
A1	0.47	0.30	0.23	0.30
A2		0.37	0.39	–
B1			0.15	0.39
B2				0.37

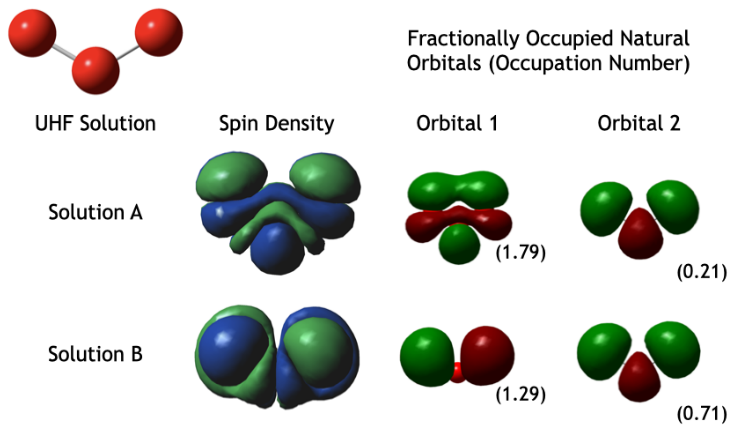


Figure 4.8. Spin densities, natural orbitals and occupation numbers of identified self-consistent field solutions that describe correlation in ozone.

4.4 Conclusions

In this work we have developed a graphical approach in an attempt to center the symmetry in describing the topology of the self-consistent field solution space. Through analysis of the minimal-basis H2 molecule, a model has been demonstrated for connecting solutions that emphasize how the topology describes orbital correlation mechanisms. The difficulty of establishing connections between solutions in a high-dimensional, nonlinear space has been described, and a technique of following eigenvectors of the orbital-rotation Hessian implemented. Additional tools to extend the analysis to larger systems have been described, including active-space basis hopping. To further demonstrating the multiple correlation mechanism, two RU solutions of ozone are examined using the orbital entropy concept, and correlation of chosen paired of orbitals inside active space are found to be larger than orbitals outside the active space through the computed mutual information. This result supports the choice of orbitals in ozone in nonorthogonal active space decomposition method.(5) Overall, it is hoped that the developments here can be extended to larger systems and provide new approaches for considering how self-consistent field solutions can represent correlation mechanisms, and how correlation mechanisms can be used to partition strong correlation into weakly correlated subsets, and so reduce the computational cost of evaluating the wavefunction in strongly correlated molecular systems.

TIME PROPAGATION OF ELECTRONIC WAVEFUNCTIONS USING NONORTHOGONAL DETERMINANT EXPANSIONS

5.1 Introduction

Non-orthogonal configuration interaction (NOCI) serves as one of the potential candidates of post-HF method due to its ability to capture the essential electron correlations within sophisticated molecular models. Two commonly studied models are conical intersection of O_3 and bond dissociation of LiF, which mainly require the accurate descriptions of static correlations when approaches the critical geometries.(100) The computational cost of NOCI is relatively reasonable compared with couple cluster method, and the cost can be further reduced if certain parameters are introduced(101) or applying the novel algorithm when evaluating the Hamiltonian matrix.(102) However, NOCI also has some issues. Lack of dynamic correlations makes NOCI hard to offer a correct correlation descriptions when dynamic correlations dominates in certain cases, but introducing of Møller-Plesset perturbation theory (MP2) allows the correction for the correlation in NOCI. (103) Non-orthogonal internally contracted multi-configurational perturbation theory (NICPT) applies the perturbative dynamic corrections for compact non-orthogonal MCSCF calculations which is a different method than NOCI that also has similar treatment.(104) Another problem of NOCI is the selection of slater determinants. Normally, those selected determinants are based on the experience that orbitals with certain properties have to be included, such as π orbitals in linear polyenes (105) and slater determinants can depict the complex energy surface.(103) Of course, experience cannot ensure every unknown condition. One proposes the novel black-box selection method, spin–flip non-orthogonal configuration interaction, to study the exchange couplings in bimetallic molecules and metal-ligand spin coupling. (106) Interestingly, NOCI method can also be extended into complex plane with holomorphic HF theory when solutions disappear at certain geometry in the real space.(107)

In addition to focusing on NOCI within time-independent context, we are more interested in introducing non-orthogonality into the time-dependent Schrodinger equation to study the electron dynamics in this chapter. Rapid development in ultrafast spectroscopy enables us to study the electron and nuclear dynamics on the femto/atto-second time scale. (108; 109) Theory-based simulation provides a feasible way to understand the linear and nonlinear response produced by strong-field laser. However, for field strength cannot the addressed by either perturbative approach or response theory, the explicit time-dependent method is required. Real-time time-dependent density functional theory (RT-TDDFT) is the mostly widely used method due to a good balance of the computational cost and accuracy,(110; 111) but this method suffers the inaccuracy in describing the Rabi oscillation,(112) resulting in the unphysical peak shifting and proved in the experiment(113), and wrong charge-transfer dynamics(114) due to the adiabatic approximation. Besides, DFT is a single reference method, and it has self-interaction error, in which RT-TDDFT is hard to manage the chemical process requires the time-resolved density updates. Resolving the issue of RT-TDDFT requires the multiconfigurational description of electron dynamics. The truncated real-time time-dependent configuration interaction (RT-TDCI) is a reliable approach for its multireference feature and less computational cost compared to other multireference methods. Real-time time-dependent configuration interaction singles (RT-TDCIS) is the most common flavor of RT-TDCI, and has been used for a wide number of applications, including photoionization,(115; 116) non-linear optical response,(117; 118) molecular switches and wires,(119) and dipole switching.(120) However, due to Brillouin's theo- rem, RT-TDCIS lacks electron correlation in the ground state, and so overestimates dynamic and static polarizabilities and hyperpolarizabilies.(121) Explicit inclusion of double and higher excitations were found to be important for correcting the deficiencies in the time-evolution of the wavefunction, although inclusion of perturbative doubles did not correct the qualitative problems of RT-TDCIS.(122; 123) Unfortunately, explicit inclusion of all double excitations is often cost prohibitive, and active space expansions are employed instead. (124; 125; 126; 127)

Rather than explicitly including higher-order excitations beyond single substitutions, an alternative is to implicitly wrap higher-order excitations into each single substituted determinant. One approach to introduce these higher-order excitations is through the use of nonorthogonality. Non-orthogonal orbitals contain different correlations within different determinants and it has potential to achieve similar accuracy as inclusion of higher order CI terms with less computational cost. In this chapter, we examine the performance of real-time time-dependent nonorthogonal configuration interaction singles (RT-TDNOCIS) in overcoming problems associated with RT-TDCIS. Specifically, we introduce the RT-TDNOCIS method before examining its performance for a number of systems with respect to the truncated RT-TDCI hierarchy to establish the extent to which nonorthogonality is able to reduce the size of the determinant expansion.

5.2 Computational Methods

Orthogonal Real-Time Time-Dependent Configuration Interaction

The time-dependent Schrödinger equation (TDSE) describes the time-evolution of the many-electron problem and so plays an important role in modeling the real-time dynamics of chemical systems when a time-dependent external perturbation is applied:

$$i\hbar\frac{\partial\Psi(t)}{\partial t} = \hat{H}(t)\Psi(t) \tag{5.2.1}$$

where $\hat{H}(t)$ is the time dependent (TD) Hamiltonian operator, $\hbar$ is reduced Planck constant, and $\Psi(t)$ is time-dependent wavefunction. For a system in a TD external electric field, the TD Hamiltonian can be written as:

$$\hat{H}(t) = \hat{H}_0 - F(t) \cdot \hat{\mu} \tag{5.2.2}$$

where $\hat{H}_0$ is the time independent (TI) field-free molecular Hamiltonian, and the TI component of the Hamiltonian couples the external electric field F(t) to the molecular dipole $\mu(t)$, which generally also evolves in time, and is evaluated using the dipole operator $\hat{\mu}$. Both the electric field and molecular dipoles are vectors in real space where F(t)=$\{F_x(t), F_y(t), F_z(t)\}$ and $\mu(t)$=$\{\mu_x(t), \mu_y(t), \mu_z(t)\}$. To simulate propagation of the wavefunction in time, a suitable ansatz is required that is simultaneously computationally efficient and sufficiently accurate to provide chemical insight. The TD wavefunction can be constructed from a linear expansion of TI solution $\Psi_A(0)$:

$$\Psi(t) = \sum_A c_A(t)\Psi_A(0) \tag{5.2.3}$$

where $c_A(t)$ are the TD coefficients that govern the time evolution of the wavefunction and the TI wavefunctions are obtained from solutions of the TI Schrödinger equation:

$$\hat{H}_0\Psi_A(0) = E_I\Psi_A(0) \tag{5.2.4}$$

Throughout the text we use the index notation A,B,C ... to refer to the basis of electronic eigenstates, $\bar{A}, \bar{B}, \bar{C}$ ... to refer to the basis of dipole eigenstates, I, J, K... to refer to the Slater determinant (SD) or configuration state function (CSF) basis in which electronic eigenstates are expressed, p, q, r ... to refer to molecular orbitals (MOs), where i, j, k ... are the set of occupied orbitals and a, b, c ... are the set of virtual MOs in the reference determinant, and finally ν, λ, σ ... refers to the atomic orbital (AO) basis set in which MOs are expressed. If substituting (5.2.3) into (5.2.1) and left-projecting by $\langle\Psi_A(0)|$ gives the equation of motion to determine the time-evolution of $\{c_A(t)\}$ as a set of coupled differential equations:

$$i\hbar\frac{\partial c_A(t)}{\partial t} = \sum_B \langle\Psi_A(0)|\hat{H}|\Psi_B(0)\rangle c_B(t) \tag{5.2.5}$$

Solving (5.2.5) above for the time evolution of the state coefficients $\{c_A(t)\}$ can make use of the split operator approach, such that:

$$c(t + \Delta t) = [\prod_{q=x,y,z} U_q^\dagger e^{iF_q(t)\tilde{\mu}_q\Delta t} U_q]e^{-iE\Delta t}c(t) \tag{5.2.6}$$

where c(t) are the state coefficients at time t, Δ t is the finite time step used in the simulation, and E are the eigenvalues associated with eigenstates of the field-free Hamiltonian $\hat{H}_0$ according to (5.2.4). To account for the role of the TD electric field, $\tilde{\mu}_q$ are the eigenvalues associated with eigenstates of the field-free dipole $\tilde{\mu}_q$

along each axis q. Note that the eigenstates of $\hat{H}_0$ and $\hat{\mu}_q$ are generally different. Calculation of the electric field interaction in the basis of dipole eigenstates simplifies computing the exponential,(128) but requires the resulting matrix to be transformed to the state basis in which $\hat{H}_0$ is diagonal, through the action of unitary matrices U, which are computed as:

$$U_q = d^\dagger D \tag{5.2.7}$$

where for the first time we have introduced an explicit basis for the TI wavefunctions, which in configuration interaction (CI) is the basis of SDs or CSFs $\{|\Phi_I\rangle\}$, such that:

$$\Psi_A(0) = \sum_I D_{IA}|\Phi_I\rangle \tag{5.2.8}$$

and

$$\Psi_{\tilde{A}}(0) = \sum_I d_{qI\tilde{A}}|\Phi_I\rangle \tag{5.2.9}$$

in which D are the CI coefficients for the Hamiltonian operator and d_q are the equivalent for the dipole operator. The TD CI coefficients D(t) that describe the TD state $\Psi(t)$ can be computed from:

$$D(t) = \sum_A c_A(t) D_A \tag{5.2.10}$$

In the CI formalism the TI wavefunction is constructed from a determinantal expansion obtained by a maximum number of orbital substitutions from a reference determinant. In the case of configuration interaction singles (CIS) the wavefunction of an excited state is:

$$\Psi_{\tilde{A}} = \sum_{ia} D_{iaA}|\Phi_i^a\rangle \tag{5.2.11}$$

where $|\Phi_i^a\rangle$ indicates a single substitution from the reference determinant $|\Phi_0\rangle$. Importantly, where $|\Phi_0\rangle$ is a minimized HF determinant, the single substitutions do not correlate to the ground state due to Brillouin's theorem. The lack of correlation in the ground state is a leading cause of error in RT-TDCIS and requires correction by including higher order substitutions, such as doubles and triples.

Real-Time Time-Dependent Nonorthogonal Configuration Interaction

To resolve the issues associated with orthogonal RT-TDCI methodologies, we modify the algorithm to use a basis of independently optimized self-consistent field (SCF) solutions. Each determinant is obtained by relaxing the orbitals to a local minimum in the vicinity of an initial determinant constructed by orbital substitutions from the reference determinant according to:

$$|\Phi_0\rangle \xrightarrow{\psi_i \to \psi_a} |\Phi_i^a\rangle \xrightarrow{\psi \to \bar{\psi}} |\bar{\Phi}_i^a\rangle \tag{5.2.12}$$

where the bar indicates the orbital or determinant results from an SCF optimization independent of that used to obtain the reference. The relaxed determinant $|\bar{\Phi}_i^a\rangle$ is related to the original substituted determinant $|\Phi_i^a\rangle$ through Thouless theorem(9):

$$|\bar{\Phi}_i^a\rangle = exp \sum_{jb} k_{jb} a_b^\dagger a_j |\Phi_i^a\rangle \tag{5.2.13}$$

where a_i is the annihilation operator that removes an electron from orbital i, $a_a^\dagger$ is the creation operator that adds an electron to orbital a, and k_{jb} is a coefficient that scales the transformation. The nonorthogonal determinant expansion is then given by:

$$|\bar{\Psi}_A(0)\rangle = \sum_I \bar{D}_{IA} |\bar{\Phi}_I\rangle \tag{5.2.14}$$

Generally the set $\{\bar{\Phi}\}$ are nonorthogonal, while the set $\{\Phi\}$ are orthogonal, and higher order substitutions are implicitly wrapped into the set $\{\bar{\Phi}\}$ by the action of the exponential, which introduces disconnected cluster operators constructed from single-substitutions up to the infinite order. The reference determinant on which substitution are performed can use any spin symmetry, although real restricted spin symmetry is most straightforward as it ensures the CI is properly symmetry adapted.

There are several additional important points to note about the use of the determinant expansion in (5.2.14): 1).The set of reoptimized single-substituted determinants is no longer orthogonal to the ground state. As a result, Brillouin's theorem no longer applies and correlation with the ground state is introduced. 2). It is straightforward to treat the aufbau determinant in the same way as excited configurations by breaking spin-symmetry, most easily achieved by following the orbital rotation Hessian eigenvectors that lower the energy. In the case that the SCF reoptimization yields a broken spin-symmetry determinant, it is possible to restore spin-symmetry by including in the expansion the set of determinants that comprise the associated Goldstone manifold. These symmetry-restoring determinants can be identified from the zero modes of the orbital rotation Hessian.(34) In this study we investigate two approaches: RT-TDNOCIS where the restricted determinant is used, which also serves as the reference for obtaining excited substitutions, and real-time time-dependent half-projected nonorthogonal configuration interaction singles (RT-TDHPNOCIS) in which half-projection of an unrestricted aufbau determinant is performed, in which the symmetry- broken determinant is obtained by following orbital-rotation Hessian eigenvectors down in energy from the restricted determinant that is used as a reference to generate substituted determinants. The half-projection is accomplished by inclusion of an additional determinant in the expansion obtained from a collective $\frac{\pi}{2}$ rotation of the axis of spin quantization according to:

$$|\bar{\Phi}_{HP}\rangle = \hat{P}_{\alpha\beta} |\bar{\Phi}_0\rangle \tag{5.2.15}$$

where $\hat{P}_{\alpha\beta} = e^{i\pi \hat{S}_y}$ is the operator that interchanges α and β spin functions, in which $\hat{S}_y$ is the operator that rotates electron spin around the y-axis of the molecular reference frame. and $|\Phi_0\rangle$ denotes the determinant obtained from symmetry breaking f the reference determinant. As a result, both the proposed methods are symmetry adapted and avoid artifactual bright spin-forbidden transitions that occur when a symmetry-broken reference is used in orthogonal methods.

Having obtained the nonorthogonal determinant basis, construction of the Hamiltonian and overlap matrix can be performed using one of a number of approaches for computing the nonorthogonal matrix elements, with advantages and disadvantages of each algorithm depending on the specific form of the determinant expansion.(129; 130; 131; 77; 132; 5; 133; 134; 135; 136; 137; 138; 139; 100) In the model of interest in this work, as there is no *a priori* knowledge of how the determinants in the expansion are related, each matrix element is constructed through contractions with transition density matrices $\boldsymbol{\rho}$ in the AO basis:

$$H_{IJ} = \tilde{N}_{IJ}\langle I|\hat{H}|J\rangle = \tilde{N}_{IJ}\left(\langle \mathbf{h}\boldsymbol{\rho}_3\rangle + \frac{1}{2}\langle \boldsymbol{\rho}_1\mathbf{G}(\boldsymbol{\rho}_2)\rangle\right) \tag{5.2.16}$$

where $\mathbf{G}(\boldsymbol{\rho}_2)$ is the contraction of two-electron integrals with $\boldsymbol{\rho}_2$, $\langle\ldots\rangle$ is the trace and $\tilde{N}_{IJ}$ is the pseudooverlap obtained from the determinant of the image of ${}^{IJ}\mathbf{M}$, where

$${}^{IJ}\mathbf{M} =^{I} \mathbf{C}^{\dagger}_{occ}\mathbf{S}^{J}\mathbf{C}_{occ} \tag{5.2.17}$$

in which $\mathbf{S}$ is the AO overlap matrix and ${}^{I}\mathbf{C}_{occ}$ is the occupied MO coefficients of determinant I. The image of ${}^{IJ}\mathbf{M}$ is the space of non-zero singular values obtained using single value decomposition (SVD)

$${}^{IJ}\mathbf{M} = \mathbf{U}^{IJ}\mathbf{\Sigma}\mathbf{V}^{\dagger} \tag{5.2.18}$$

in which $\mathbf{U}$ and $\mathbf{V}$ are right and left singular vectors, and $\mathbf{\Sigma}$ are the singular values. The SVD allows the MOs to be transformed to ${}^{I}\tilde{\mathbf{C}} = {}^{I}\mathbf{C}\mathbf{U}$ and ${}^{J}\tilde{\mathbf{C}} = {}^{J}\mathbf{C}\mathbf{V}$, which is the basis that separates the null space and image of the occupied-occupied (oo) orbital overlap. As a result, the transition density matrices can be computed as

$$\boldsymbol{\rho}_1^{IJ} = \begin{cases} {}^{J}\mathbf{C}^{IJ}\mathbf{M}^{-1I}\mathbf{C} \text{ or } {}^{J}\tilde{\mathbf{C}}^{IJ}\mathbf{\Sigma}^{-1I}\tilde{\mathbf{C}} & \text{for } \dim(\ker({}^{IJ}\mathbf{M})) = 0 \\ \sum_j {}^{J}\tilde{\mathbf{C}}_j{}^{I}\tilde{\mathbf{C}}_j^{\dagger}/\sigma_{jj} & \text{for } \dim(\ker({}^{IJ}\mathbf{M})) = 1 \\ {}^{J}\tilde{\mathbf{C}}_j{}^{I}\tilde{\mathbf{C}}_j^{\dagger} & \text{for } \dim(\ker({}^{IJ}\mathbf{M})) = 2 \\ 0 & \text{for } \dim(\ker({}^{IJ}\mathbf{M})) > 2 \end{cases} \tag{5.2.19}$$

$$\boldsymbol{\rho}_2^{IJ} = \begin{cases} \boldsymbol{\rho}_1^{IJ} & \text{for } \dim(\ker({}^{IJ}\mathbf{M})) = 0 \\ {}^{I}\tilde{\mathbf{C}}_i{}^{J}\tilde{\mathbf{C}}_i^{\dagger} & \text{for } \dim(\ker({}^{IJ}\mathbf{M})) \leq 2 \\ 0 & \text{for } \dim(\ker({}^{IJ}\mathbf{M})) > 2 \end{cases} \tag{5.2.20}$$

$$\boldsymbol{\rho}_3^{IJ} = \begin{cases} \boldsymbol{\rho}_1^{IJ} & \text{for } \dim(\ker({}^{IJ}\mathbf{M})) = 0 \\ {}^{I}\tilde{\mathbf{C}}_i{}^{J}\tilde{\mathbf{C}}_i^{\dagger} & \text{for } \dim(\ker({}^{IJ}\mathbf{M})) = 1 \\ 0 & \text{for } \dim(\ker({}^{IJ}\mathbf{M})) > 1 \end{cases} \tag{5.2.21}$$

The computational cost of building the calculation is a Fock matrix build for each matrix element, which for nonorthogonal configuration interaction singles (NOCIS) scales as $\mathcal{O}(N_{occ}^2 N_{virt}^2 N_{basis}^4)$, although in practice, techniques for accelerating construction of the Fock matrix can be applied to reduce the scaling. As a result, the calculation of the time-independent Hamiltonian is more expensive in the nonorthogonal basis than the orthogonal basis. However, the time-propagation component of the calculation has the same cost, which is matrix-vector multiplications and so scales as $\mathcal{O}(N_{occ}^2 N_{virt}^2)$, regardless of whether the basis is orthogonal or nonorthogonal. The main difference between orthogonal and nonorthogonal methods in the time-propagation step is that D and d are not necessarily unitary, where generally the determinant overlap matrix N gives the identify $D^\dagger N D = 1$ and $d^\dagger N d = 1$. As a result, the transformation matrix between the dipole eigenstate basis and the Hamiltonian eigenstate basis in (5.2.7) is not valid and instead is written as:

$$U_q = d_q^{-1} D \tag{5.2.22}$$

Furthermore, in the nonorthogonal basis, the transformation matrix is also not unitary, and so it is not necessarily true that $U^{-1} = U^\dagger$, and (5.2.6) must be modified to account for non-unitary properties. However, although explicit calculation of the inverse is more computationally expensive than taking the conjugate transpose, these matrices are fixed based on the time-independent component of the calculation, and so only need to be computed once. As a result, provided that the simulation time is sufficiently long, the computational cost is the same regardless of whether an orthogonal or nonorthogonal basis is used.

Dynamic Field Response Calculations

To determine the field response of the molecule, the time-evolution of the dipole can be used to extract the polarizability and hyperpolarizabilities. The time-dependent dipole moment can be expressed as(140; 6)

$$\begin{aligned}
\mu_i(t) =& \mu_i^{(0)} + \mu_{ij}^{(1)}(t) + \mu_{ijk}^{(2)}(t) + \mu_{ijkl}^{(3)}(t) + \dots \\
=& \mu_i^0 + \sum_j \int_{-\infty}^{\infty} dt_1 \alpha_{ij}(t - t_1) F_j(t_1) \\
&+ \frac{1}{2!} \sum_{jk} \int \int_{-\infty}^{\infty} dt_1 dt_2 \beta_{ijk}(t - t_1, t - t_2) F_j(t_1) F_k(t_2) \\
&+ \frac{1}{3!} \sum_{jkl} \int \int \int_{-\infty}^{\infty} dt_1 dt_2 dt_3 \gamma_{ijkl}(t - t_1, t - t_2, t - t_3) F_j(t_1) F_k(t_2) F_l(t_3) + \dots .
\end{aligned} \tag{5.2.23}$$

where F is the applied field, and α, β and γ are the parameters that define the polarizability, first hyperpolarizability and second hyperpolarizability respectively.

The time-dependent electric field in terms of Fourier components

$$F_i(t) = F_i^{(0)} + \frac{1}{2}\sum_I F_i(\omega_I)\left(e^{-i\omega_I t} + c.c.\right) \tag{5.2.24}$$

where ω_I is the frequency of each Fourier component and $F_i^{(0)}$ is the static field. For a monochromatic field, there is a single term in the sum of eq. 5.2.24, which leads to the following terms for the field

$$F_j(t) = \frac{1}{2}F_j(\omega)\left(e^{-i\omega t} + c.c.\right) \tag{5.2.25}$$

$$F_j(t)F_k(t) = \frac{1}{4}F_j(\omega)F_k(\omega)\left(e^{-i2\omega t} + 2e^0 + c.c.\right) \tag{5.2.26}$$

$$F_j(t)F_k(t)F_l(t) = \frac{1}{8}F_j(\omega)F_k(\omega)F_l(\omega)\left(e^{-i3\omega t} + 3e^{-i\omega t} + c.c.\right) \tag{5.2.27}$$

Substituting the expressions for the frequency domain field into eq. 5.2.23 and using Euler's formula gives the following expressions for the time-dependent dipole moment at each order

$$\mu_{ij}^{(1)}(t) = \alpha_{ij}(-\omega;\omega)\cos(\omega t)F_j(\omega) \tag{5.2.28}$$

$$\mu_{ijk}^{(2)}(t) = \frac{1}{4}\left[\beta_{ijk}(-2\omega;\omega,\omega)\cos(2\omega t) + \beta_{ijk}(0;\omega,-\omega)\right]F_j(\omega)F_k(\omega) \tag{5.2.29}$$

$$\mu_{ijkl}^{(3)}(t) = \tfrac{1}{24}\left[\gamma_{ijkl}(-3\omega;\omega,\omega,\omega)\cos(3\omega t) + 3\gamma_{ijkl}(-\omega;\omega,\omega,-\omega)\cos(\omega t)\right]F_j(\omega)F_k(\omega)F_l(\omega) \tag{5.2.30}$$

where the response parameters α, β and γ are labelled by the total exponent in the frequency domain field, as well as the component frequencies that were summed together in the expansion. The different induced dipole moments can be connected with the various processes to which they contribute. In particular, eq. 5.2.28 is responsible for the linear refractive index, the first term in eq. 5.2.29 leads to second harmonic generation, the second term in eq. 5.2.29 is optical rectification, the first term in eq. 5.2.30 give rise to third harmonic generation, while the second term in eq. 5.2.30 is degenerate four-wave mixing. For a given field size, the values of $\alpha_{ij}(-\omega;\omega)$, $\beta_{ijk}(-2\omega;\omega,\omega)$ and $\gamma_{ijkl}(-3\omega;\omega,\omega,\omega)$ are directly related to the peaks in the high harmonic generation (HHG) spectrum at the first, second and third harmonic orders, and so HHG spectra can provide further insight into the extent to which different wavefunction theories correctly describe polarizability and hyperpolarizability.(141)

To obtain the response parameters α, β and γ from real-time simulations, the difference in the dipole evolution with respect to changes in amplitude can be used(6)

$$\mu_{ij}^{1} = \frac{8[\mu_i(t,A_j) - \mu_i(t,-A_j)] - [\mu_i(t,2A_j) - \mu_i(t,-2A_j)]}{12A_j} \tag{5.2.31}$$

$$\mu^2_{ijj} = \frac{16[\mu_i(t, A_j) + \mu_i(t, -A_j)] - [\mu_i(t, 2A_j) + \mu_i(t, -2A_j)] - 30\mu^0_i}{24A^2_j} \quad (5.2.32)$$

$$\mu^3_{ijjj} = \frac{-13[\mu_i(t,A_j)+\mu_i(t,-A_j)]+8[\mu_i(t,2A_j)-\mu_i(t,-2A_j)]-[\mu_i(t,3A_j)-\mu_i(t,-3A_j)]}{48A^3_j} \quad (5.2.33)$$

By equating eqs. 5.2.31-5.2.33 with eqs. 5.2.28-5.2.30, the only unknown terms are the response parameters, which can be obtained by fitting the curves described by eqs. 5.2.28-5.2.30 to the simulation data.

Computational Details

Both orthogonal and nonorthogonal RT-TDCI method were implemented in a stand-alone code using the MQCPack library(61) interfaced with Gaussian 16.(62) The nonorthogonal determinant basis was obtained using a modified version of Gaussian 16.(62)

In order to obtain the relaxed singly-excited determinants, substitutions were performed from the ground state restricted reference orbitals, followed by orbital optimization using a modified initial maximum overlap method,(13; 14; 142) in which the occupied orbitals in each SCF iteration are the set with the largest value P

$$P_p = \max_{i'} \left(\text{abs} \left(\sum_{\mu\nu} C_{\mu i'} S_{\mu\nu} C_{\nu p} \right) \right) \quad (5.2.34)$$

where the index i' labels the initial orbitals occupied in the substituted determinant, $C_{\mu p}$ is an element of the MO coefficients, and $S_{\mu\nu}$ is the atomic orbital overlap. As only the set of occupied orbitals selected is modified in the optimization, a standard direct inversion of the iterative subspace (DIIS) optimization can be used and there is no increase computational cost per iteration compared to a ground state SCF optimization. However, the number of SCF optimizations required scales as $\mathcal{O}(N_{\text{occ}}N_{\text{virt}})$. As the orthogonal substituted determinant provides a good initial guess, especially for single substitutions, convergence issues in reoptimizing the orbitals are generally avoided, and were not observed in the current study.

The pulse shapes used in this study along with the name used in the rest of the text are shown in table 5.1. The parameters that govern the different pulse shapes are the frequency ω, and the pulse width σ. Two different pulse shapes were used. For the field response calculations the continuous pulse was used with a 5 optical cycle ramp, with frequency ω=0.072 a.u., amplitude 0.001 a.u. and duration 21 fs aligned along the internuclear axis for the dipole response of H_2 (fig. 5.1a),(6) and frequency ω=0.056 a.u., amplitude 0.089 a.u., and duration 21 fs aligned in the plane along the axis connecting the CH_2 groups in butadiene (fig. 5.1b). Additionally, H_2 hyperpolarizabilities were computed using field amplitude 0.1 a.u. due to numerical noise that occurs when small amplitudes are used with larger basis sets. Results were compared with response theory results computed using Dalton.(143; 144) For the HHG spectrum calculations, the cosine squared pulse was used with frequency ω=0.1102 a.u., amplitude 0.007 a.u., and duration 60 fs for H_2 (fig. 5.1c),(141) frequency ω=0.1102 a.u., amplitude 0.007 a.u., and duration 52.8 fs for butadiene, and frequency ω=0.03675 a.u., amplitude 0.05 a.u., and duration 52.8 fs for water. The orientation of the field was the same as for the field response calculations for H_2 and butadiene, and for water it was along the C_2 axis. Further analysis of the different response orders are determined by computing harmonic generation spectra. Fourier transformation of the time-dependent induced dipole moment is performed giving $\mu_z(\omega) = \int \mu_z(t)e^{-i\omega t}dt$. The spectrum intensity $I(\omega)$ is then computed as $I(\omega) = \text{abs}\,(\mu_z(\omega))$.

Table 5.1. Names and equations of pulse shapes used in this work.

Name	Equation	
Continuous	$\text{F(t)}=\text{F}_{max}\sin(\omega(t-t_0))$	(1.1)
Cosine Squared	$\text{F(t)}=\text{F}_{max}\cos^2(\frac{\pi}{2\sigma}(t-t_0))\sin(\omega(t-t_0))$	(1.2)

F_{max} – maximum field vector; ω – frequency; σ – pulse width

5.3 Results and Discussion

In this section, we examine the orthogonal RT-TDCI hierarchy and the position in which RT-TDNOCIS can be placed within the hierarchy. To establish the performance of the model, the polarizability and hyperpolarizabilities are modeled using both the fitting of the time-dependent dipole moment response and calculation of the HHG spectra. Molecular geometries for the molecules studied are shown in figs. B.1-B.3.

Dipole Response Fitting

H_2

The polarizability obtained from applying the continuous pulse to H_2 with various basis sets for CIS, configuration interaction singles and doubles (CISD) and NOCIS is shown in table 5.2, while the fit quality (determined as the coefficient of determination R^2 between the simulated data and the analytical function in eq. 5.2.28) is shown in table B.14. Using 0.001 a.u. amplitude field, the polarizability mean fit (mean of all R^2 values for different methods and basis sets) was 0.999 with 0.000 standard deviation (standard deviation of all R^2 values for different methods and basis sets), while for second-order hyperpolarizabilites, the mean fit is 0.358 with standard deviation 0.397, figures that show the simulation data and fitted dipole response function can be found in the published paper. While the 0.001 a.u. amplitude field gave good fit statistics for polarizability with all basis sets, and hyperpolarizability with 6-31G, numerical noise when calculating $\mu^{(3)}_{ijkl}$ led to poor fit statistics, especially for triple zeta basis sets and basis sets including diffuse functions. Therefore, results are also presented using a 0.1 a.u. field, which was found to give lower numerical noise with larger basis sets, although at the cost of introducing intrinsic noise (small high frequency oscillations due to excessive population of excited states) and possibly numerical truncation error in the dipole evolution which limits the fit with the large basis sets. We emphasize that this numerical noise is present when both orthogonal and nonorthogonal determinant expansions are used and so is not a consequence of the nonorthogonal basis. Several different field frequencies were tested to ensure that any noise in the simulated data did not result from excitation energies in close proximity to the field frequency. The 0.1 a.u. amplitude field resulted in a polarizability mean fit of 0.897, with standard deviation 0.169.

As H_2 has two electrons, CISD is equivalent to the exact result for a given basis set. The real-time propagation results are consistent between simulations with 0.001 a.u. and 0.1 a.u. amplitude fields, except when diffuse functions are included in the basis set. Because of the better fitting statistics with 0.001 a.u. field, we consider the results with 0.001 a.u. amplitude field to be the most reliable across all basis sets. Comparing the real-time propagation results (table 5.2, top) with the results from response theory, CISD provides identical results between the two approaches, while CIS gives higher polarizability across all basis sets by 0.6-0.8 a.u. when using real-time propagation than when using response theory. Relative to CISD, CIS overestimates the polarizability using both real-time propagation and response theory, although by a smaller amount when using response theory. NOCIS also overestimated the polarizability with respect to CISD, and in fact gave higher polarizability than CIS, which is the opposite of the expected behavior, as nonorthogonality should implicitly include higher order substitutions leading to closer agreement with CISD. Comparing the results with the best estimate for the polarizability, which is the experimental α_{zz}(-ω;ω) (ω=0.072 a.u.) obtained from refs. 145 and 146, reveals that with response theory gives better results than CISD, although this result reflects the fact that the basis set is still not complete. However, based on results from real-time simulations, the best estimate lies between CIS and CISD. While the basis set is found to have a small effect on polarizability, the polarizability increases with basis set size, which indicates that CISD will give the closest result to the best estimate as the complete basis set limit is reached.

Table 5.2. Frequency-dependent polarizability $\alpha_{zz}(\omega;\omega)$ (ω=0.072 a.u.) of H_2 in a.u. computed using the dipole real-time evolution to a continuous field with frequency 0.072 a.u., duration 21 fs, amplitude 0.001 a.u. (top) and 0.1 a.u. (middle) and response theory (bottom).

Basis	CIS	CISD	NOCIS
Real-Time Propagation			
A=0.001 a.u.			
6-31G	7.6	6.2	7.8
cc-pVDZ	7.5	6.4	7.9
cc-pVTZ	7.7	6.7	8.3
aug-cc-pVDZ	7.6	6.7	8.3
aug-cc-pVTZ	7.5	6.5	8.2
A=0.1 a.u.			
6-31G	7.4	6.2	7.6
cc-pVDZ	7.4	6.5	7.8
cc-pVTZ	7.7	6.7	8.3
aug-cc-pVDZ	11.8	13.4	12.8
aug-cc-pVTZ	12.7	12.3	9.9
Response Theory			
6-31G	6.8	6.2	
cc-pVDZ	6.8	6.4	
cc-pVTZ	7.0	6.7	
aug-cc-pVDZ	7.0	6.7	
aug-cc-pVTZ	6.9	6.5	
Best Estimate(145; 146)			6.9

To understand the observed trends in polarizability the excitation energies and transition moments from the ground state of each method can be analyzed (tables B4-B13). Generally, CIS has lower excitation energies than CISD due to the fact that double substitutions are the leading correlation term in the ground state, so the ground state is stabilized more than the single-excited states by the inclusion of double substituted determinants (although not always, especially for larger basis sets). For H_2, CISD is equivalent to full CI and so the excitation energies are exact, but for more than two-electron systems, truncation at double substituted determinants is likely to overestimate excitation energies and so underestimate the polarizability. As CIS has lower excitation energies, a given pulse size populates excited states to a greater extent and perturbs the system further from the equilibrium, leading to a larger change in dipole moment. Despite the fact that NOCIS correlates the ground state, which is uncorrelated in CIS, it gives even lower excitation energies than CIS. This observation results from the fact that NOCIS is effectively carrying out symmetry breaking and restoration on the excited states where strong correlation is significantly more important than the ground state, and so the energy of the excited states are decreased by a greater extent than the ground state, leading to lower excitation energies and an increase in the polarizability. As a result, the increase in polarization in NOCIS results from an imbalance of dynamic and static correlation, and incorporation of dynamic correlation in NOCIS wavefunction would be expected to yield results closer to CISD. The increase in polarizability with basis set size can be explained by low-lying Rydberg states appear, which increase the number of states that are populated by a given laser pulse size. In terms of transition dipole moments, CIS generally has larger values than CISD, which leads to greater population of excited states and so larger polarizability. The transition dipole moments of NOCIS are slightly smaller than CIS, although remain larger than CISD. As a result, the larger polarizability from NOCIS calculations is a consequence of the smaller excitation energies, while transition dipole moments serve to reduce the polarizability.

Results for second-order hyperpolarizability are shown in Table 5.3, with the fit quality shown in table B.15. Additionally, figs. S31-S60 show the fitted response function and the simulation data inside the corresponding published version. As a result of a center of inversion, there is no first-order hyperpolarizability. For the second-order hyperpolarizability, with 0.001 a.u. amplitude field the mean fit is 0.358 with standard deviation 0.397, while with 0.1 a.u. amplitude field the mean fit is 0.715 with standard deviation 0.260. Regardless of the field amplitude, the fit decreases with basis set size, but is significantly better with 0.1 a.u. field at larger basis sets, and so the results using the 0.1 a.u. field are considered most reliable for discussing the second-order hyperpolarizability. However, despite the improved fit, inspection of figs. S54 and S59 shows that even with the larger amplitude field, the results for aug-cc-pVTZ are still not sufficiently reliable to draw conclusions.

Table 5.3. Frequency-dependent second-order hyperpolarizability $\gamma_{zzzz}(3\omega;\omega,\omega,\omega)$ (ω=0.072 a.u.) of H_2 in a.u. computed using the dipole real-time evolution to a continuous field with frequency 0.072 a.u., duration 21 fs, amplitude 0.001 a.u. (top) and 0.1 a.u. (middle) and response theory (bottom)

Basis	CIS	CISD	NOCIS
Real-Time Propagation			
A=0.001 a.u.			
6-31G	543.1	63.9	657.3
cc-pVDZ	346.8	26.1	533.0
cc-pVTZ	80.4	234.6	244.5
aug-cc-pVDZ	951.2	1183.1	1189.5
aug-cc-pVTZ	1027.2	850.4	1189.7
A=0.1 a.u.			
6-31G	326.1	73.0	345.0
cc-pVDZ	267.1	40.4	339.1
cc-pVTZ	139.7	169.9	264.0
aug-cc-pVDZ	754.7	723.9	1229.6
aug-cc-pVTZ	1763.3	3392.3	4263.3
Response Theory			
6-31G	173.0	63.2	
cc-pVDZ	96.1	25.9	
cc-pVTZ	126.5	187.5	
aug-cc-pVDZ	1099.6	970.5	
aug-cc-pVTZ	1155.3	1012.6	
Best Estimate(147; 148)			1056.4

The real-time propagation results show similar trends between CIS, CISD and NOCIS regardless of the field amplitude, although values can duffer by several hundred a.u. The results of real-time propagation also show a similar trend to results from response theory, and shows quantitative agreement for small basis sets with CISD when the 0.001 a.u. field is used (even though the CCSD/cc-pVDZ simulation only has 0.042 fit at 0.001 a.u.). Generally, the second order hyperpolarizability shows a similar trend to the polarizability, in which NOCIS and CIS generally have higher values than CISD, especially at small basis sets, which is observed in both real-time propagation and response theory results. NOCIS gives similar, although slightly higher values than CIS. As orthogonal methods appear to underestimate the best estimate value, NOCIS gives results in better agreement with the best estimate value, although there is no clear evidence that NOCIS improves on CIS and gives results in closer agreement with CISD. The best estimate for H_2 second-order hyperpolarizability is from $\gamma_{zzzz}(-3\omega;\omega,\omega,\omega)$ ($\omega = 0.072$ a.u.) computed using an explicitly correlated wavefunction optimized for second-order hyperpolarizabilities. (149) Regardless of method, diffuse functions are required to give results in qualitative agreement with the best estimate, and CISD/aug-cc-pVTZ using response theory gives a reasonably close quantitative agreement with the best estimate. Unfortunately, the numerical and intrinsic noise when using diffuse functions in real-time propagation simulations makes it difficult to conclude whether NOCIS improves on CIS. It is possible that, due to the effective basis set effect if NOCIS,(130) in which DODC means that there is greater variational flexibility in determining the form of the overall natural orbitals, the larger values in closer agreement with the best estimate are a reflection of the use of DODC, which would explain why NOCIS is different from CIS but not closer to CISD. A few results provide indications that NOCIS is capturing more correlation through implicit inclusion of doubles, such as results from cc-pVTZ and aug-cc-pVDZ using 0.001 a.u. amplitude field, which give close agreement between NOCIS and CISD, but the fit quality is relatively poor so it is not so clear if these results are fortuitous.

Butadiene

To test NOCIS on a larger system with lower symmetry, the polarizability and second-order hyperpolarizability of trans-butadiene is explored. Trans-butadiene has been a widely used test molecule for establishing the performance of different methods in calculating hyperpolarizabilities. (150; 151; 152) In addition, the model uses a four-electron in four-orbital active space to assess how nonorthogonality not only captures higher-order substitutions within the strongly-correlated orbital subspace, but also can include substitutions to orbitals outside of the active space. Applying the continuous pulse with amplitude 0.089 a.u. and frequency 0.056 a.u. with different basis sets provided the polarizabilities and second-order hyperpolarizabilities shown in Table 5.4 and 5.5, respectively. As for H_2, the first-order polarizability is zero due to the inversion symmetry in the molecule. Since the active space contains the four π orbitals up to quadruple substitutions can be obtained in the orthogonal CI, and so the convergence with substitution level can be compared to NOCIS. An additional feature of butadiene is that due to strong correlation in the ground state, the lowest energy SCF solution has real unrestricted (RU) spin symmetry. As a result, either the unstable real restricted (RR) determinant can be used for the ground state, or the pair of RU half-projected determinants can be used to restore symmetry. While nonorthogonality allows substituted determinants to correlate the ground state, symmetry breaking of the reference determinant itself provides another avenue for improving on orthogonal CIS and should provide a better balance of correlation between ground and excited states. The use of the RR is termed NOCIS, while the use if the half-projection RU determinants us termed half-projected nonorthogonal configuration interaction singles (HPNOCIS). The fit quality for the polarizability was mean 0.959 with standard deviation 0.053 (table B.16), and mean 0.756 with standard deviation 0.224 for the second-order hyperpolarizabilities (table B.18). Plots of the simulation data and fitted dipole response are shown in figs. S61-S144 in the published version. A field amplitude of 0.089 a.u. was chosen as it provided for NOCIS and HPNOCIS and has been previously used on real-time time-dependent studies.(112) Unfortunately, it was not possible to obtain results with diffuse functions owing to numerical and intrinsic noise that prevented reasonable quality fitted data.

Examining the polarizabilities (table 5.4) shows that the real-time propagation methods give smaller polarizability than the response theory methods. No best estimate is given because of sensitivity to the exact molecular geometry and field frequency, but based on studies of how polarizability changes with geometry(153) and field frequency,(154) we estimate $\alpha_{zz}(-\omega;\omega)$ (ω=0.056 a.u.) to be 75-90 a.u. The discrepancy between values reported in table 5.4 is most likely due to the lack of diffuse functions in the simulations. Orthogonal CI response theory results indicate that polarizability increases with basis set size, and decreases with CI truncation order. CISD shows a significant change in polarizability compared to CIS, while configuration interaction singles, doubles and triples (CISDT) adjusts the CISD polarizability values by around 1 a.u. and are the same as the configuration interaction singles, doubles, triples and quadruples (CISDTQ) values. The only outlier in the orthogonal CI convergence is with the 6-311G(d) basis set, in which adding quadruple substitutions changes the polarizability from 65.2 a.u. to 79.3 a.u., although it is not clear why as the frequency (0.056 a.u.) is not close to resonant with any excitation that could lead to divergence in the response equations. Orthogonal CI real-time propagation results systematically underestimate the response theory results and are further away from the best estimate range. Except for STO-3G and 6-31G(d) basis sets, the polarizability had converged by the inclusion of double substitutions, and even the inclusion of double substitutions only changed the polarizability by around 1 a.u. compared to CIS. As a result, unlike in response theory where CIS leads to overpolarization compared to larger determinant expansions, real-time propagation approaches do not show the expected decrease in polarizability upon inclusion of double substitutions. Performing CIS calculations using the full orbital space, rather than just the four-electrons in four-orbital active space (column labelled CIS (full) in table 5.4) shows that use of an active space leads to underpolarization. Both NOCIS and HPNOCIS show a larger polarizability than the orthogonal CI methodologies, indicating that orbitals outside of the active space are included to an extent in these methods. HPNOCIS gives lower polarizability than NOCIS, which is consistent with the hypothesis that greater correlation of the ground state leads to increased excitation energies, which in turn decrease the polarizability, even though the same effect was not evident in the orthogonal CI expansion. Overall, although it is difficult to judge if the nonorthogonal methods perform better than orthogonal methods, it is clear that the nonorthogonal results are different from those using orthogonal methods.

Table 5.4. Frequency-dependent polarizability $\alpha_{zz}(\omega;\omega)$ (ω=0.056 a.u.) of butadiene in a.u. computed using the dipole real-time evolution to a continuous field with frequency 0.056 a.u., duration 21 fs and amplitude 0.089 a.u. (top) and response theory (bottom)

Basis	CIS	CISD	CISDT	CISDTQ	NOCIS	HPNOCIS
Real-Time Propagation						
STO-3G	41.4	37.9	35.9	32.9	42.2	35.8
6-31G(d)	59.1	59.0	60.6	55.4	51.5	46.2
6-311G	44.9	45.8	45.8	45.8	54.8	46.4
6-311G(d)	43.8	44.7	44.7	44.7	51.4	48.7
SVP	43.7	44.5	44.5	44.5	52.6	48.5
cc-pVDZ	43.5	44.3	44.2	44.2	51.7	48.6
Response Theory						
STO-3G	51.0	35.0	34.1	34.1		
6-31G(d)	79.5	62.9	62.0	62.0		
6-311G	83.5	63.6	62.5	62.5		
6-311G(d)	82.8	66.1	65.2	79.3		
SVP	81.7	64.5	63.5	63.5		
cc-pVDZ	82.4	66.2	65.3	65.3		

The second-order hyperpolarizabilities displayed in Table 5.5 show that, as for polarizabilities, inclusion of double substitutions gives close to the converged CISDTQ values for both response theory and real-time propagation. However, polarizabilities calculated with real-time propagation are much smaller than those computed with response theory, except using STO-3G. Regardless of whether response theory or real-time propagation is used, CIS results generally underestimate hyperpolarizabilities with respect to higher order truncated CI, as do the nonorthogonal methods. However, while NOCIS results do not significantly differ from CIS results (polarizabilities are smaller by 0.1-0.8 a.u.). NOCIS gives hyperpolarizabilities that are larger and closer to the CISDTQ results. However, as for H_2, the fitting for the second-order hyperpolarizabilities is challenging due to the requirement to balance a convergent field size with avoiding numerical noise, and so it is not obvious the extent to which the fitted dipole results are reliable for indicating the performance of nonorthogonal methods in capturing higher order substituted determinants. Table 5.6 shows the values for second order hyperpolarizabilities computed using NOCIS and HPNOCIS as a function of field amplitude, to show the sensitivity of the fitting protocol to simulation parameters. It can be seen that a small reduction in field amplitude leads to results that are much closer to response theory, for example at 0.01 a.u. field amplitude, while larger fields may result in distortion of the wavefunction causing a decrease in the hyperpolarizability. However, as shown in table B.18, the fit quality with lower field amplitudes can deteriorate, so it is difficult to determine if the lower second-order hyperpolarizability values are because of numerical noise or related to the method of calculation.

Table 5.5. Frequency-dependent second-order hyperpolarizability $\gamma_{zzzz}(3\omega; \omega, \omega, \omega)$ (ω=0.056 a.u.) of butadiene in 10^3 a.u. computed using the dipole real-time evolution to a continuous field with frequency 0.056 a.u., duration 21 fs and amplitude 0.089 a.u. (top) and response theory (bottom)

Basis	CIS	CISD	CISDT	CISDTQ	NOCIS	HPNOCIS
Real-Time Propagation						
STO-3G	5.7	5.8	6.6	7.0	5.6.	7.4
6-31G(d)	3.0	23.4	20.9	24.8	2.8	6.6
6-311G	3.7	8.0	8.0	8.0	2.1	4.4
6-311G(d)	2.3	7.6	7.6	7.6	1.7	5.0
SVP	2.0	7.6	7.6	7.6	0.9	4.4
cc-pVDZ	2.1	7.2	7.2	7.2	0.3.	4.9
Response Theory						
STO-3G	4.8	6.9	7.0	7.0		
6-31G(d)	13.8	21.3	21.8	21.7		
6-311G	18.9	27.5	27.8	27.9		
6-311G(d)	18.1	25.6	26.0	31.0		
SVP	17.1	25.0	25.5	25.5		
cc-pVDZ	16.6	24.0	24.5	24.5		

Table 5.6. Frequency-dependent second-order hyperpolarizability $\gamma_{zzzz}(3\omega; \omega, \omega, \omega)$ (ω=0.056 a.u.) of butadiene in 10^3 a.u. computed using the dipole real-time evolution of nonorthogonal methods with the 6-31G(d) basis set to a continuous field with frequency 0.056 a.u. and duration 21 fs with various field amplitudes (a.u.)

Field	NOCIS	HPNOCIS
0.001	18.4	12.9
0.005	21.9	13.0
0.010	20.1	13.4
0.050	3.0	12.2
0.089	2.8	6.6
0.100	3.6	5.5
0.300	1.3	0.1
0.500	1.8	0.0

High Harmonic Generation Spectra

Rather than relying on the difference of the dipole response with respect to changes in field strength, which can be similar in magnitude to numerical noise, calculation of the HHG spectrum provides an assessment of different orders of field response based on the dipole response itself, and so is less susceptible to numerical noise. In HHG, which is used to generate ultrashort attosecond laser pulses, strong laser fields induce ionization through tunneling followed by acceleration of the electron away from the molecule. The subsequent change in orientation of the oscillating field causes the ionized electron to accelerate back to the molecule and emit a photon as it decays back to the ground state. The peaks at the various harmonic orders are related to the polarizability and different order hyperpolarizabilities, such that HHG spectra provide a sensitive test of the performance of a given model for correctly reproducing the field-response of the electronic structure. As a result, HHG spectra are a good test of the ability of NOCIS to capture the effect of double and higher order substitutions. In this section, the HHG spectrum of H2 and butadiene are examined to compare with the results in the previous section, while water is also studied to provide an example of a molecule without a center of inversion that has non-zero even-order field response.

H_2

The HHG spectrum of H_2 was computed using the cosine squared pulse (eq. 1.2 in Table 5.1). Each method (CIS, CISD, NOCIS) was tested using five different basis sets (fig. 5.2). Due to the inversion symmetry, H_2 only shows peaks at odd number of harmonic order. Examining the first harmonic order, which corresponds to the polarizabilty, it can be seen that for all basis sets, the NOCIS HHG spectrum (black) has a larger intensity than CIS (red), which in turn has a larger intensity than CISD (green), in agreement with the findings from the fitted polarizability from the dipole response. Therefore, the HHG spectrum provides further corroboration that, in contrast to the expected result, NOCIS increases the polarizability with respect to CIS. For the third harmonic order, which corresponds to the second-order hyperpolarizability, small basis sets (6-31G and cc-pVDZ) show that NOCIS and CIS are in close in value and larger than CISD, while using cc-pVTZ, NOCIS gives closer agreement to CISD than CIS. The larger cc-pVTZ basis set gives greater flexibility for orbital relaxation than smaller basis sets, so NOCIS is better able to capture double substitutions and so more closely reproduces CISD. When diffuse functions are used, the CIS and CISD results become indistinguishable, while NOCIS shows a very slightly more intense peak, as the diffuse functions dominate the description of higher-order hyperpolarizabilities. Finally, the HHG spectrum is able to provide insight into fourth-order hyperpolarizabilities, which are described by the fifth harmonic and are only present when diffuse functions are included. As for the third harmonic with diffuse functions, the CIS and CISD intensities are essentially equal, while NOCIS gives a slightly larger value. One explanation for the larger high-order dipole response is the DODC framework provides a larger effective one-electron basis set for NOCIS than the orthogonal methods, and so generally results in a larger field response because of the greater flexibility in the basis. Finally, the peaks between the third and fifth harmonics that are not integer multiples of the field frequency correspond to the frequency of transition to excited electronic states.

Butadiene

The HHG spectrum of butadiene computed using the four-electron in four-orbital active space with CIS, CISD, CISDT, CISDTQ, as well as NOCIS and HPNOCIS using different basis sets are shown in fig. 5.3. As for H_2, the inversion symmetry of butadiene leads to intensity at only odd-order harmonics. Examining the first harmonic order shows that CIS has the largest polarizability at all basis sets. Additionally, it can be seen that triple and quadruple substituted determinants do not lead to any significant change in the intensity or the first-harmonic peak. At the STO-3G level of theory, NOCIS and HPNOCIS give lower intensity first-harmonic peaks than CISDTQ, implying smaller polarizability, although the difference is small. However, for the remaining, larger basis sets, the remaining models have very similar polarizability, while with other basis sets NOCIS leads to a slight reduction in intensity with respect to CIS, which remains larger than the CISD, CISDT and CISDTQ peaks, while HPNOCIS leads to a much greater reduction in the intensity and indicates a smaller polarizability than calculated by CISD, CISDT and CISDTQ. For the second- order hyperpolarizabilities at the third harmonic, while STO-3G and 6-31G(d) do not give clear trends, the larger basis sets all show a difference between nonorthogonal and orthog- onal CI, but very little difference within each type of method. As a result, it is concluded that configurations involving substitutions outside of the active space lead to the different second-order hyperpolarizabilities. Finally, for the fifth harmonic, only at small basis sets (STO-3G and 6-31G(d)) is there any noticeable difference between CISD, CISDT and CIS- DTQ. With larger basis sets, CIS generally shows lower intensity than the other orthogonal CI methods, HPNOCIS gives a similar intensity to CISDTQ, and NOCIS shows higher intensity. Unfortunately, the conclusions regarding the relative magnitudes of polarizability and hyperpolarizability using simulated HHG spectra and fitted dipole differences disagree. Regardless, both the HHG spectra and the fitted dipole-evolution data demonstrate that the use of nonorthogonality differs from orthogonal expansions, and so must be implicitly capturing the effect of substitutions to orbitals external to the active space.

Water

To analyze a molecule with even-order harmonics, HHG spectrum of water with C_{2v} symmetry is shown in fig. 5.4, where different basis sets are compared for CIS in panel a, and NOCIS in panel b. The general trend when using CIS is that the largest basis set (aug-cc-pVDZ) with diffuse functions leads to greater intensity at every harmonic order, while the smallest basis set (STO-3G) has lower intensity at every harmonic order. The remaining split-valence basis sets, regardless of whether they are double or triple zeta, or include polarization functions, lead to similar intensity. In NOCIS, for odd harmonic orders, the inclusion of the diffuse function still leads to greater intensity, but there is less difference with other basis sets at even harmonic orders. Additionally, STO-3G gives higher intensity peaks with NOCIS compared to CIS, which are in closer agreement with other basis sets. The conclusion is that, while diffuse functions are still required to properly model HHG spectra, use of a nonorthogonal basis can enable improved performance of small basis sets.

5.4 Conclusions

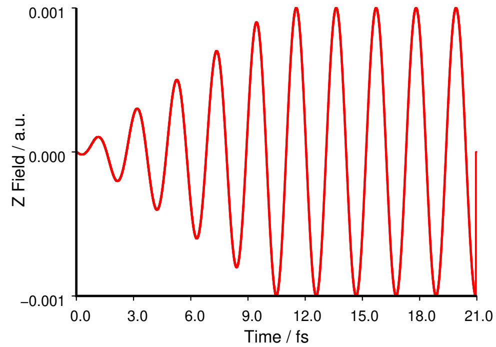
In this paper, we developed the theory of nonorthogonal determinant expansions for use in real-time time-dependent electronic structure simulations. The use of nonorthogonality was hypothesized to improve the description of the electronic field response, on the basis that it implicitly includes higher order orbital substitutions. The hypothesis was tested by comparing a nonorthogonal CIS model, in which orbitals of single substituted reference determinant are allowed to relax in response to the change in electron configuration, to the hierarchy of orthogonal CI methods. As a result, the nonorthogonal CIS approach acts as an effective Hamiltonian, in which higher order determinant substitutions are wrapped into each matrix element to allow correlation with the reference determinant.

The hypothesis was tested using two approaches. In the first approach, the polarizability of H2 and butadiene were determined by fitting the dipole difference with respect to simulations with different field amplitudes to different order dipole response functions. In the second approach, the HHG spectrum of H2, butadiene and water was computed and the peak intensities compared as a measure of different order field response parameters. For H2 polarizations it was found that NOCIS led to larger polarizability than CIS, which contra- dicted the hypothesis that nonorthogonality would lead to reduced polarizability. Analysis of excitation energies suggested that NOCIS increased the polarizability because the excita- tion energies were lower than CIS due to an imbalance in the dynamic correlation recovered, against static correlation recovered when correlating the ground state. Evaluation of the second-order hyperpolarizability in H2 revealed that NOCIS provides different results to CIS, but the technical challenges of fitting with basis sets containing diffuse functions meant it was difficult to assess if nonothogonality improves the performance of CIS. Butadiene has an unrestricted lowest-energy SCF solution, allowing the HP-NOCIS model to be investigated. Both nonorthogonal models resulted in higher polarizability than orthogonal methods, but in this case shifted the result closer to the best estimate range. Additionally, higher-order substituted orthogonal CI expansions led to greater polarizability than predicted with CIS, which illustrates that use of CIS does not necessarily lead to overpolarization, and that the problem is system specific. For hyperpolarizabilites, orthogonal CIS and nonorthogonal underestimated the value with respect to higher-order substituted CI and response theory,although use of half-projection of the reference gave values closer to CISDTQ.

Overall, the results in this work indicate that nonorthogonality is able to capture higher order substitutions, and so leads to different results than orthogonal CIS. However, the results are not necessarily closer to higher-order truncated orthogonal

CI because NOCIS tend to give a good description of excited states, while missing dynamic correlation in the ground state. Technical challenges in fitting the hyperpolarizabilities, especially with larger basis sets may require improvements in the numerical accuracy of the time propagation, for example through the use of Trotter factorization instead of the split-operator approach.

[t]0.3



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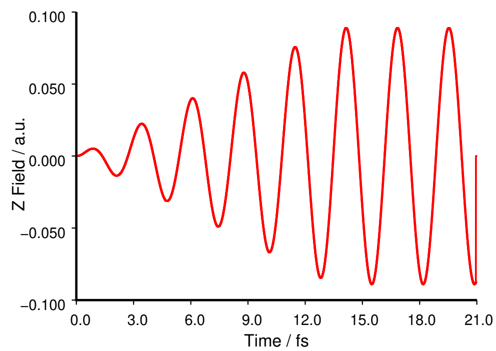

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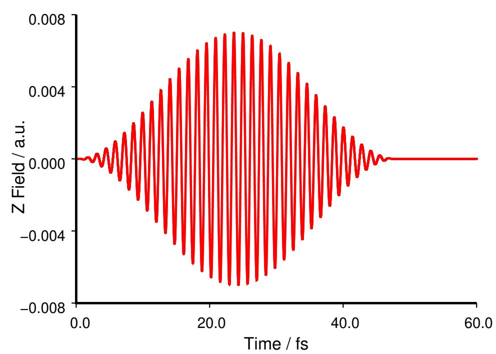

Figure 5.1. Pulse shapes used in this study. Panel a shows the continuous pulse with 0.001 a.u. amplitude, 0.072 a.u. frequency, and 21 fs duration used to determine the field response of H_2. The same field with 0.1 a.u. amplitude was also used. Panel b) shows the continuous pulse with amplitude 0.089 a.u., frequency 0.056 a.u., and duration 21 fs used for the dipole response of butadiene. Panel c shows the cosine squared pulse with amplitude 0.007 a.u., frequency 0.1102 a.u. and duration 60 fs used to compute the high harmonic generation spectrum of H_2.

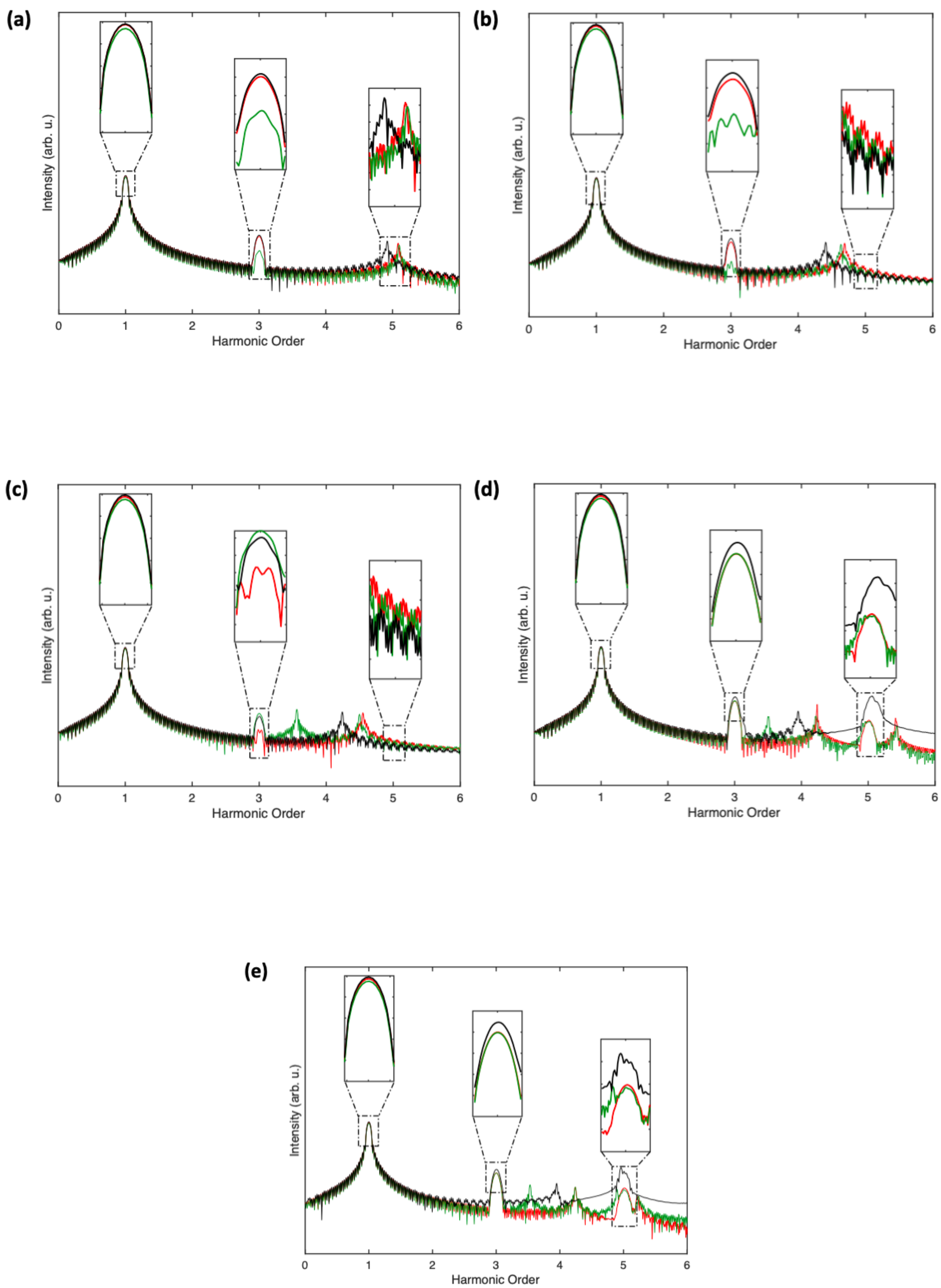

Figure 5.2. Cosine squared pulse (0.007 (a.u),ω=0.1102 (a.u.)) in z-direction within 60 fs are shown in 6-31G, cc-pVDZ, cc-pVTZ, aug-cc-pVDZ and aug-cc-pVTZ basis sets. In each basis set, it contains CIS(red), CISD(dark-green),NOCIS(black).

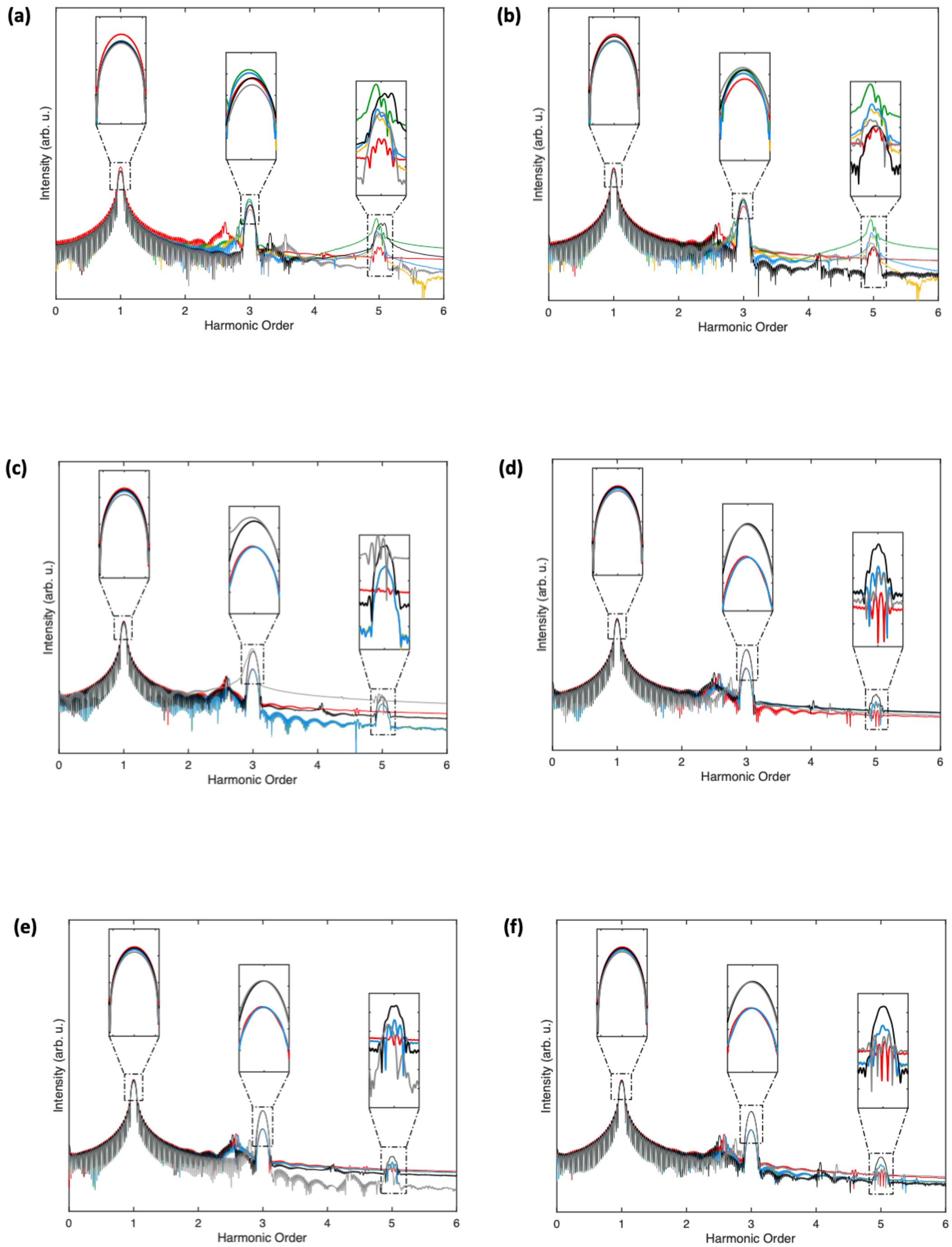

Figure 5.3. HHG spectrum of butadiene computed with STO-3G, 6-31G(d), 6311G, 6-311G(d) ,SVP and cc-pVDZ basis sets are plotted in (A)-(G) under cosine squared pulse (0.007 (a.u),ω=0.1102 (a.u.)) in z-direction within 52.8 fs. For each basis set, CIS(red), CISD(dark-green), CISDT (gold), CISDTQ (blue), NOCIS (black) and HPNOCIS (gray) are included and analyzed in (4,4) active space.

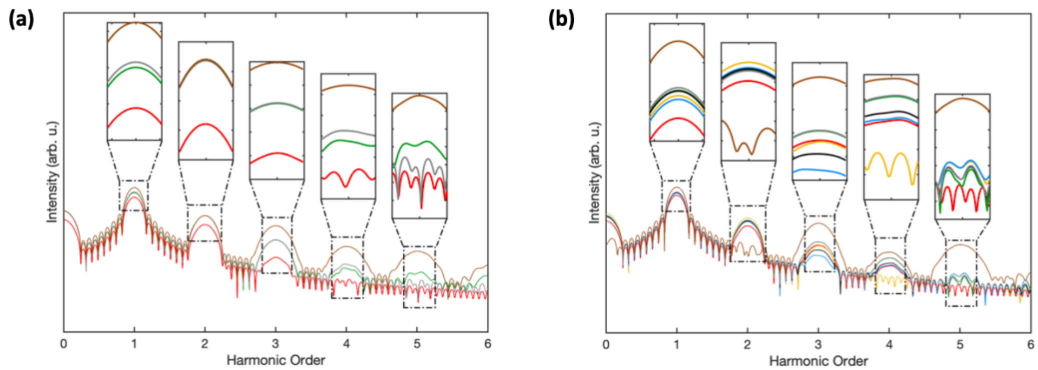

Figure 5.4. HHG spectrum of water computed using CIS and NOCIS methods under cosine squared pulse (0.05 (a.u),ω=0.03675 (a.u.)) in z-direction within 52.8 fs. In both methods, STO-3G(red), 6-31G(d)(dark-green), 6-311G(gold), 6-311G(d)(blue), SVP(black), cc-pVDZ(gray), and aug-cc-pVDZ(brown) are discussed here.

OVERALL CONCLUSIONS

In this thesis we have explored wavefunctions constructed from linear combinations of self-consistent field solutions. The necessary theory to understand the developments in subsequent chapters was described in chapter 2. In chapters 2-5, the developments that comprise the main research of the thesis work were provided. To construct wave function theories based on linear combinations of self-consistent field solutions, there were three projects that touched on three different areas of development. In chapter 3, algorithms for searching and locating sets of self-consistent field solutions were investigated. In chapter 4, the connection of self-consistent field solutions to the specific electron-electron interactions present in molecular systems were analyzed, along with development of tools for establishing the specific combination of self-consistent field solutions needed to describe particular electronic states. Finally in chapter 5, the performance of a wave function constructed from a linear combination of self-consistent field solutions was examined, in the context of real-time time-dependent wavefunction theory.

Global elucidation of SCF solution among high-dimensional wavefunction parameter space is facillated by Lie-algebraic description of wavefunction parameters and basin hopping searching algorithm. They allow to generate different symmetries of wavefunction as the initial guess and determine the next starting point based on the Metropolis-Hasting criteria. H_4 is used to explore the features of parameters inside basin hopping code. The optimal conditions are applied to benzene and NO_2, and antiaromaticity and spin crossover are identified based on the located SCF solutions. This method is effective up to 100 basis functions.

The hypothesis of whether the use of nonorthogonality improves orthogonal configuration interactions method are tested through polarizability, second-order hyperpolarizability and HHG spectrum in three small molecules. These three chemical quantities are obtained from the superposition's dipole information via finite field method. NOCIS method correlates with ground state so that giving the larger polarizability and second-order hyperpolarizability for H_2. HPNOCIS and NOCIS methods provide relative higher fitting values in polarizability in all tested basis sets in butadiene due to the extra correlations captured. However, diffused function required for both polarizability and second-order hyperpolarizability compared with response theory, but the requirement of balance fitting percentage and convergent field are crucial to consider. Nonorthogonality is able to capture the higher order substitutions, but not necessary close to the higher order orthogonal configuration interactions method, since NOCIS tends to give a good description of excited states but missing some of the dynamic correlation in the ground state.

The nonorthogonal highlights the importance of capturing the correct amount of correlations. Since each SCF serves as the indicator of the correlation mechanism, the collection of SCF solutions based on the wavefunction's symmetry can interpret the multiple correlation mechanisms. The disconnectivity graph is used to connect SCF solutions according to their symmetry and energy values using the home-made stability code. H2 is used to define the rules for connections for the SCF solutions within high-dimensional wavefunction parameter space. The multiple correlation mechanisms occur when solutions have same energy but connecting to the different RR solution nodes. In this case, the orbital entropy and mutual information are introduced to further differentiate how same or how different they are from the orbital levels.

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APPENDIX A: GLOBAL SEARCHING METHOD ON ENERGY LANDSCAPE CHAPTER SUPPLEMENTARY INFORMATION

Effect of Temperature on Acceptance Ratio

Regardless of wavefunction symmetry or number of iterations, the simulation temperature was not found to have an effect on the number of solutions located. One possible explanation for this result is that the typical energy gap between the self-consistent field (SCF) solutions is relatively small in a highly multireference molecule such as square H_4 and so all temperatures tested gave a very small $\Delta E/k_bT$ ratio. To test this hypothesis, we examined how the energy of the initial solution at each step changed during the simulation. In fig. S1 the energy of the K0 solution is shown for the first 100 iterations of five different real unrestricted simulation runs using simulation temperatures of 10^4, 10^5 and 10^6 K. As the temperature increases from fig. S1a to fig. S1c it indicates that the $\Delta E/k_bT$ ratio is significantly smaller at 10^6 K than at 10^4 K and so the temperatures investigated spanned the range from low to high acceptance of higher energy solutions. The acceptance ratios shown in table S1 further indicate that the temperatures tested span the acceptance probability range of the Metropolis-Hastings criteria, with less than $<$ 5% of iterations accepted at 10^4 K and rejection of around 5% of iterations at 10^6 K.

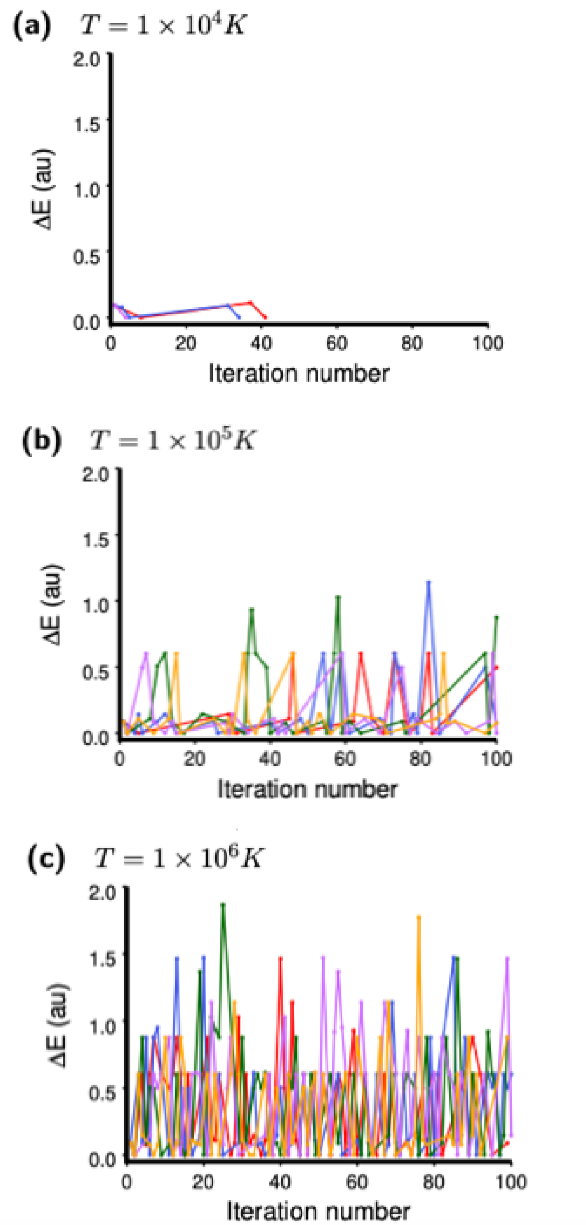

Figure A.1. Energy of initial (identity) solution at each iteration of basin hopping simulation over real unrestricted solutions in minimal basis H_4 for the first 100 iterations using a simulation temperature of a) 10^4 K, b) 10^5 K and c) 10^6 K. Five different simulation runs are show with different color lines and only points where the identity solution changes from the previous iteration are shown. Energy is relative to lowest identified solution.

Table A.1. SCF optimization failure ratio, ratio of convergence to same solution as previous iteration, and ratio of acceptance (to different solution from previous iteration) ratios for basin hopping simulation over real unrestricted solutions in minimal basis H_4 for the first 100 iterations.

Temperature (K)	SCF Failure Ratio	Reconvergence Ratio	MH Acceptance Ratio	Average MH Acceptance Ratio
	0.03	0.15	0.04	
	0.01	0.22	0.05	
10^4	0.04	0.25	0.02	0.03
	0.03	0.18	0.02	
	0.01	0.27	0.02	
	0.02	0.26	0.18	
	0.03	0.26	0.30	
10^5	0.01	0.20	0.30	0.28
	0.01	0.16	0.30	
	0.03	0.25	0.31	
	0.01	0.25	0.72	
	0.02	0.15	0.76	
10^6	0.03	0.19	0.71	0.70
	0.03	0.22	0.69	
	0.03	0.23	0.64	

Molecular Coordinates

Table A.2. H_4 molecular coordinates

H	0.000000	0.000000	0.000000
H	0.000000	0.000000	1.058354
H	1.058224	0.000000	-0.016624
H	1.058224	0.000000	1.074978

Table A.3. Benzene molecular coordinates

C	-1.39434120	-0.79695288	0.00000000
C	0.00081880	-0.79695288	0.00000000
C	0.69835680	0.41079812	0.00000000
C	0.00070280	1.61930712	-0.00119900
C	-1.39412220	1.61922912	-0.00167800
C	-2.09172320	0.41102312	-0.00068200
H	-1.94410020	-1.74926988	0.00045000
H	0.55032680	-1.74946588	0.00131500
H	1.79803680	0.41087812	0.00063400
H	0.55090280	2.57145012	-0.00125800
H	-1.94424420	2.57151012	-0.00263100
H	-3.19132720	0.41120612	-0.00086200

APPENDIX B: EXPLORATION OF ElECTRON DYNAMICS WITH TIME-DEPENDENT NON-ORTHOGONAL METHOD CHAPTER SUPPLEMENTARY INFORMATION

Molecular coordinates

Table B.1. H_2 molecular coordinates

H	0.00000000	0.00000000	0.37068900
H	0.00000000	0.00000000	-0.37068900

Table B.2. Butadiene molecular coordinates

C	0.44024230	0.00000000	0.63180884
C	-0.12999056	0.00000000	1.86119948
C	-0.44024230	0.00000000	-0.63180884
H	0.48158470	0.00000000	2.73897420
H	-1.19511977	0.00000000	1.96126209
H	1.50537151	0.00000000	0.53174623
C	0.12999056	0.00000000	-1.86119948
H	-1.50537151	0.00000000	-0.53174623
H	-0.48158470	0.00000000	-2.73897420
H	1.19511977	0.00000000	-1.96126209

Table B.3. Water molecular coordinates

O	0.00000000	0.00000000	-0.11081188
H	0.00000000	-0.78397589	0.44324751
H	-0.00000000	0.78397589	0.44324751

H_2 lowest energy singlet excitation energies and associated transition dipole moments

Table B.4. H_2 singlet excitation energies (a.u.) computed using 6-31G.

CIS	CISD	NOCIS
0.559802	0.562168	0.542473
1.057562	1.047503	1.052773
1.610503	1.109248	1.606917
	1.416644	
	1.758530	
	1.970276	
	2.103842	
	2.618166	
	3.075885	

Table B.5. H_2 singlet excitation energies (a.u.) computed using cc-pVDZ.

CIS	CISD	NOCIS
0.516777	0.511194	0.486130
0.788421	0.786267	0.771061
1.186104	1.078952	1.186269
1.481008	1.138528	1.481892
2.078875	1.403045	2.078025
2.148526	1.490974	2.143686
3.653051	1.562786	3.651452
	1.714393	
	1.803023	

Table B.6. H_2 singlet excitation energies (a.u.) computed using cc-pVTZ.

CIS	CISD	NOCIS
0.501445	0.496139	0.467790
0.646579	0.640239	0.617571
0.930146	0.927476	0.912726
0.937002	0.928846	0.930128
1.307344	1.051639	1.289115
1.313534	1.241612	1.289770
1.739631	1.252910	1.712175
2.474797	1.315424	2.475150
2.525954	1.414508	2.525185

Table B.7. H_2 singlet excitation energies (a.u.) computed using aug-cc-pVDZ.

CIS	CISD	NOCIS
0.466620	0.464880	0.435344
0.480729	0.481243	0.450028
0.579432	0.577167	0.548019
0.596039	0.595692	0.568569
0.743681	0.737345	0.715093
0.759595	0.754693	0.729460
0.894287	0.893569	0.874927
0.988536	0.973170	0.977326
1.476486	1.071531	1.479771

Table B.8. H_2 singlet excitation energies (a.u.) computed using aug-cc-pVTZ.

CIS	CISD	NOCIS
0.467929	0.467860	0.435222
0.479428	0.482480	0.447636
0.533071	0.530037	0.498535
0.576296	0.578437	0.498656
0.653762	0.655323	0.546346
0.659078	0.661889	0.622763
0.762688	0.758518	0.737419
0.789476	0.793030	0.764914
1.031501	1.020551	1.002604

Table B.9. H_2 transition dipole moments with ground state (a.u.) computed using 6-31G.

CIS	CISD	NOCIS
-1.437308	-1.298826	1.429076
0.000000	0.000000	0.000000
0.316641	0.000000	-0.298991
	0.206613	
	0.163604	
	0.000000	
	0.000000	
	0.031350	
	0.000000	

Table B.10. H_2 transition dipole moments with ground state (a.u.) computed using cc-pVDZ.

CIS	CISD	NOCIS
-1.337488	-1.237845	-1.333533
0.000000	0.000000	0.000000
0.504981	0.000000	0.488086
0.000000	0.415482	0.000000
0.000000	0.137625	0.000000
0.000000	0.000000	0.000000
0.032831	0.000000	0.032632
	0.000000	
	0.000000	

Table B.11. H_2 transition dipole moments with ground state (a.u.) computed using cc-pVTZ.

CIS	CISD	NOCIS
-1.293439	-1.209235	-1.305697
0.000000	0.000000	0.000000
0.000000	0.000000	0.000000
0.625309	0.537735	0.605816
0.000000	0.000000	0.000000
0.000000	0.000000	0.000000
0.051739	0.011315	-0.051585
0.066347	0.000000	-0.065415
0.000000	0.000000	0.000000

Table B.12. H_2 transition dipole moments with ground state (a.u.) computed using aug-cc-pVDZ.

CIS	CISD	NOCIS
-1.012489	-0.993558	1.052096
0.000000	0.000000	0.000000
0.000000	0.000000	0.000000
-0.858467	-0.743801	0.829267
0.000000	0.000000	0.000000
0.000000	0.000000	0.000000
0.470407	0.407636	0.461957
0.000000	0.000000	0.000000
-0.269809	0.000000	-0.266078

Table B.13. H_2 transition dipole moments with ground state (a.u.) computed using aug-cc-pVTZ.

CIS	CISD	NOCIS
-0.986308	-0.973543	1.025751
0.000000	0.000000	0.000000
0.000000	0.000000	0.000000
0.817342	0.702727	0.793789
0.000000	0.000000	0.000000
0.000000	0.000000	0.000000
0.000000	0.000000	0.000000
-0.480371	-0.410584	-0.456536
0.000000	0.000000	0.000000

Polarizability and second-order hyperpolarizability fitting statistics

Table B.14. H_2 polarizability fit quality (R^2) for continuous field with frequency 0.072 a.u., duration 21 fs, and amplitude 0.001 a.u. (top; mean 0.999; standard deviation 0.000) and amplitude 0.1 a.u. (bottom; mean 0.897; standard deviation 0.169).

	CIS	CISD	NOCIS
A=0.001 a.u.			
6-31g	0.999	0.999	0.999
cc-pVDZ	0.999	0.999	0.999
cc-pVTZ	0.999	0.999	0.999
aug-cc-pVTZ	0.999	0.999	0.999
aug-cc-pVTZ	0.999	0.999	0.999
A=0.1 a.u.			
6-31g	0.999	0.999	0.999
cc-pVDZ	0.999	0.999	0.999
cc-pVTZ	0.999	0.999	0.999
aug-cc-pVDZ	0.801	0.726	0.789
aug-cc-pVTZ	0.907	0.842	0.395

Table B.15. H_2 second-order hyperpolarizability fit quality (R^2) for continuous field with frequency 0.072 a.u. and duration 21 fs, with amplitudes 0.001 a.u. (mean 0.358; standard deviation 0.397) and 0.1 a.u. (mean 0.715; standard deviation 0.260).

	CIS	CISD	NOCIS
A=0.001 a.u.			
6-31g	0.870	0.960	0.652
cc-pVDZ	0.998	0.042	0.922
cc-pVTZ	0.014	0.001	0.050
aug-cc-pVDZ	0.177	0.129	0.328
aug-cc-pVTZ	0.070	0.033	0.127
A=0.1 a.u.			
6-31g	0.911	0.875	0.937
cc-pVDZ	0.997	0.598	0.978
cc-pVTZ	0.915	0.912	0.964
aug-cc-pVDZ	0.586	0.348	0.477
aug-cc-pVTZ	0.517	0.462	0.245

Table B.16. Butadiene polarizability fit quality (R^2) for continuous field with amplitude 0.089 a.u., frequency 0.056 a.u. and duration 21 fs (mean 0.959; standard deviation 0.053).

	CIS	CISD	CISDT	CISDTQ	NOCIS	HPNOCIS	CIS (full)
STO-3G	0.996	0.810	0.846	0.823	0.830	0.982	0.992
6-31G(d)	0.987	0.922	0.928	0.909	0.971	0.965	0.972
6-311G	0.965	0.998	0.998	0.998	0.971	0.951	0.948
6-311G(d)	0.963	0.998	0.998	0.998	0.950	0.979	0.949
SVP	0.965	0.998	0.998	0.998	0.966	0.974	0.969
cc-pVDZ	0.965	0.998	0.998	0.999	0.960	0.980	0.968

Table B.17. Butadiene second-order hyperpolarizability fit quality (R^2) for continuous field with amplitude 0.089 a.u., frequency 0.056 a.u. and duration 21 fs (mean 0.756; standard deviation 0.224).

	CIS	CISD	CISDT	CISDTQ	NOCIS	HPNOCIS	CIS (full)
STO-3G	0.766	0.715	0.497	0.555	0.847	0.757	0.958
6-31G(d)	0.855	0.840	0.832	0.855	0.774	0.775	0.948
6-311G	0.880	0.706	0.707	0.707	0.932	0.873	0.963
6-311G(d)	0.873	0.659	0.659	0.659	0.783	0.831	0.966
SVP	0.868	0.636	0.635	0.636	0.810	0.841	0.955
cc-pVDZ	0.872	0.620	0.617	0.617	0.782	0.826	0.959

Table B.18. Butadiene second-order hyperpolarizability fit quality (R^2) for continuous field with various amplitudes, frequency 0.056 a.u. and duration 21 fs applied to nonorthogonal methods with the 6-31G(d) basis set.

	NOCIS	HPNOCIS
0.001	0.014	0.477
0.005	0.152	0.504
0.010	0.343	0.737
0.050	0.928	0.428
0.089	0.774	0.775
0.100	0.794	0.836
0.300	0.412	0.254
0.500	0.051	0.075

CURRICULUM VITAE

Xinju (Spancer) Dong

EDUCATION

B.S.	Western Kentucky University Major: Chemisrty (ACS certificate) GPA:3.8	2014 Fall – 2018 Spring
M.S.	University of Louisville Major: Computational Chemistry GPA:3.8	2018 Fall – 2020 Fall
Ph.D	University of Lousiville Major: Computational Chemistry	2020 Spring – 2023 Fall

RESEARCH WORK EXPERIENCE

Graduate Research Assistant: University of Louisville, Louisville, KY (Aug 2018 – Dec 2023)

Global mapping of self-consistent field solution space (2018-2023): Advisor: Lee M. Thompson

- Algorithms to locate solutions to self-consistent field equations for use as a basis state in post-Hartree-Fock method;
- Writing efficient code (Fortran 90/77) for characterization of the self-consistent field energy landscape;
- Implementation of single orbital entropy and mutual information to differentiate multiple correlation mechanisms from disconnectivity graph project.

Time-dependent non-orthogonal method vs orthogonal method study (2021-2023): Advisor: Lee M. Thompson

- Data analysis in orthogonal and non-orthogonal configuration interaction methods;
- Using the patterns identified to explore whether the non-orthogonal method is better than orthogonal method that can achieve accuracy with lower computational cost under real-time time-dependent context.

Undergraduate Research Assistant: Western Kentucky University, Bowling Green, KY (Aug 2015 – May 2018)

Creation of an Interdisciplinary Platform for Renewable Solar Fuel Production (2015-2017): Advisor: Cao Yan

- Primary assistant synthesizing TiO2 nanotube arrays with photo-responsive properties on titanium foil;
- Instrumental in evaluating, testing, and analyzing complex data for trends and statistical significance;
- Maintained and troubleshot laboratory equipment individually, or with customer support center. disconnectivity graph project.

Enhancement of Solar Fuel Productivity and Energy Consumption Saving (2015-2017): Advisor: Cao Yan

- Coordinated lab responsibilities among team members to forward research agenda and solve problems;
- Synthesized improved photocatalytic nanocomposites of TiO2 with reduced graphene oxide;
- Achieved experiment results resulting in a 20% increase of performance of photosensitive material;
- Provided clear quantitative analytical chemistry conclusions through extensive analyzing and testing.

Laboratory Support Activities:

- Aided current projects with the use of software including Gaussian in analyzing molecule's vibrations and rotation modes with different simulation methods;
- Interpreted and clarified graphs generated from UV-vis, IR, MS, LCMS, GCMS, and HPLC for team;
- Prepared samples for laboratory partners IAW complex, exacting, procedures;
- Transposed experiment raw data, into readable sketch graphs, and analyses for the team to use.

SKILLS

Professional:

- Skilled and experienced with acquiring and interpreting analytical data, and preparing reports.
- Trained in materials characterization techniques including: optical techniques, SEM, ICP, FTIR, TGA, LCMS, GC, UV-vis, and building effective photometric titration instruments on-site in lab.

Computational:

- Computer programming: Mathematica, Gaussian, Origin, Fortran, Python, Perl, MATLAB, DALTON, GAMESS, Microsoft Office (Word, Excel, PowerPoint, and associated Office Suite technology).

Relational:

- Mentored seamless integration of new students to university and chemistry lab;
- Bilingual: English, Chinese;
- Able to exhibit strong relationship, self-management, and social awareness abilities.

Teaching:

- Experienced in teaching undergraduate chemistry labs; training two undergraduates' research in computational chemistry; training the first-year doctoral student.

PUBLICATIONS

- Dong, X., Thompson, L. M. Time propagation of electronic wavefunctions using nonorthogonal determinant expansions. **2023**, Submitted.
- Dong, X., Thompson, L. M. Connections Between the Self-Consistent Field Energy Landscape and Electron Correlation Mechanisms Using a Symmetry-Based Graphical Approach, **2023**, In Progress.

- Dong, X., Megan J. Mackintosh., Piotr Lodowski., Lee M. Thompson Pawel M. Kozlowski. Electronic and photolytic properties of carbonmonoxy hemoglobin. **2023**, In progress.

- Dong, X., Mahler, A. D., Kempfer-Robertson, E. M., Thompson, L. M. Global Elucidation of Self-Consistent Field Solution Space Using Basin Hopping. Journal of Chemical Theory and Computation, **2020**, 16(9), 5635-5644.

- Chen, Y., Gao, H., Wei, D., Dong, X., Cao, Y. Langmuir-Blodgett assembly of visible light responsive TiO2 nanotube arrays/graphene oxide heterostructure. Applied Surface Science, **2017**, 392, 1036-1042.

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PRESENTATIONS

- Xinju,Dong, Lee.Thompson. (**2023**). Understanding of Correlation Mechanisms Through Disconnectivity Graph, San Francisco, CA.

- Xinju,Dong, Lee.Thompson. (**2023**). Real-time Time-dependent Non-orthogonal Configuration Interaction, ACS Meeting, Indianapolis, IN.

- Xinju,Dong, Lee.Thompson. (**2022**). Exploring the Self-Consistent Field (SCF) Energy Landscape as a Tool for Identifying New Forms of Reference Wavefunctions, ACS Meeting, Chicago, IL.

- Xinju,Dong, Lee.Thompson. (**2021**). Global Searching of Self-Consistent Field Solutions Extended to Large Systems, SERMACS 2021 Regional Meeting, Birmingham, AL.

- Xinju,Dong, Lee.Thompson. (**2021**). Global Elucidation Approached for the Self-Consistent Field Solution Space of Molecules, ACS Oral Presentation online.

- Xinju,Dong, Lee.Thompson. (**2019**). Basin hopping approach to global exploration of self-consistent field solution space, SERMACS 2019 Regional Meeting, Knoxville, TN.